

NUMERICAL INVESTIGATION OF TURBULENCE MODEL ACCURACY ON LOW-REYNOLDS-NUMBER E387 AIRFOIL AERODYNAMICS

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We the undersigned declare that the work presented in this project is extremely our effort and any information used from other sources has been referenced.

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NOMENCLATURE

The principal symbols and abbreviations used throughout this report are listed below. Where a quantity is dimensional, representative SI units are given in parentheses; model constants and coefficients are dimensionless unless stated otherwise.

Roman symbols

Symbol	Description
a_1	Constant in the SST eddy-viscosity limiter (= 0.31)
c	Airfoil chord length (m)
C_d	Drag coefficient (also written c_d in the setup)
C_l	Lift coefficient (also written c_L in the setup)
C_l / C_d	Lift-to-drag ratio
Ca_1, Ca_2, Ce_1, Ce_2	Calibration constants of the intermittency transport equation
$C_{\theta t}$	Constant in the transported-Reynolds-number source term (= 0.03)
$CD_{k\omega}$	Positive part of the cross-diffusion term in the ω -equation
D_k	Destruction term of the SST k -equation
E_γ	Destruction (relaminarisation) source of intermittency
e^N	Amplification-ratio (e-to-the-N) transition-prediction method
F_1, F_2	SST blending functions
F_{length}	Correlation controlling the streamwise length of transition
F_{onset}	Function triggering the onset of transition
$F_{reattach}$	Reattachment function in the separation correction
F_{turb}	Function controlling relaminarisation in laminar regions
$F_{\theta t}$	Blending function in the transported-Reynolds-number equation
k	Turbulent kinetic energy (m^2/s^2)
N	Amplification factor of the e-to-the-N method
$\bar{p}$	Mean (time-averaged) static pressure (Pa)
P_k	Production of turbulent kinetic energy
$\tilde{P}_k$	Limited production of turbulent kinetic energy used in the SST model
P_γ	Production source of intermittency
$P_{\theta t}$	Source term of the transported-Reynolds-number equation
Re, Re_c	Chord (chord-based) Reynolds number
Re_ν	Vorticity (strain-rate) Reynolds number
Re_θ	Momentum-thickness Reynolds number
$Re_{\theta c}$	Critical momentum-thickness Reynolds number
$Re_{\theta t}$	Transition-onset momentum-thickness Reynolds number
$R\tilde{e}_{\theta t}$	Transported transition-onset momentum-thickness Reynolds number
s_1	Constant of the separation-induced transition correction (= 2)
S	Magnitude of the mean strain-rate tensor
S_{ij}	Mean strain-rate tensor
t	Time; also, the relaxation time scale in the $R\tilde{e}_{\theta t}$ source term
Tu	Freestream turbulence intensity
$\bar{u}_i$	Mean (time-averaged) velocity component
u_i'	Fluctuating velocity component

U	Local velocity magnitude
U_{∞}	Freestream velocity (m/s)
x_i	Cartesian coordinate directions (tensor index notation)
y	Distance to the nearest wall (m)
y^+	Dimensionless wall distance (first-cell wall spacing)

Greek symbols

Symbol	Description
α	Constant in the ω -equation production term
β, β^*	SST destruction-term constants ($\beta^* = 0.09$)
γ	Intermittency (fraction of time the flow is turbulent)
γ_{eff}	Effective intermittency applied to the k-equation
γ_{sep}	Separation intermittency of the bubble correction
δ_{ij}	Kronecker delta
ε	Turbulent dissipation rate
λ_{θ}	Pressure-gradient parameter
μ	Dynamic (molecular) viscosity (Pa·s)
μ_t	Turbulent (eddy) viscosity
ν	Kinematic viscosity (m ² /s)
ν_t	Kinematic eddy viscosity
ρ	Fluid density (kg/m ³)
$\sigma_k, \sigma_{\omega}$	Diffusion (turbulent Prandtl-number) constants for k and ω
$\sigma_{\gamma}, \sigma_{\theta t}$	Diffusion constants of the two transition equations
φ	Generic SST model constant, blended via F_1
ω	Specific dissipation rate (1/s)
Ω	Magnitude of the vorticity

Abbreviations and acronyms

Abbreviation	Meaning
2D	Two-dimensional
3D	Three-dimensional
AoA	Angle of attack
CFD	Computational Fluid Dynamics
LCTM	Local Correlation-based Transition Model
LSAT	Low-Speed Airfoil Test (UIUC programme)
LSB	Laminar separation bubble
LTPT	Low-Turbulence Pressure Tunnel (NASA Langley)
NASA	National Aeronautics and Space Administration
RANS	Reynolds-Averaged Navier–Stokes
SST	Shear Stress Transport

UAV	Unmanned aerial vehicle
UIUC	University of Illinois at Urbana–Champaign

ABSTRACT

This research includes investigation and comparison of the standard $k-\omega$ SST model and the Transition SST ($\gamma-Re_\theta$) model accuracy and efficiency on low Reynolds number $Re = 200,000-300,000$ UAV airfoil performance, through a CFD assessment and validating the results against UIUC wind tunnel data, the airfoil used in the experiment is the E387.

1.INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Since the creation of the small, unmanned ariel vehicles (UAV) an exponential improvement in its design, and manufacturing processes are being constantly observed, what began as a single purpose technology limited to military reconnaissance has evolved into a global industry that includes precision agriculture, infrastructure inspection, environmental monitoring, cinematography, parcel delivery, search-and-rescue operations, and border surveillance (Mueller and DeLaurier, 2003). This rapid spread has created a pressing demand for aerodynamic prediction tools that are simultaneously accurate, computationally affordable, and dependable in the specific flight regime that small UAVs occupy, a regime that differs fundamentally from the one for which most conventional aerodynamic design methods were developed.

Unlike commercial transport aircraft, cargo aircraft, or military fighters, which operate at chord-based Reynolds numbers of the order of 10^7 to 10^8 , small fixed-wing UAVs typically fly at chord Reynolds numbers between approximately 1×10^5 and 5×10^5 (Lissaman, 1983; Mueller and DeLaurier, 2003). Within this range, and particularly within the range of $Re = 200,000$ to $300,000$ that is fundamental to this project, the boundary layer that develops over a lifting surface is neither fully laminar nor fully turbulent. Instead, it exists in a *transitional* state: the flow leaves the leading edge laminar, is destabilized by the opposing pressure gradient on the aft portion of the suction surface and frequently separates from the surface before transition to turbulence occurs. The separated laminar shear layer then transitions in the free shear region, gains enough momentum through turbulent mixing, and may reattach to the surface as a turbulent boundary layer, enclosing a region of recirculating fluid known as a *laminar separation bubble* (LSB) (Tani, 1964; Gaster, 1967; Horton, 1968).

The appearance of the laminar separation bubble including its proportions and behavior controls the aerodynamic performance of airfoils in this regime. The bubble changes the effective shape of the airfoil, alters the surface pressure distribution, increases pressure drag greatly, and controls the character of stall (O'Meara and Mueller, 1987). Carmichael (1981) famously described the Reynolds number range of roughly 70,000 to 200,000 as one of the most difficult regimes in applied aerodynamics specifically because of this sensitivity. Tiny fluctuations in Reynolds number, angle of attack, freestream turbulence intensity, or surface condition can cause the bubble to grow, shrink, move, or burst entirely with intense effects on lift, drag, controllability, and stability. For UAV designers, this

means the fully turbulent flow can be inaccurate to the degree that it is no longer usable, using performance predictions tools that are made specifically for high Reynolds numbers.

Computational Fluid Dynamics (CFD), and specifically the Reynolds-Averaged Navier–Stokes (RANS) approach, is the pillar of industrial aerodynamic analysis because it provides an acceptable margin between physical reliability and computational expense. However, the most widely used RANS turbulence models including the industry standard $k-\omega$ Shear Stress Transport (SST) model of Menter (1994) were created under the assumption that the boundary layer is fully turbulent from the leading edge onward. They have no function to represent a laminar region, no mechanism to predict where transition occurs, and therefore no mechanism to capture the laminar separation bubble that controls low-Reynolds-number airfoil behavior. Despite this well-known limitation, the standard $k-\omega$ SST model is still the default choice of many university groups and small UAV developers, largely because of its low computational cost, numerical accuracy, and familiarity (Menter, Kuntz and Langtry, 2003).

In order to fix this specific flaw, Menter, Langtry and co-workers developed the Transition SST model – also known as the $\gamma-Re_\theta$ model – which augments the baseline $k-\omega$ SST model with two additional transport equations that allow the location and extent of laminar-to-turbulent transition to be predicted using only local flow variables (Menter et al., 2006; Langtry and Menter, 2009). The model was designed particularly to capture events such as natural transition, bypass transition, and significantly for this project, separation-induced transition of the type that produces laminar separation bubbles.

This report presents a numerical investigation, conducted in ANSYS Fluent, into the accuracy of these two turbulence modelling approaches when applied to the Eppler E387 airfoil – one of the most thoroughly documented low-Reynolds-number benchmark airfoils in the open literature – at chord Reynolds numbers of 200,000 and 300,000, across a range of angles of attack from 0° to beyond stall. This chapter establishes the scientific context for that investigation: it reviews the physics of low-Reynolds-number aerodynamics and the laminar separation bubble (Section 1.2), introduces the E387 airfoil and its documented characteristics (Section 1.3), explains why the two turbulence models are being compared (Section 1.4), presents and explains the governing equations of each model (Sections 1.5 and 1.6), and closes by stating the aim, objectives, and research questions of the project (Sections 1.7 and 1.8).

1.2 LOW-REYNOLDS-NUMBER AERODYNAMICS AND THE LAMINAR SEPARATION BUBBLE

1.2.1 THE SIGNIFICANCE OF REYNOLDS NUMBER

The chord Reynolds number,

$$Re_c = \frac{\rho U_\infty c}{\mu},$$

where ρ is the fluid density, U_∞ the freestream velocity, c the airfoil chord, and μ the dynamic viscosity, expresses the ratio of inertial to viscous forces and is the single most

important similarity parameter governing boundary-layer behaviour. At high Reynolds numbers, viscous effects are confined to a thin boundary layer that transitions to turbulence close to the leading edge; the turbulent boundary layer is energetic, resistant to separation, and comparatively insensitive to small surface imperfections. At the low Reynolds numbers characteristic of small UAVs, the picture changes entirely: the laminar boundary layer persists over a large fraction of the chord, and laminar boundary layers are notoriously incapable of negotiating adverse pressure gradients (Lissaman, 1983). A laminar boundary layer can typically only withstand a small pressure recovery before separating, whereas a turbulent boundary layer – by virtue of the cross-stream momentum exchange provided by turbulent mixing – can remain attached against far stronger adverse gradients.

The consequence is that, over the aft portion of the suction surface of an airfoil at moderate incidence, the laminar boundary layer separates before natural transition has time to occur on the surface. What happens next determines the aerodynamic character of the airfoil.

1.2.2 FORMATION AND STRUCTURE OF THE LAMINAR SEPARATION BUBBLE

The classical description of the laminar separation bubble is due to Tani (1964), Gaster (1967) and Horton (1968), and proceeds as follows. Downstream of the suction peak, the laminar boundary layer encounters an adverse pressure gradient and separates from the surface at a laminar separation point. The detached shear layer, now removed from the damping influence of the wall, is highly unstable: Kelvin–Helmholtz-type instabilities amplify rapidly, and the shear layer undergoes transition to turbulence a short distance downstream of separation. The turbulent shear layer drags high momentum fluid from the freestream, thickens, and curves back to the surface and reattach as the turbulent boundary layer, with the condition that the adverse pressure gradient is not exceedingly high. The region between the separation point, the free shear layer, and the reattachment point contains slowly recirculating fluid and is named the laminar separation bubble.

The bubble induces a particular signature on the surface pressure distribution, a plateau of near constant pressure over the laminar front part of the bubble followed by a prompt pressure recovery over the transition and reattachment zone. This sign is one of the key diagnostic properties used to recognize bubbles in wind-tunnel experiments and in CFD solutions and is a feature that a turbulence model represents if it is to be trustworthy in this regime.

Gaster (1967) distinguished between short bubbles, which occupy only a small percentage of the chord and act as a localized transition activating device with modest global effect on the pressure distribution, and long bubbles, which spread over a substantial fraction of the chord, significantly alter the effective airfoil geometry, diminish the suction peak, and severely reduce lift and drag. A short bubble may burst into a long bubble, or fail to reattach altogether, as the AoA is increased or the Reynolds number is reduced, creating an abrupt loss of lift, a phenomenon related with thin-airfoil or combined stall behavior and with the clear hysteresis loops measured on many low-Reynolds-number airfoils (O'Meara and Mueller, 1987).

1.2.3 AERODYNAMIC CONSEQUENCES

The practical consequences of bubble dominated flow, all related to UAV design, may be summarized as follows. **Drag penalty:** The bubble displaces the outer flow, thickens the boundary layer at reattachment, and produces a substantial pressure-drag increment. For E387, drag measurements at $Re = 100,000$ are distinctly higher than at $Re = 300,000$ mainly because of the larger bubble at the lower Reynolds number (Selig et al., 1995; McGhee, Walker, and Millard, 1988). **Non-linear lift behavior:** Movement of the bubble with incidence alters the lift-curve slope and can give readings in the C_l - α curve that no fully turbulent calculation will predict. **Stall character:** Whether the airfoil stalls smoothly from the trailing edge or suddenly burst through bubble depends on the bubble dynamics. **Sensitivity to surface condition.**

It follows that any numerical technique used for low-Reynolds-number UAV aerodynamics must be able to embody, a laminar boundary layer region, a predicted transition location receptive to the local pressure gradient and turbulence environment, and a separation induced transition with turbulent reattachment. These are exactly the capabilities that separate the Transition SST model from the standard k - ω SST model, and they identify the axis of comparison of this project.

1.3 THE EPPLER E387 AIRFOIL

1.3.1 ORIGIN AND GEOMETRIC CHARACTERISTICS

The E387 airfoil was created by Richard Eppler in the early 1960s, initially for high performance model sailplanes, using his inverse conformal mapping design method in which the velocity distribution is specified and the geometry recovered from it. The section is thin and moderately cambered, it has a maximum thickness of approximately 9.1% of chord located near 31% chord, and a maximum camber of approximately 3.2–3.3% of chord located near mid-chord (Selig et al., 1995). The design philosophy behind the section is characteristic of Eppler's low Reynolds number work, the upper surface pressure distribution features a steady, extended pressure recovery intended to manage the laminar boundary layer and limit the severity of laminar separation, while the lower surface remains benign across the effective range of lift coefficients.

The E387 gets its best lift-to-drag performance at moderate lift coefficients ($C_l \approx 0.5$ – 0.9) and at the Reynolds numbers of interest here, displays maximum lift coefficients of the order of 1.1–1.3 with stall angles in the region of 10 – 12° , the exact values depending strongly on Reynolds number and on the measuring facility (McGhee, Walker and Millard, 1988; Selig et al., 1995).

1.3.2 THE E387 AS A BENCHMARK: EXPERIMENTAL HERITAGE

The principal reason the E387 was selected for this project is not its performance but its documentation. The E387 is debatably the most tested low-Reynolds-number airfoil in existence, and it functions within the research community as a de facto calibration standard.

The one vital data sets used to validate the experimental results:

The second is **UIUC Low-Speed Airfoil Tests (LSATs)**, The University of Illinois at Urbana–Champaign program led by Selig and co-workers (Selig et al., 1995) measured the E387 repeatedly across its multi volume test campaign, regarding it explicitly as a benchmark section used to validate the wind tunnel methodology itself. Lift was acquired from a force balance and drag from wake momentum deficit studies at Reynolds numbers spanning 100,000 to 460,000, and the results have been shown to agree well with the LTPT data. The UIUC campaign also documented the behavior of the E387 with boundary layer trips and surface disturbances, providing an experimental example for the surface irregularity study done in this project.

The recorded behavior of the E387 across $Re = 100,000\text{--}300,000$ shows exactly the phenomena this project seeks to model. At $Re = 100,000$, a long laminar separation bubble takes over the upper surface at low-to-moderate incidence, producing an explicit drag rise and visible distortion of the drag polar. At $Re = 200,000\text{--}300,000$ the bubble shortens and moves forward with increasing angle of attack, drag falls significantly, and the polar becomes progressively cleaner. This strong, well-documented Reynolds-number dependence between 200,000 and 300,000 makes the E387 an ideal discriminating test case: a turbulence model that cannot represent the bubble will fail in a predictable, measurable way at the lower Reynolds number while appearing more plausible at the higher one.

1.3.3 SUITABILITY FOR THE PRESENT INVESTIGATION

In summary, the E387 was selected because of multiple reasons including, high-quality experimental force data exist at precisely the Reynolds numbers needed which are $Re = 200,000\text{--}300,000$, from independent facilities (UIUC), allowing to perform numerical validation rather than qualitative comparison. the airfoil's aerodynamics in this regime are known to be bubble-dominated, so the physics that differentiates the two turbulence models is guaranteed to be present and influential, the geometry coordinates are freely available through the UIUC Airfoil Coordinates Database and its widespread use in the literature allows the results of this project to be positioned against, and contribute to, an existing body of validation work.

1.4 WHY COMPARE THE $k\text{--}\omega$ SST AND TRANSITION SST MODELS?

RANS turbulence modelling rests on the Reynolds decomposition of the instantaneous flow variables into mean and fluctuating components. Substituting this decomposition into the Navier–Stokes equations and time-averaging yield the RANS equations, which for incompressible mean flow may be written as

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0,$$

$$\rho \left(\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} \right) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial \bar{u}_i}{\partial x_j} - \rho \overline{u_i' u_j'} \right),$$

where the overbar denotes time-average quantities and primes denote fluctuations. The averaging process introduces the Reynolds-stress tensor $-\overline{\rho u_i' u_j'}$, an unknown that must be modelled – this is the turbulence closure problem. Eddy-viscosity models close the system through the Boussinesq approximation,

$$-\overline{\rho u_i' u_j'} = \mu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{2}{3} \rho k \delta_{ij},$$

in which the entire effect of turbulence on the mean flow is represented by a scalar turbulent (eddy) viscosity μ_t , and $k = 1/2 \overline{u_i' u_i'}$ is the turbulent kinetic energy. Two-equation models such as $k-\omega$ SST compute μ_t from two transported turbulence quantities.

The fundamental point for this project is that a conventional two-equation model activates turbulence production wherever the mean flow contains shear, which is everywhere in a boundary layer, from the stagnation point onward. The model therefore behaves as though the boundary layer is turbulent from the leading edge. There is no representation of a laminar region, no transition point, and consequently no possibility of a laminar separation bubble, laminar separation cannot occur because the modelled boundary layer is never laminar. At $Re = 10^7$ this assumption is nearly true and has little to no effect, at $Re = 1.5 \times 10^5$ it removes the dominant physics of the problem.

Two model families are compared in this project (with a laminar calculation as an additional reference case in the methodology):

- The **standard $k-\omega$ SST model** (Menter, 1994) represents the fully turbulent modelling approach that dominates industrial practice. Its selection is deliberate: it is precisely the model that UAV developers and student groups reach for by default, so quantifying *its* error – rather than that of some strawman model – is what gives the project practical value.
- The **Transition SST ($\gamma-Re_\theta$) model** (Menter et al., 2006; Langtry and Menter, 2009) represents the transition-sensitive approach. It is built directly *on top of* the $k-\omega$ SST model, sharing its two turbulence transport equations, wall treatment, and numerical infrastructure, and adding two further transport equations that control where turbulence is allowed to develop.

This architectural relationship is what makes the comparison scientifically clean. Because the Transition SST model reduces to behavior closely resembling the standard SST model once the flow is fully turbulent, any systematic difference between the two solutions can be attributed to the transition of physics alone, rather than to unrelated differences in model formulation, near-wall treatment, or calibration. The comparison therefore isolates a single question: at $Re = 200,000$ – $300,000$ on the E387, how much predictive accuracy is gained by resolving transition, and is that gain worth the additional computational cost of solving four turbulence-related transport equations instead of two?

1.5 THE k - ω SST TURBULENCE MODEL

1.5.1 CONCEPT AND RATIONALE

The Shear Stress Transport model of Menter (1994) was created to combine the complementary strengths of the two classical two-equation models. The k - ω model of Wilcox (1988) integrates accurately through the viscous sublayer to the wall without damping functions and performs well in adverse-pressure-gradient boundary layers but is notoriously sensitive to the freestream value of ω . The k - ε model is insensitive to freestream conditions but performs poorly near walls and in adverse pressure gradients. Menter's solution was to use the k - ω formulation in the inner boundary layer and mixes smoothly into a k - ε formulation (rewritten in k - ω variables) in the outer boundary layer and freestream, the switch being controlled by a blending function F_1 . A second ingredient, the shear-stress transport limiter, bounds the eddy viscosity in adverse-pressure-gradient regions to enforce Bradshaw's observation that the principal turbulent shear stress is proportional to k , which greatly improves the prediction of flow separation under adverse pressure gradients.

1.5.2 GOVERNING EQUATIONS

The model solves two transport equations, one for the turbulent kinetic energy k (dimensions m^2/s^2), representing the energy contained in the turbulent velocity fluctuations, and one for the specific dissipation rate ω ($1/\text{s}$), representing the rate at which that energy is converted to heat per unit of turbulent kinetic energy – effectively an inverse turbulent time scale. In conservation form (Menter, 1994; ANSYS, 2021):

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho u_j k)}{\partial x_j} = \tilde{P}_k - \beta^* \rho k \omega + \frac{\partial}{\partial x_j} \left[(\mu + \sigma_k \mu_t) \frac{\partial k}{\partial x_j} \right],$$

$$\frac{\partial(\rho \omega)}{\partial t} + \frac{\partial(\rho u_j \omega)}{\partial x_j} = \frac{\alpha}{\nu_t} \tilde{P}_k - \beta \rho \omega^2 + \frac{\partial}{\partial x_j} \left[(\mu + \sigma_\omega \mu_t) \frac{\partial \omega}{\partial x_j} \right] + 2(1 - F_1) \rho \sigma_{\omega 2} \frac{1}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j}.$$

Each term has a clear physical interpretation, considered from left to right:

- The **unsteady and convection terms** on the left-hand side describe the accumulation and transport of k and ω by the mean flow.
- The **production term** $\tilde{P}_k$ represents the generation of turbulent kinetic energy by mean-flow shear working against the Reynolds stresses. It is computed as $P_k = \mu_t S^2$, where $S = \sqrt{2\overline{S_{ij}S_{ij}}}$ is the magnitude of the mean strain rate is, and in practice is limited, $\tilde{P}_k = \min(P_k, 10 \beta^* \rho k \omega)$, to prevent spurious turbulence build-up in stagnation regions. In the ω -equation the corresponding production is scaled by α/ν_t .
- The **destruction terms** $\beta^* \rho k \omega$ and $\beta \rho \omega^2$ model the dissipation of turbulent kinetic energy into heat by viscosity at the smallest scales, and the corresponding destruction of ω .

- The **diffusion terms** model the spatial spreading of k and ω by molecular viscosity and by turbulent transport, the latter through the eddy viscosity weighted by the model constants σ_k and σ_ω .
- The **cross-diffusion term** (final term of the ω -equation) arises from transforming the ε -equation into ω variables. It is active only where $F_1 \rightarrow 0$ (outer layer and freestream) and is what confers k - ε -like freestream insensitivity on the model there.

1.5.3 BLENDING FUNCTIONS AND THE SST LIMITER

The blending function F_1 equals 1 deep inside the boundary layer (activating the k - ω branch) and falls to 0 towards the boundary-layer edge (activating the transformed k - ε branch):

$$F_1 = \tanh \left\{ \left[\min \left(\max \left(\frac{\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega} \right), \frac{4\rho\sigma_{\omega 2} k}{CD_{k\omega} y^2} \right) \right]^4 \right\},$$

where y is the distance to the nearest wall and $CD_{k\omega}$ is the positive part of the cross-diffusion term. Every model constant ϕ is blended between its inner (ϕ_1) and outer (ϕ_2) values as $\phi = F_1 \phi_1 + (1 - F_1) \phi_2$.

The eddy viscosity is defined by the shear-stress transport limiter:

$$\mu_t = \frac{\rho a_1 k}{\max(a_1 \omega, SF_2)}, \quad F_2 = \tanh \left\{ \left[\max \left(\frac{2\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega} \right) \right]^2 \right\}.$$

In regions of adverse pressure gradient, where production of k exceeds its dissipation, the limiter caps the modelled shear stress at $\rho a_1 k$ (with $a_1 = 0.31$), preventing the over-prediction of eddy viscosity that causes standard two-equation models to delay or miss separation. The principal calibration constants are $\beta^* = 0.09$; inner set $\sigma_{k1} = 0.85$, $\sigma_{\omega 1} = 0.5$, $\beta_1 = 0.075$; outer set $\sigma_{k2} = 1.0$, $\sigma_{\omega 2} = 0.856$, $\beta_2 = 0.0828$ (Menter, 1994).

1.5.4 STRENGTHS AND LIMITATIONS AT LOW REYNOLDS NUMBER

The SST model is an excellent general-purpose aerodynamic model: it resolves the viscous sublayer directly (given a mesh with $y^+ \leq 1$), predicts turbulent separation under adverse pressure gradients more accurately than either parent model, and has two decades of industrial validation behind it (Menter, Kuntz and Langtry, 2003). Its limitation in the present context is structural rather than incidental: because turbulence production is active from the leading edge, the model develops a turbulent boundary layer immediately. The predicted boundary layer is therefore too resistant to separation at low incidence, no laminar separation bubble can form, the characteristic pressure plateau is absent, drag at low Reynolds number gives a wrong prediction (typically through the wrong mixture of skin-friction and pressure drag), and the stall mechanism defaults to turbulent trailing-edge separation regardless of what the real flow does. Quantifying the size of these errors on the E387 – rather than merely asserting their existence – is a core purpose of this project.

1.6 THE TRANSITION SST (γ - Re_θ) MODEL

1.6.1 CONCEPT: LOCAL, CORRELATION-BASED TRANSITION MODELLING

Historically, engineering transition prediction relied either on empirical correlations, such as that of Abu-Ghannam and Shaw (1980) relating the transition-onset momentum-thickness Reynolds number to freestream turbulence and pressure gradient, or on stability-theory methods such as the e^N method used in codes like XFOIL (Drela, 1989). Both approaches depend on *non-local* information – integrated boundary-layer quantities or instability growth along the surface – which is straightforward in boundary-layer codes but fundamentally incompatible with modern unstructured, parallel, general-purpose CFD solvers, in which no cell “knows” where the boundary layer starts or what its integrated thickness is.

The breakthrough of Menter et al. (2006) and Langtry and Menter (2009) – the Local Correlation-based Transition Model (LCTM) – was to reformulate empirical transition correlations entirely in terms of *local* variables. The key vehicle is the vorticity (strain-rate) Reynolds number,

$$Re_v = \frac{\rho y^2 S}{\mu},$$

a quantity computable in every cell from wall distance and local strain rate. Within a Blasius-type boundary layer, the maximum of Re_v across the layer is proportional to the momentum-thickness Reynolds number ($\max(Re_v) \approx 2.193 Re_\theta$), so a local field quantity can stand in for the non-local integral quantity on which the classical correlations depend. This allows transition physics to be introduced through standard transport equations that any modern solver can manage.

The Transition SST model adds two such equations to the SST framework: one for the **intermittency** γ and one for the **transported transition-onset momentum-thickness Reynolds number** $\tilde{Re}_{\theta t}$. Together with the k and ω equations, the model therefore solves four turbulence-related transport equations – the origin of its higher computational cost.

1.6.2 THE INTERMITTENCY TRANSPORT EQUATION

The intermittency γ is the fraction of time the flow at a point is turbulent: $\gamma = 0$ denotes fully laminar flow and $\gamma = 1$ fully turbulent flow. Its transport equation is (Langtry and Menter, 2009):

$$\frac{\partial(\rho\gamma)}{\partial t} + \frac{\partial(\rho u_j \gamma)}{\partial x_j} = P_\gamma - E_\gamma + \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\gamma} \right) \frac{\partial \gamma}{\partial x_j} \right],$$

with production and destruction sources

$$P_\gamma = F_{\text{length}} c_{a1} \rho S (\gamma F_{\text{onset}})^{0.5} (1 - c_{e1} \gamma),$$

$$E_\gamma = c_{a2} \rho \Omega \gamma F_{\text{turb}} (c_{e2} \gamma - 1),$$

where S and Ω are the strain-rate and vorticity magnitudes. The behaviour of this equation is governed by two dimensionless functions:

- F_{onset} is the **transition trigger**. It is constructed from the ratio $Re_v/(2.193 Re_{\theta c})$, where $Re_{\theta c}$ is the critical momentum-thickness Reynolds number is obtained by correlation from the transported $\widetilde{Re}_{\theta t}$. Upstream of the transition point $F_{\text{onset}} = 0$, production of γ is switched off, and the boundary layer remains laminar. Where the local Re_v exceeds the critical value, F_{onset} activates and intermittency is produced.
- F_{length} is an empirical correlation that controls the **streamwise length of the transition region**, i.e., how rapidly γ grows from 0 to 1 once onset has occurred.

The destruction term E_γ , active in laminar regions through F_{turb} , drives γ back towards zero and enables the model to *relaminarize* the boundary layer where conditions warrant – for example under strong favorable pressure gradients.

1.6.3 THE $\widetilde{Re}_{\theta t}$ TRANSPORT EQUATION

Empirical correlations give the transition-onset momentum-thickness Reynolds number $Re_{\theta t}$ as a function of the freestream turbulence intensity Tu and the pressure-gradient parameter λ_θ – quantities that are well defined in the freestream but not inside a boundary layer. The second transport equation exists to solve this “communication” problem: it takes the correlation value computed in the freestream and diffuses it into the boundary layer as a transported scalar $\widetilde{Re}_{\theta t}$ (Langtry and Menter, 2009):

$$\frac{\partial(\rho \widetilde{Re}_{\theta t})}{\partial t} + \frac{\partial(\rho u_j \widetilde{Re}_{\theta t})}{\partial x_j} = P_{\theta t} + \frac{\partial}{\partial x_j} \left[\sigma_{\theta t} (\mu + \mu_t) \frac{\partial \widetilde{Re}_{\theta t}}{\partial x_j} \right],$$

with the source term

$$P_{\theta t} = c_{\theta t} \frac{\rho}{t} (Re_{\theta t} - \widetilde{Re}_{\theta t}) (1 - F_{\theta t}), \quad t = \frac{500\mu}{\rho U^2}.$$

Outside the boundary layer, the blending function $F_{\theta t} = 0$ and the source term relaxes $\widetilde{Re}_{\theta t}$ towards the local correlation value $Re_{\theta t}(Tu, \lambda_\theta)$ over the time scale t ; inside the boundary layer, $F_{\theta t} = 1$, the source is switched off, and $\widetilde{Re}_{\theta t}$ simply carries the freestream-derived value inward by diffusion. In this way, every cell inside the boundary layer acquires a locally available estimate of the transition-onset Reynolds number that reflects the *upstream* freestream turbulence and pressure-gradient history – restoring, through a transport equation, the non-local information that the classical correlations required. This term is what makes the model responsive to freestream turbulence intensity, which is essential when comparing CFD against wind tunnels of differing flow quality (e.g., the exceptionally low turbulence LTPT versus the UIUC facility).

1.6.4 SEPARATION-INDUCED TRANSITION

Because laminar separation bubbles are the dominant transition mechanism at UAV Reynolds numbers, Langtry and Menter (2009) added a dedicated modification for separation-induced transition. Under a laminar separation, the standard intermittency

production is too slow, the shear layer transitions too late, and the predicted bubble becomes unrealistically long. The correction defines separation intermittency.

$$\gamma_{\text{sep}} = \min \left(s_1 \max \left[0, \frac{Re_v}{3.235 Re_{\theta c}} - 1 \right] F_{\text{reattach}}, 2 \right) F_{\theta t},$$

which is allowed to exceed unity locally (up to 2), and the effective intermittency becomes $\gamma_{\text{eff}} = \max(\gamma, \gamma_{\text{sep}})$. Physically, this allows an intense, rapid burst of turbulence production in the separated shear layer, hastening transition and letting the shear layer to reattach over a realistic length – exactly the mechanism that sets the size of the laminar separation bubble on the E387. The quality of this sub-model is therefore central to Research Question 2 of this project.

1.6.5 COUPLING TO THE SST EQUATIONS

The transition equations influence the flow solution solely by gating the k -equation of the underlying SST model. The production and destruction terms of the k -equation are modified as

$$\tilde{P}_k = \gamma_{\text{eff}} P_k, \quad \tilde{D}_k = \min(\max(\gamma_{\text{eff}}, 0.1), 1.0) D_k,$$

where P_k and D_k are the original SST production and destruction terms? Where $\gamma_{\text{eff}} \approx 0$, turbulence production is suppressed and the boundary layer remains laminar; where $\gamma_{\text{eff}} \rightarrow 1$, the model recovers standard SST behaviour. The blending function F_1 is also altered to guarantee it does not incorrectly switch to the k - ε branch inside thin laminar layers. The principal model constants are $c_{e1} = 1.0$, $c_{a1} = 2.0$, $c_{e2} = 50$, $c_{a2} = 0.06$, $\sigma_\gamma = 1.0$, $c_{\theta t} = 0.03$, $\sigma_{\theta t} = 2.0$, and $s_1 = 2$ (Langtry and Menter, 2009).

1.6.6 EXPENSES AND CAPABILITIES

The Transition SST model captures natural transition, avoiding transition, and separation-induced transition using only local variables, is fully compatible with unstructured parallel solvers, and has been validated across flat plates, airfoils, turbomachinery cascades, and full aircraft configurations (Langtry and Menter, 2009). Its costs and limitations are equally important to state honestly. It solves twice as many turbulence transport equations as standard SST, converges more slowly, and is more demanding of mesh quality: the correlations assume the boundary layer is well resolved, in practice requiring $y^+ \leq 1$ and adequate streamwise resolution through the transition region. Its predictions are restricted to the stated freestream turbulence intensity and turbulent viscosity ratio, whose decay from the inlet must be controlled carefully. Finally, it is a correlation-based model, not a stability solver, it reproduces the outcomes of transition as calibrated against experimental data, and its accuracy outside the calibration space is not ensured. Whether these costs are justified for low-Reynolds-number UAV analysis is exactly the trade-off this project quantifies.

1.7 AIM AND OBJECTIVES

1.7.1 AIM

The principal aim of this project is to compute, through validated 2D steady-state RANS simulation in ANSYS Fluent, the accuracy of the standard $k-\omega$ SST turbulence model and the Transition SST ($\gamma-Re_\theta$) model in predicting the aerodynamic performance of the Eppler E387 airfoil at chord Reynolds numbers of 200,000 and 300,000, with all predictions validated against the UIUC experimental benchmark datasets.

The intended output is to present comparisons and validated engineering recommendations, a defensible statement of which turbulence modelling standard should be regarded as the minimum acceptable for low-Reynolds-number UAV airfoil CFD analysis, and of the manufacturing acceptance consequences of leading-edge surface quality.

1.7.2 OBJECTIVES

First objective is to **Create the theoretical foundation** through a structured literature review of low Reynolds number aerodynamics, laminar separation bubble physics, transitional flow theory, and the formulation and known constraints of the $k-\omega$ SST and Transition SST models. Second objective is to **Develop and validate the computational model**, construct a 2D C-type computational domain around the E387 geometry, generate an inflation layer mesh reaching $y^+ \leq 1$, and validate laminar, $k-\omega$ SST, and Transition SST solutions against the UIUC experimental data from 0° angle of attack to stall. Third objective is to **Compare turbulence model performance across Reynolds numbers** by running steady-state RANS simulations at $Re = 200,000$ and $300,000$ across the full angle-of-attack range, extracting C_l , C_d , C_l/C_d , drag polars, pressure and skin-friction distributions, and compute each model's prediction error relative to experiment. Last objective is to **Incorporate the findings into engineering recommendations** on turbulence model selection and surface finish tolerance for low Reynolds number UAV airfoil analysis, documented in a structured final report.

1.8 RESEARCH QUESTIONS

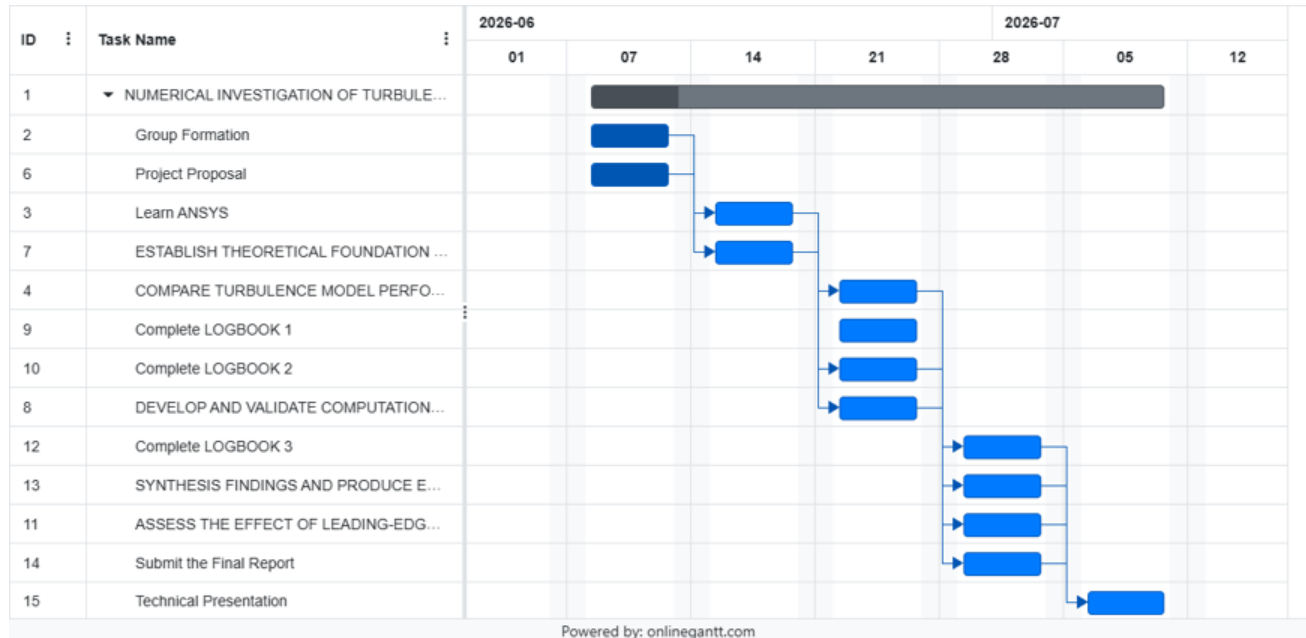
The project is structured to answer four testable questions:

1. How much does the Transition SST $\gamma-Re_\theta$ model improve the prediction accuracy of lift coefficient, drag coefficient, and stall angle of the E387 airfoil relative to the $k-\omega$ SST model, when validated against UIUC wind-tunnel data at $Re = 200,000$ and $300,000$?
2. How do the pressure coefficient distribution, skin-friction behavior, wake development, and flow separation characteristics of the E387 airfoil change with Reynolds numbers and angles of attack at the optimum range of operation?
3. How does Reynolds number affect the predicted aerodynamic performance and stall related behavior of the E387 airfoil across the tested angle of attack range?

4. What turbulence modelling standard should be recommended for low-Reynolds-number UAV airfoil CFD analysis, based on the measured prediction error and the computational cost of each model?

1.9 PROJECT TIME MANAGEMENT

1.9.1 PROJECT PLAN



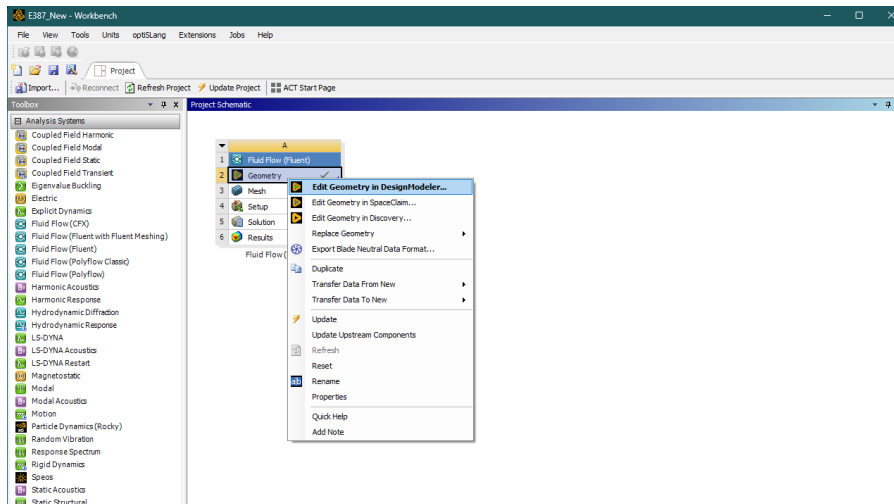
2. METHODOLOGY

2.1 SIMULATION PROCEDURE

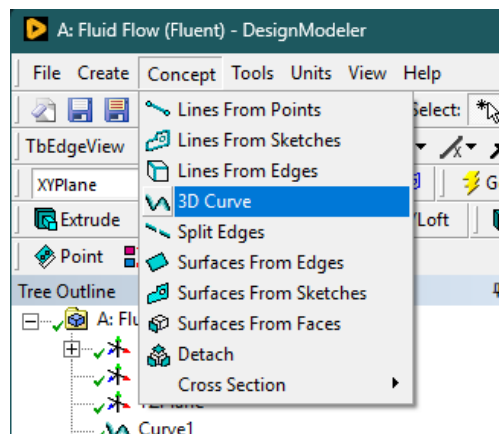
The process used to collect the data was done using simulation through Ansys Fluent Student edition, the first step of using the software is to extract the E387 coordinates file from the UIUC database and import it into the geometry in design modular, then create a C-section domain around the airfoil. The next step is to go into meshing and define each part of the mesh such as the Inlet, outlet, airfoil and walls, finally in the Setup section select the model starting with the standard $k-\omega$ SST turbulence model and then Transition SST ($\gamma-Re_\theta$) model, calculate the X and Y velocity components for the inlet, and the X and Y components for c_L and c_d to simulate different angels of attack.

2.2 GEOMETRY

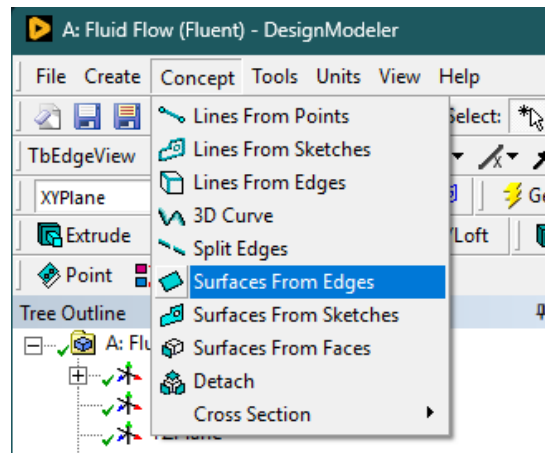
1. To begin creating the geometry for the airfoil open the Ansys Workbench and select Fluid Flow from the Analysis Systems menu, then click on the geometry cell and select **Edit Geometry in Design Modeler**, as shown in the picture.



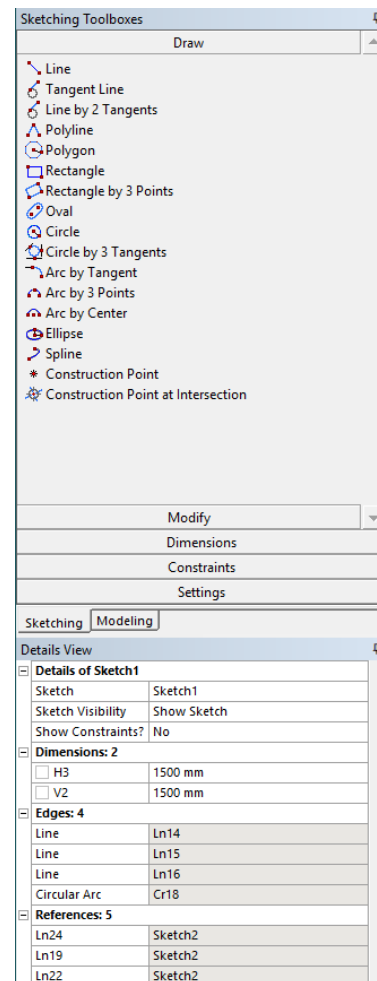
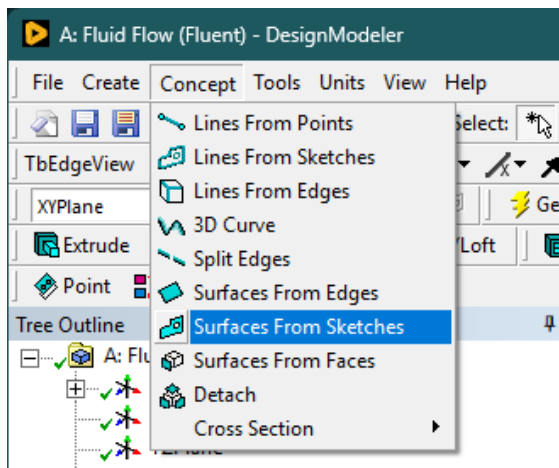
2. After opening the Design Modeler window select the **Concept**, then **3D Curve** from the top taskbar, when prompted to select a file choose the downloaded E387 coordinates text file.



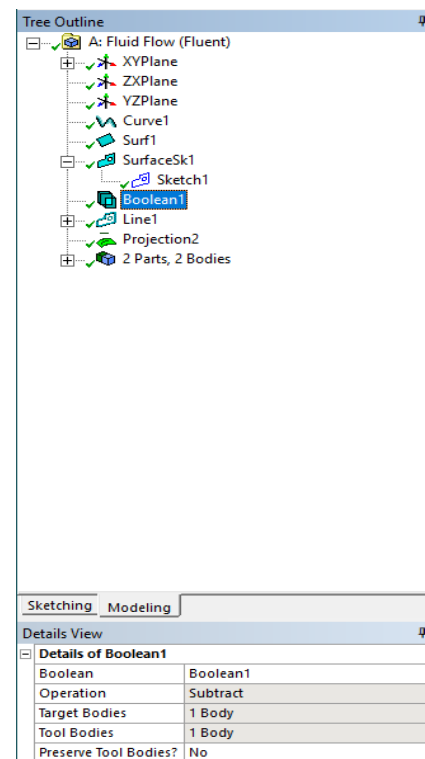
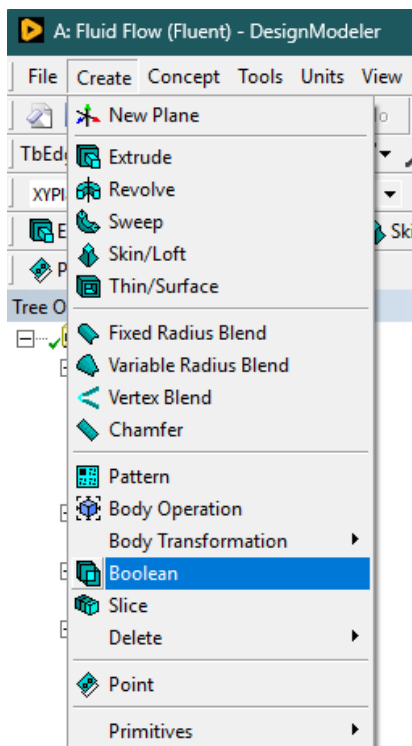
3. An airfoil should appear in the work area, select the airfoil and from the **Concept** taskbar select **Surface from Edges**, and press generate.



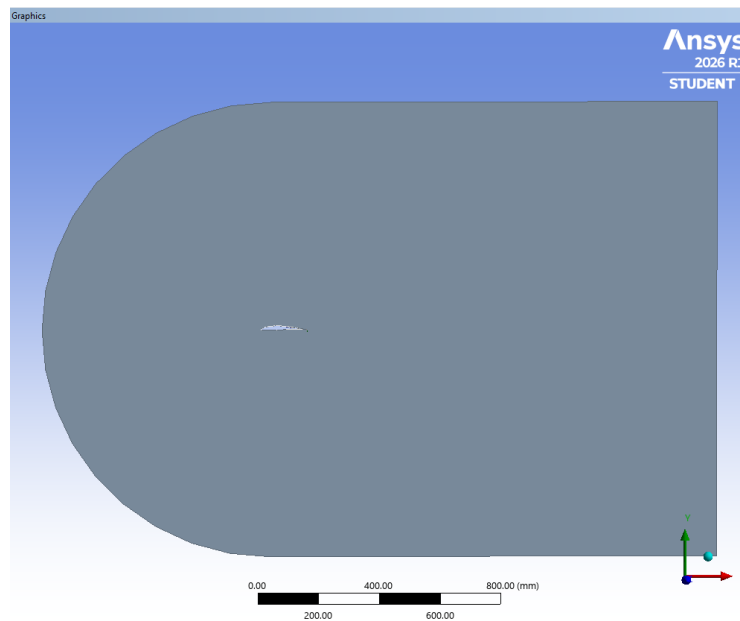
- To create the far-field domain, use the sketching tab in the bottom left and draw a half circle in front of the airfoil and a rectangle connected to the half circle.



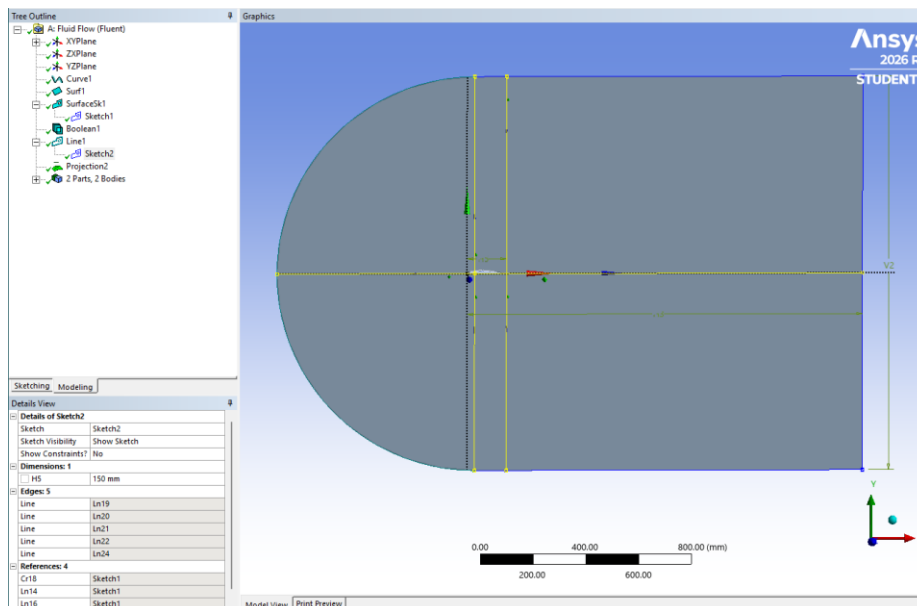
- Select the top and bottom lines and make them symmetrical around the origin and change the dimensions on the lines sketched. As the chord length is 0.15m, the height of the right wall should be chord length times 30 which is 4.5 m, and from the origin to the right-side wall should be chord length times 25 which is 3.75 m, and the radius of the half circle is chord length times 15 which is 2.25. The domain created is a C-Domain which gives higher accuracy results during simulations, Select **Surface from Sketches** from the **Concept** in the taskbar and press generate.
- Now subtract the airfoil from the domain using the **Boolean** function from the **Create** dropdown menu in the task bar.
- From the detailed view in the bottom left corner select the **Subtract** Operation, choose the domain as the target body and the airfoil as the tool body, and choose not to preserve the tool bodies and press generate.



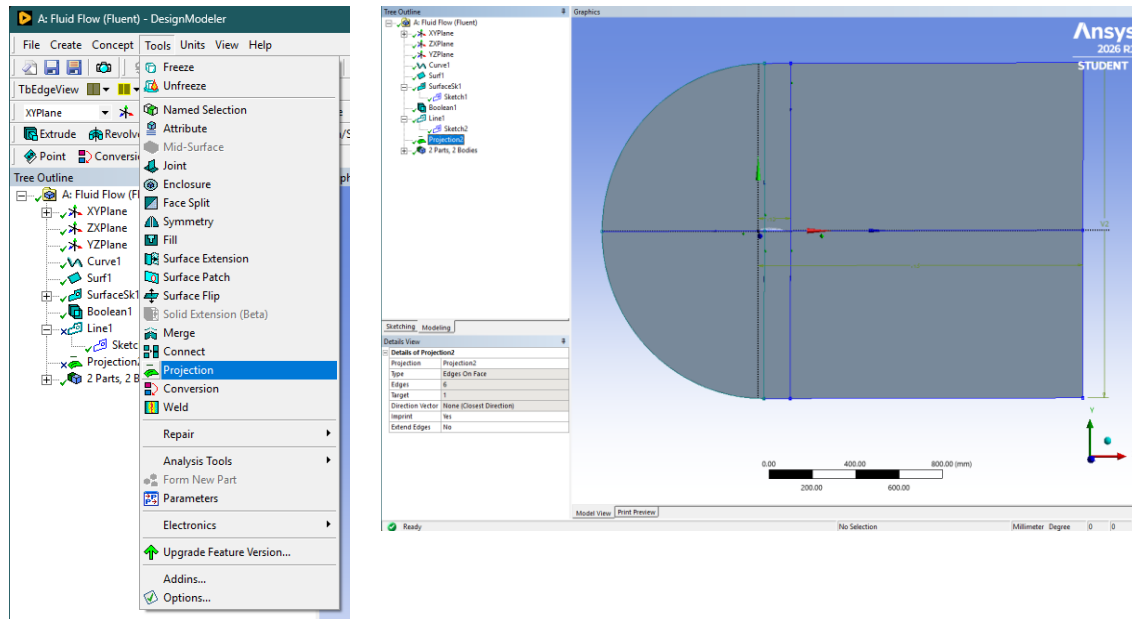
- The graphics window should look like the picture below after pressing generate.



- Create another sketch with lines going from the airfoil to the walls of the domain, to better segment the bottom and top part of the airfoil.



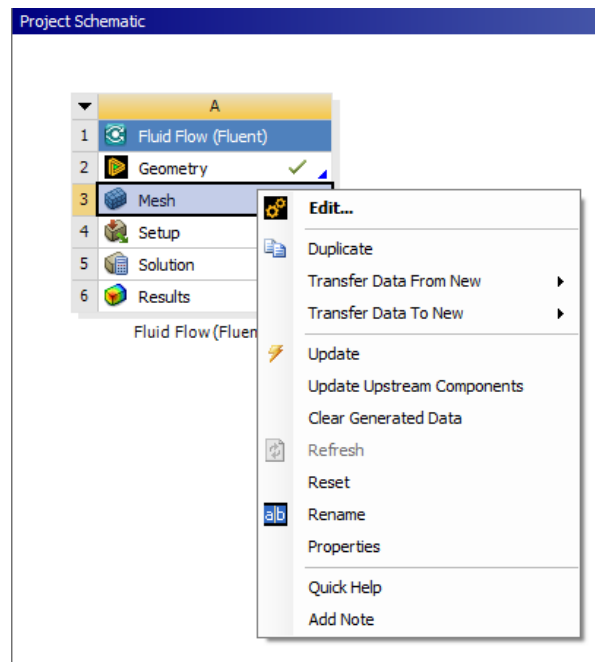
- Select **Tool** from the taskbar, then **Projection** selects all the lines created in the previous



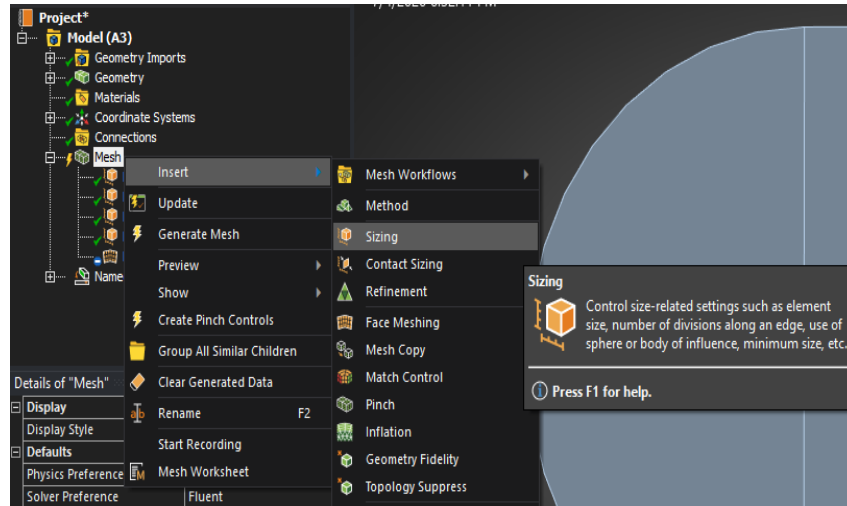
- step and press generate.

2.3 MESH

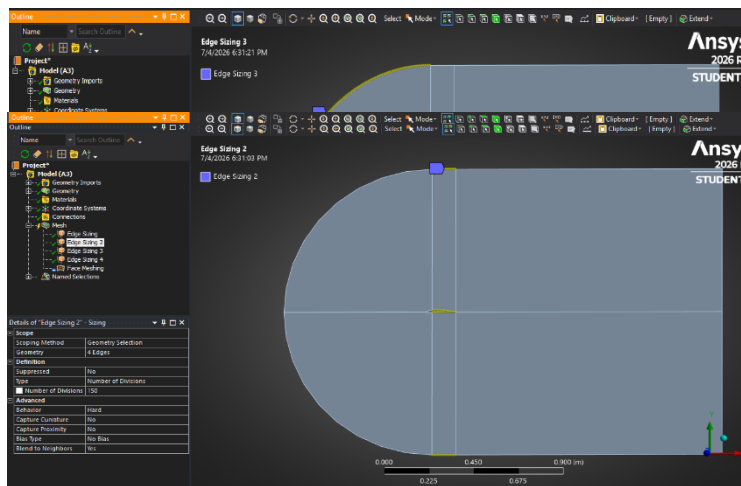
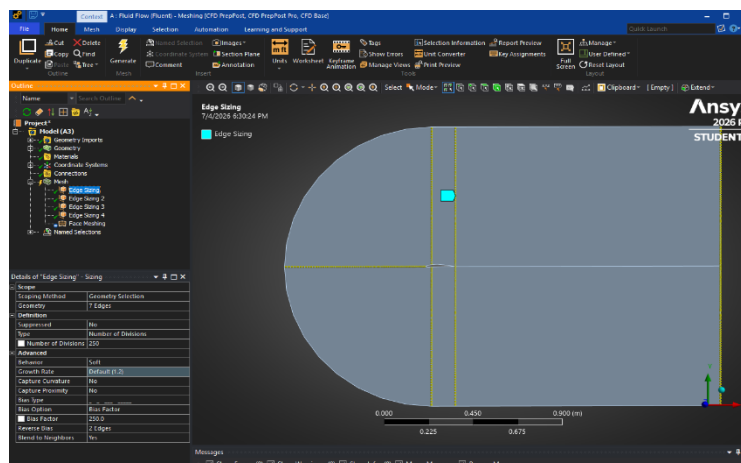
- Open the Mesh to do edge sizing, face meshing, and creation of named selections, in the Ansys Workbench right click the **Mesh** cell and press edit.



- Right click the **Mesh** on the left side of the screen then **Insert** and **Sizing**.



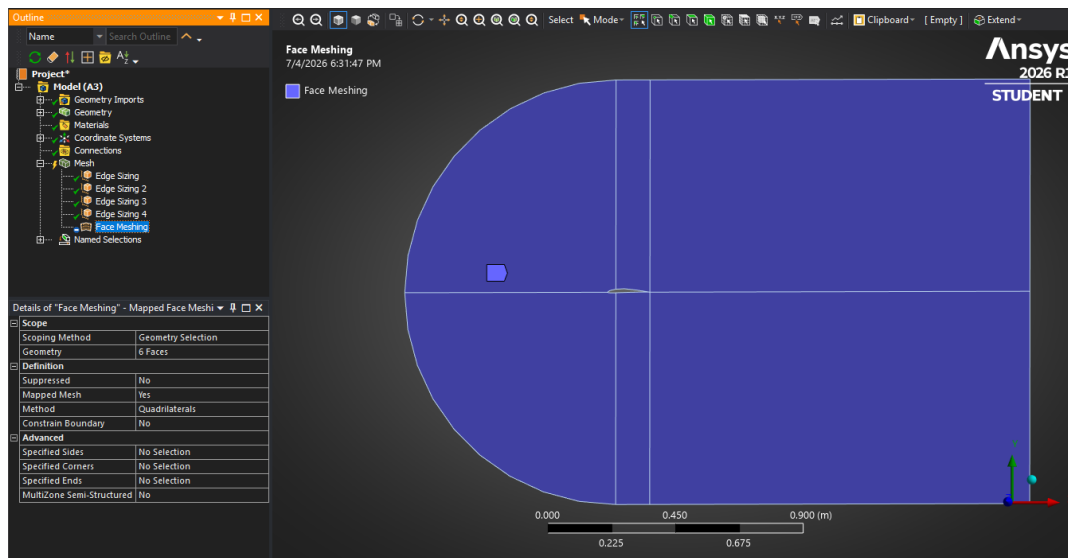
- In the details of edge sizing on the left side of the screen, select the shown geometry and select the settings as shown in the pictures for each group of edges.



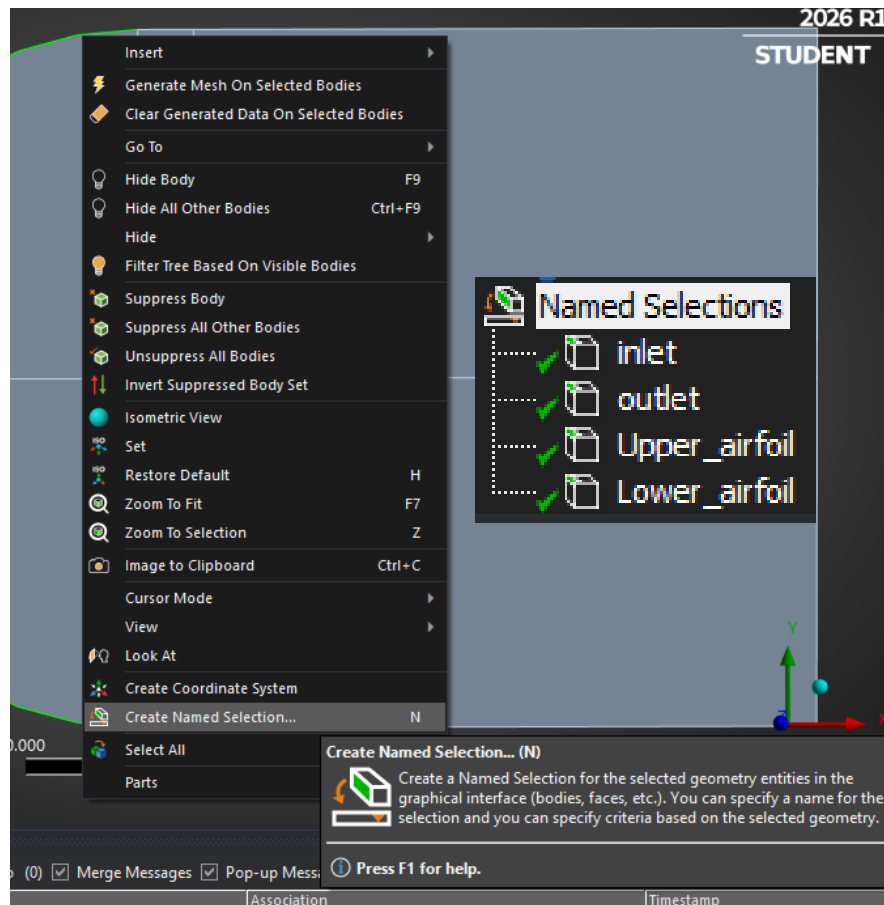
- Next step is to create face meshing, right click **Mesh** on the left side of the screen, then **Insert** and **Face Meshing**.



- In the details of Face Meshing screen on the left side select the six faces on the domain by pressing CTRL and selecting each one, as shown in the picture.



- Finally in meshing create named selection for inlet, outlet, walls, upper and lower airfoil to ensure smooth and clear process while obtaining the results, to create named selection choose a group of edges such as the inlet edges press **CTRL** and select both edges, right click the **Create Named Selection**.



2.4 SETUP AND RESULTS

Before starting the CFD setup, the inlet velocity components and the force coefficient transformation equations are defined for each angle of attack. Since the airfoil geometry remain fixed and the angle of attack was simulated by changing the inlet flow direction.

For an angle of attack, the inlet velocity components were calculated as:

$$U_x = U_\infty \cos(\alpha)$$

$$U_y = U_\infty \sin(\alpha)$$

where U_∞ is the free-stream velocity.

For $Re=200,000$, the free stream velocity was: $U_\infty=19.47646$ m/s.

Therefore, the inlet components were calculated using:

$$U_x = 19.47646 \cos(\alpha) \text{ \& } U_y = 19.47646 \sin(\alpha)$$

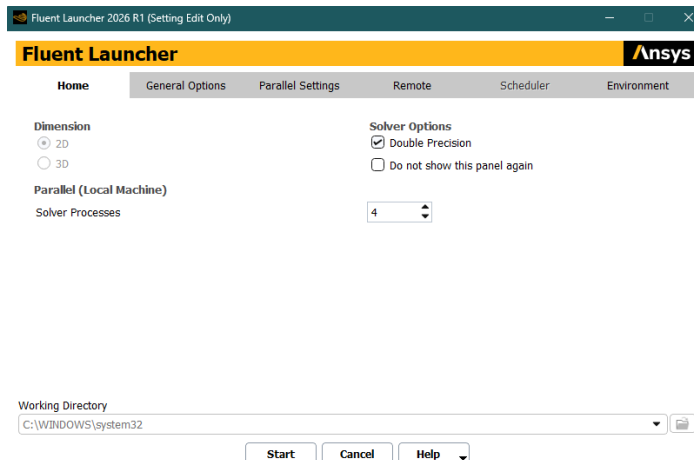
Because fluent reports force coefficients relative to fixed global x and y directions, the calculated force coefficients are transformed into drag and lift coefficients relative to the incoming flow direction as follows:

$$C_d = C_x \cos(\alpha) + C_y \sin(\alpha) \text{ \& } C_l = -C_x \sin(\alpha) + C_y \cos(\alpha)$$

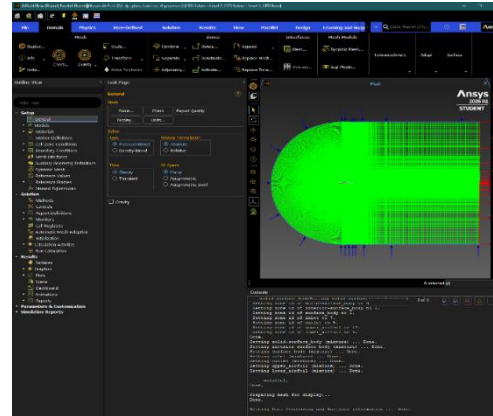
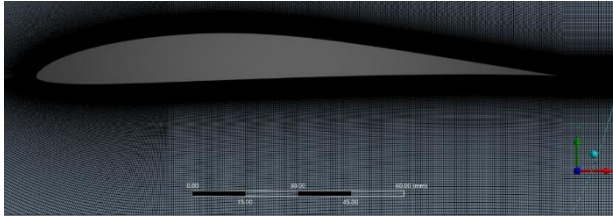
Where, C_x is the force coefficient in the global x direction and C_y is the force coefficient in the global y direction.

AoA / Deg	cd		Re = 200,000		cl		Re = 300,000	
	X-comp	Y-comp	Vx	Vy	X-comp	Y-comp	Vx	Vy
0	1	0	19.47646	0	0	1	29.2147	0
2	0.999390827	0.034899497	19.46459547	0.679718652	-0.034899497	0.999390827	29.21469458	0.017794998
4	0.99756405	0.069756474	19.42901632	1.358609171	-0.069756474	0.99756405	29.21467835	0.035568308
6	0.994521895	0.104528463	19.36976591	2.035844434	-0.104528463	0.994521895	29.21465138	0.053298271
8	0.990268069	0.139173101	19.28691643	2.710599334	-0.139173101	0.990268069	29.21461381	0.070963279
10	0.984807753	0.173648178	19.18056881	3.382051786	-0.173648178	0.984807753	29.21456583	0.088541803
12	0.978147601	0.207911691	19.05085262	4.04938373	-0.207911691	0.978147601	29.21450765	0.106012422
14	0.970295726	0.241921896	18.8979259	4.711782123	-0.241921896	0.970295726	29.21443958	0.123353843

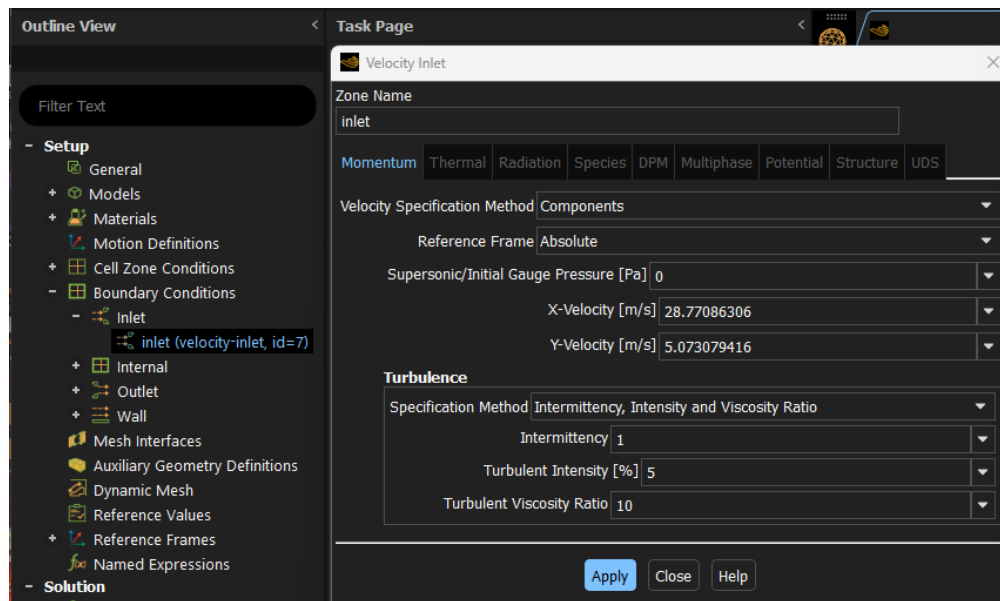
- To open the setup and result workspace press on the setup cell in the Workbench, a Fluent launcher window will appear select **Double Precision** in Solver Options and set the **Solver Processes** to **four** which is the maximum in the student version.

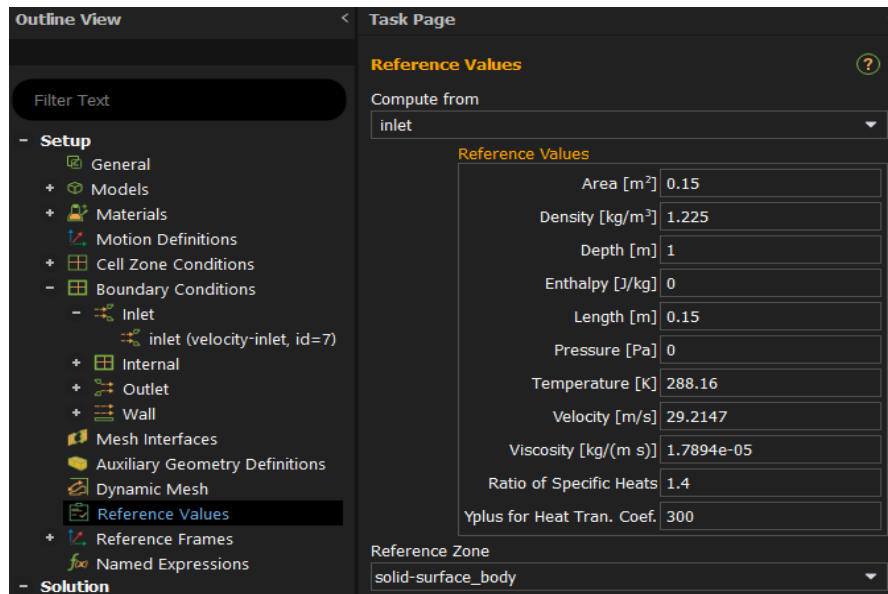
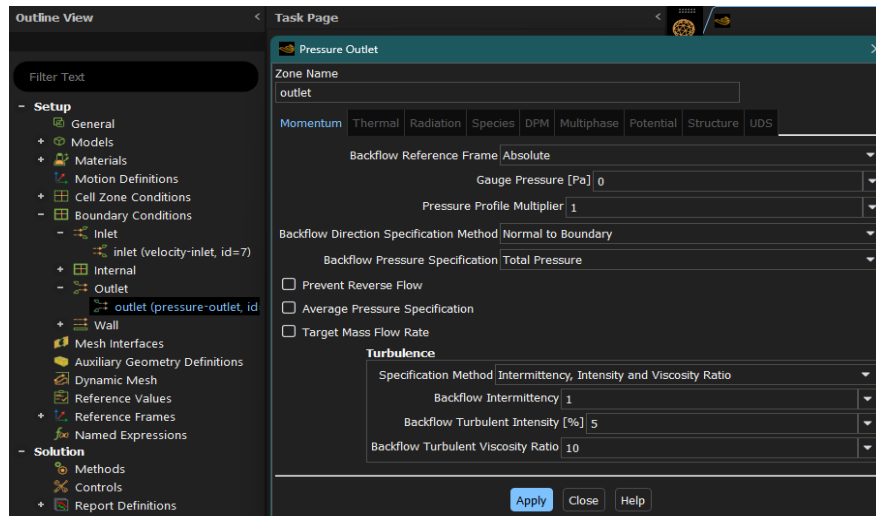


- The settings used in the setup are shown in the following pictures, the shown simulation is for $Re = 300,000$ with $AoA 10^\circ$. The following pictures may be used as a reference to highlight the overall workflow of the project in ANSYS.

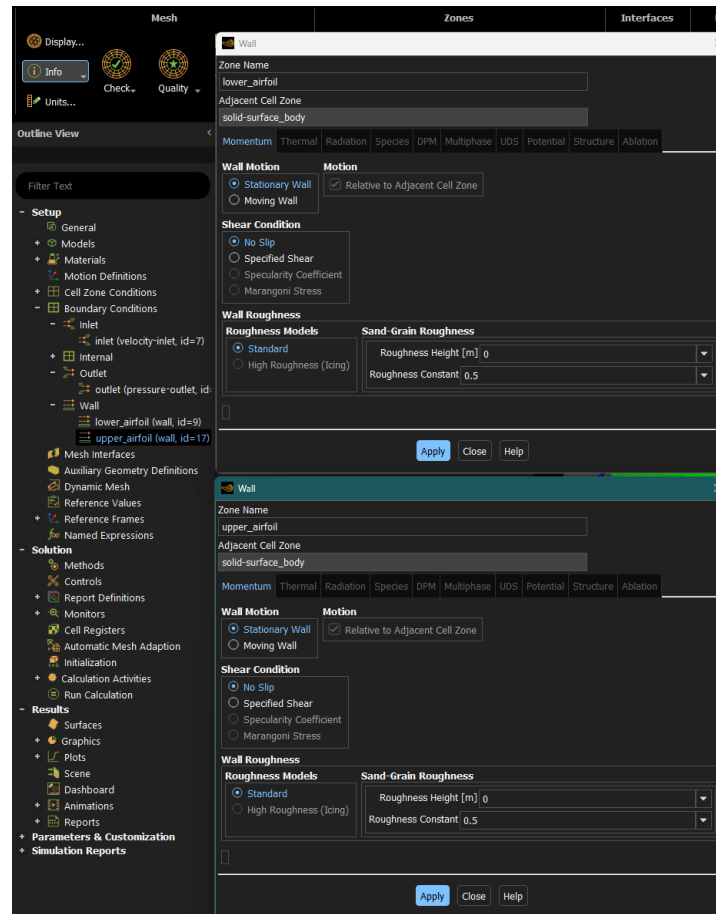


- Further inspection, the first layer height is verified to be less than $y^+ > 1$ which is the condition we need to get better numerically accurate data readings.
- In this page, the relevant velocity inlets are setup, where $X = 28.770 \text{ m/s}$ and $Y = 5.073 \text{ m/s}$. These steps are taken in accordance with the change in Angle of Attack of the airfoil. Since ANSYS does not allow physically to change the Angle of Attack, the angles components are used to simulate the angle of attack.

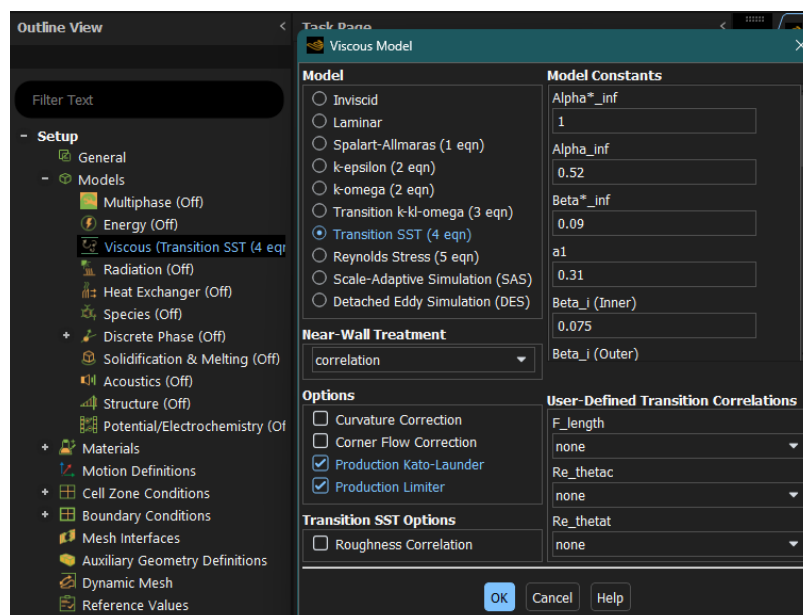


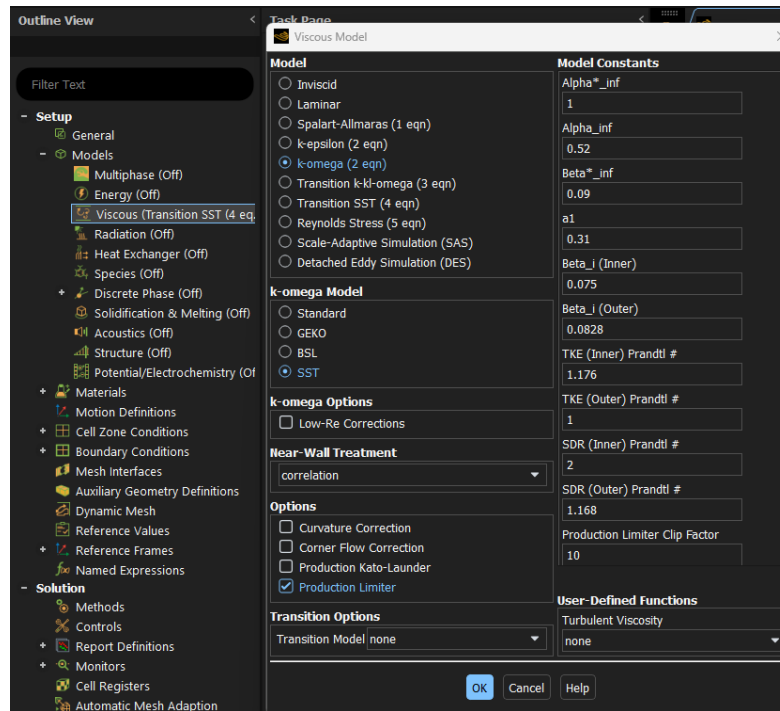


- Turbulence intensity and backflow viscosity ratio is set at 5% and 10. The referenced values are setup according to requirements as per the experiment.

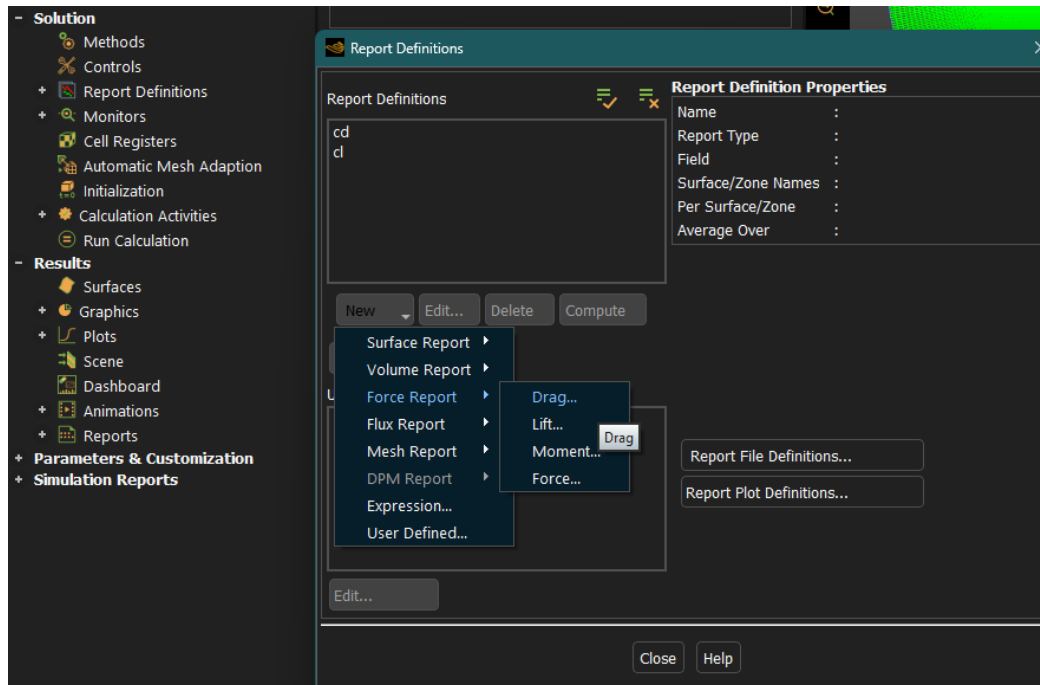


- To select a model to use in the simulation, select **Models** then **Viscous**, A window will appear prompting the user to choose a Viscous Model, these are the settings used for both the standard $k-\omega$ SST turbulence model and the Transition SST ($\gamma-Re_\theta$) model.

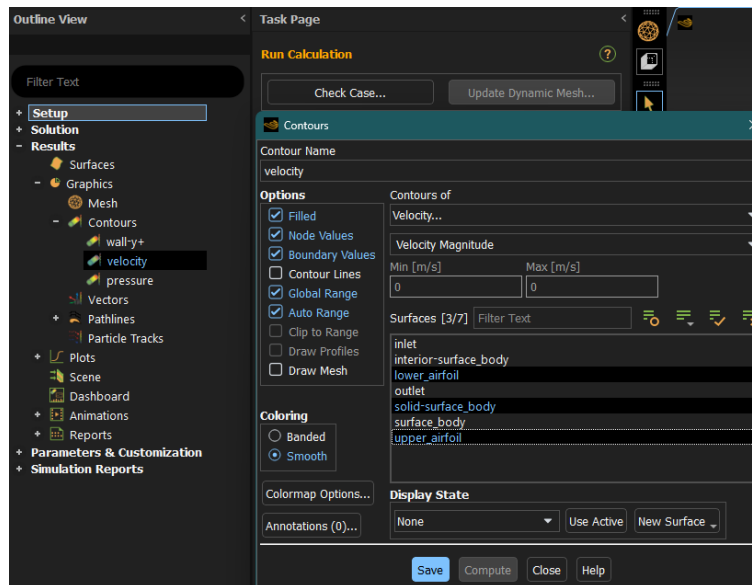
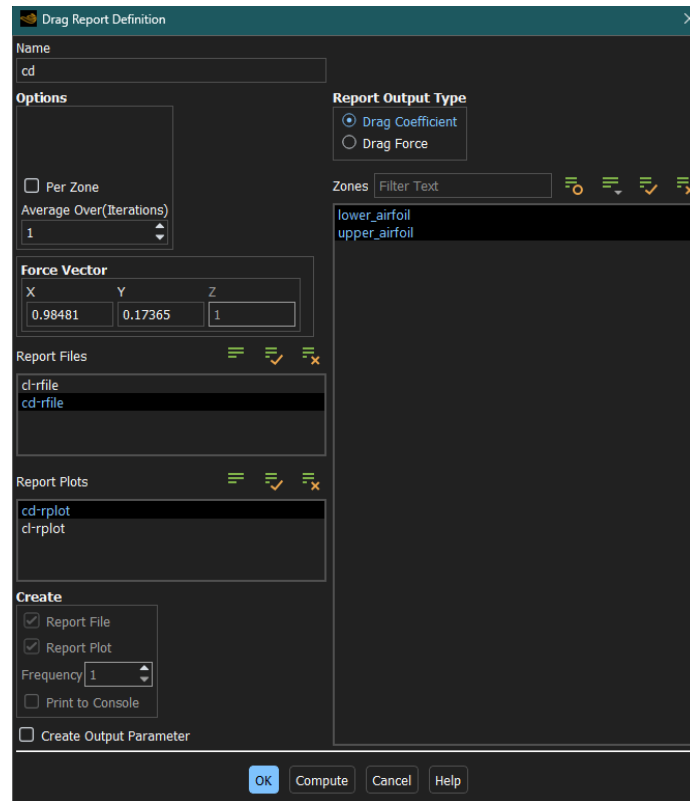




- To set up the values and graphs to be recorded. Right click on report definitions, click new force report, and select lift/drag coefficients.

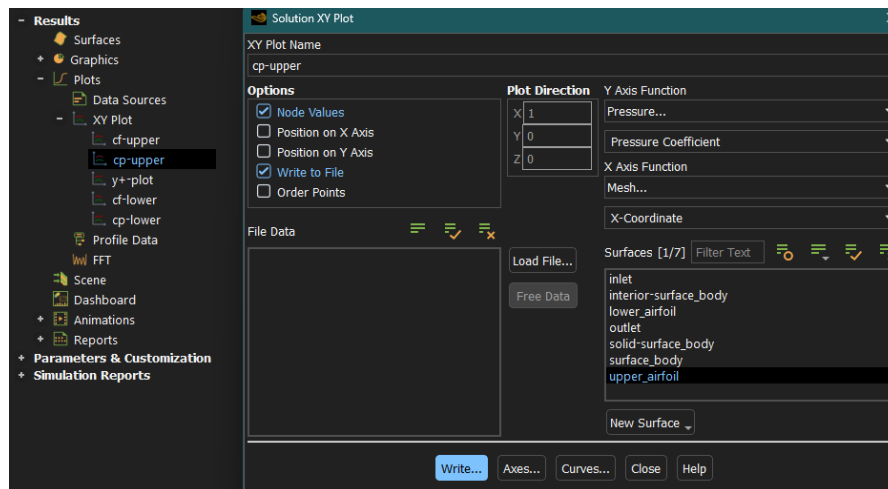
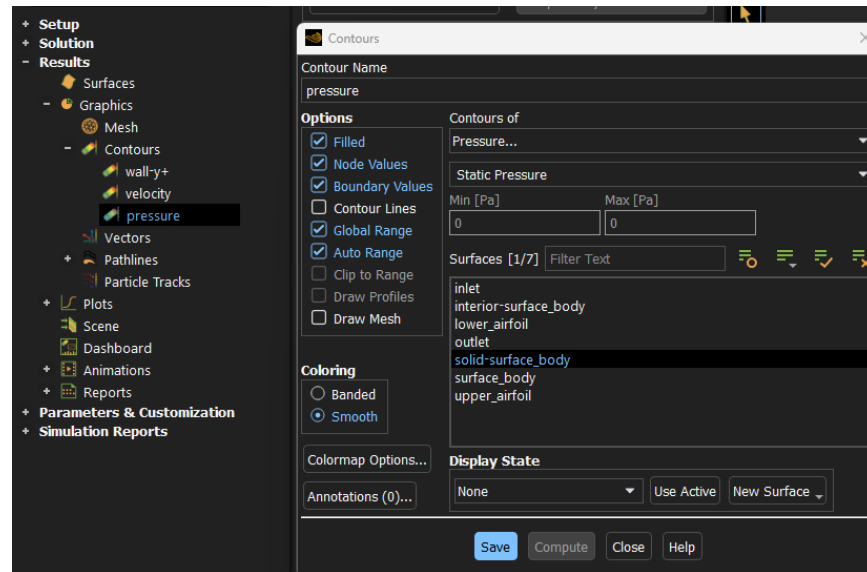


- Report output is drag/lift coefficient and zones are lower and upper airfoil surfaces.

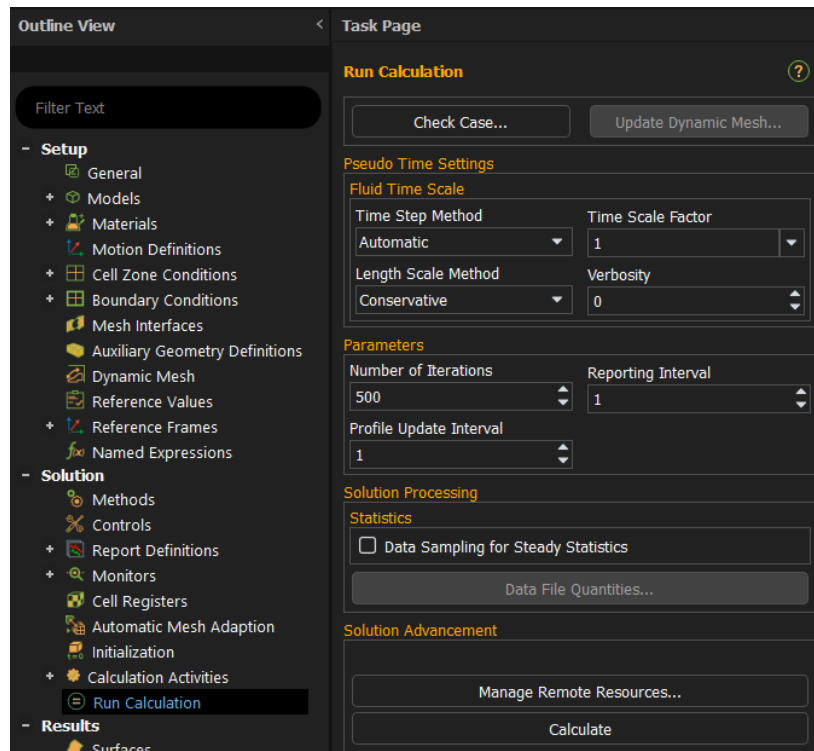
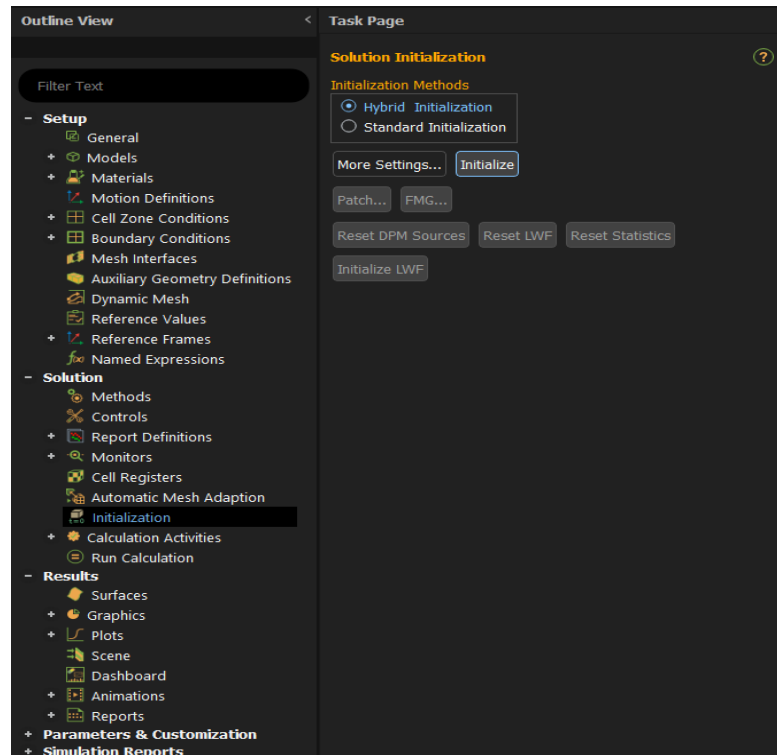


- Contours are used to display velocity and pressure flow definitions over the upper and lower surfaces of the airfoil. The picture above used is to set up the velocity contours.

- The picture below is a reference to show settings used to record the pressure contours.



- After finishing setting up all the values and settings, press on **Initialization** then **Initialize**, the initialization process must be done every time after changing the angle of attack to get the correct results. After initializing press on run calculations and set the number of iterations to 500 or more if there is no convergence after five hundred iterations, finally press on calculation and the simulation process will begin, and all the values and graphs will be generated.



3. RESULTS AND ANALYSIS

3.1 K- ω SST RESULTS AND ANALYSIS

This section presents the aerodynamic results obtained using the k- ω SST turbulence model. The E387 airfoil was analyzed at Reynolds number of 200,000 and 300,000 over an angle of attack at a range from 0° to 14°. The results were evaluated using lift coefficient, drag coefficient, aerodynamic efficiency, pressure coefficient, skin friction coefficient, and flow field contours.

For angle of attacks between 0° and 11°, the simulation produced stable lift and drag coefficient quantities. However, the cases at 12° and 14° showed large and persistent oscillations in the lift and drag monitors. These high-angle cases were therefore treated as separation dominated flow conditions and were analyzed only based on skin-friction distribution, pressure distributions, velocity, and pressure contours.

The k- ω SST model was analysed separately before comparison with the Transition SST model.

3.1.1 RE = 200,000 K- ω SST

Table 1. k- ω SST Coefficient of lift, drag, lift to drag ratio and Angle of attack.

ANGLE OF ATTACK	COEFFICIENT OF LIFT, C_l	COEFFICIENT OF DRAG, C_d	C_l/C_d
0	0.35842	0.01484	24.15229
2	0.5705	0.0152	37.53289
4	0.776	0.01685	46.05341
6	0.9718	0.0195	49.8359
8	1.1442	0.0242	47.28099
10	1.2613	0.0334	37.76347
11	1.2798	0.0421	30.4352

At $R_e = 200,000$ The velocity is of the flow field based on the chord length of the E387 airfoil is taken as follows.

$$R_e = \frac{\rho \cdot V \cdot c}{\mu} \text{ where } \rho = 1.225 \frac{\text{kg}}{\text{m}^3}, c = 0.15 \text{ m}, \mu = 1.7894 \times 10^{-5} \frac{\text{kg}}{\text{m.s}}$$

$$V = \frac{R_e \cdot \mu}{\rho \cdot c}$$

Therefore $V = 19.476 \text{ m/s}$ at $R_e = 200,000$

3.1.2 LIFT AND DRAG COEFFICIENT VARIATION WITH ANGLE OF ATTACK

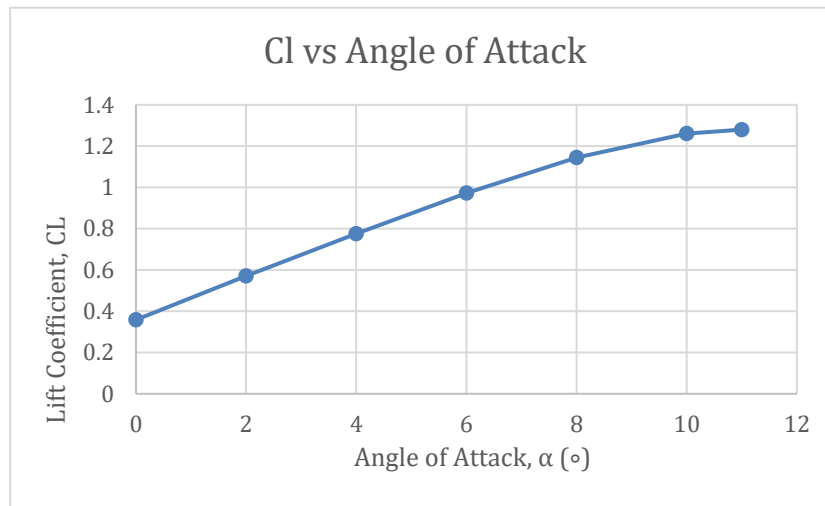


Fig 1. Variation of Coefficient of Lift vs Angle of Attack at $Re = 200,000$

Fig 1 presents the variation of lift coefficient with angle of attacks for the E387 Airfoil at Reynolds Number 200,000. The lift coefficient increased at 0° to $C_L = 1.14$ at 8° . This near linear increase indicates that the flow remained predominantly attached over the Airfoil surface through the low to moderate angle of attack. Between 8° and 10° , the lift coefficient continued to increase, reaching at $C_L = 1.26$ 10° . However, the lift coefficient increased only slightly to approximately $C_L = 1.28$ at 11° . This flattening of the lift curve suggests that the Airfoil was approaching a near-stall condition, where the increasing angle of attack produced a smaller increase in lift due to the onset of adverse pressure gradient effects and possible upper surface separation. Based on Fig 2. The lift coefficient histories stabilized after approximately 200 iterations. The small remaining fluctuations were limited, indicating that a reliable steady state aerodynamic solution was obtained. Therefore, the corresponding C_L values were used in quantitative aerodynamic analysis. The positive lift coefficient at 0° is consistent with cambered geometry of the E387 airfoil, which produces lift even when the geometric angle of attack is zero.

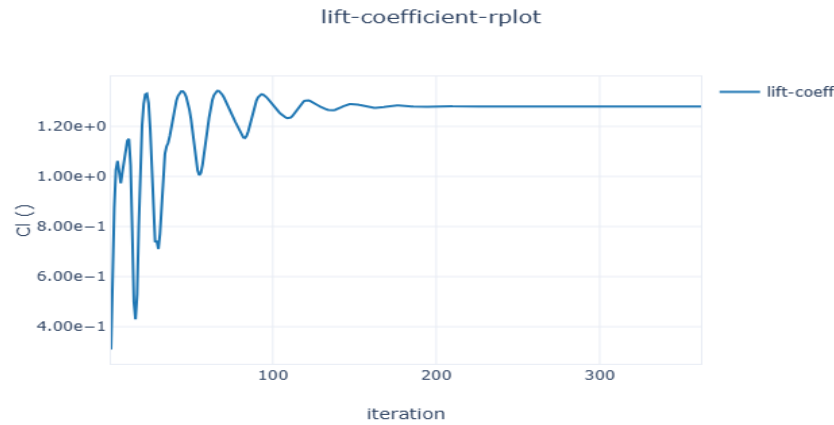


Fig 2. coefficient convergence history for the $k-\omega$ SST model at $Re = 200,000$ and $\alpha = 10^\circ$

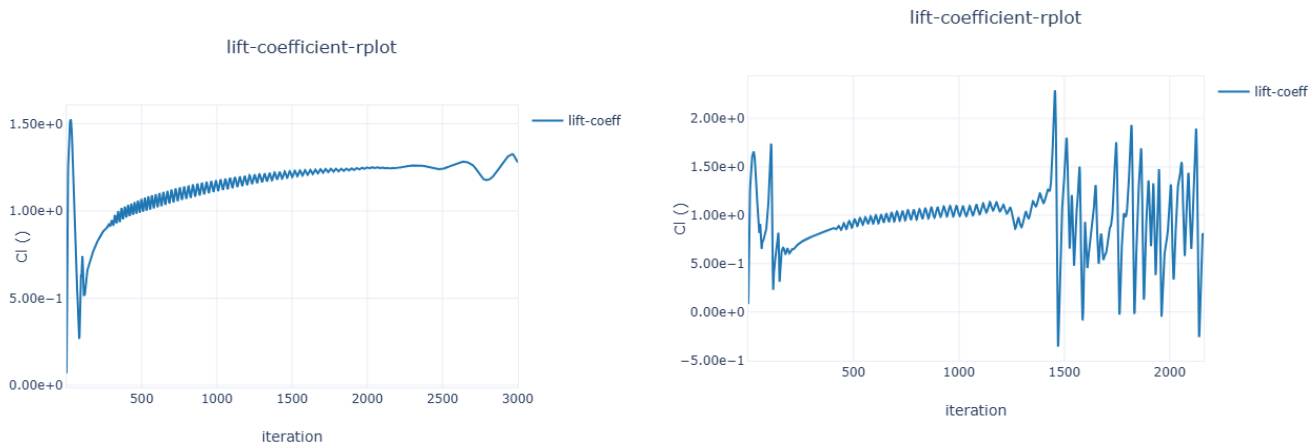


Fig.3 Lift-Coefficient monitor for $k-\omega$ SST model at $Re = 200,000$ for $\alpha = 12^\circ$ and 14°

Based on Fig 3. simulations conducted at 12° and 14° did not produce stable lift coefficient trend. Severe oscillations were observed in the monitor, and therefore no dependable steady-state lift coefficient was reported for these cases. This behavior indicated that the flow has become strongly separation dominated. At these higher angles of attacks, the upper surface boundary layer is likely to separate, producing enlarged wake and time dependent aerodynamic loading. Since the present simulations were performed using a steady RANS equation, a single reliable steady state C_L values could not be extracted.

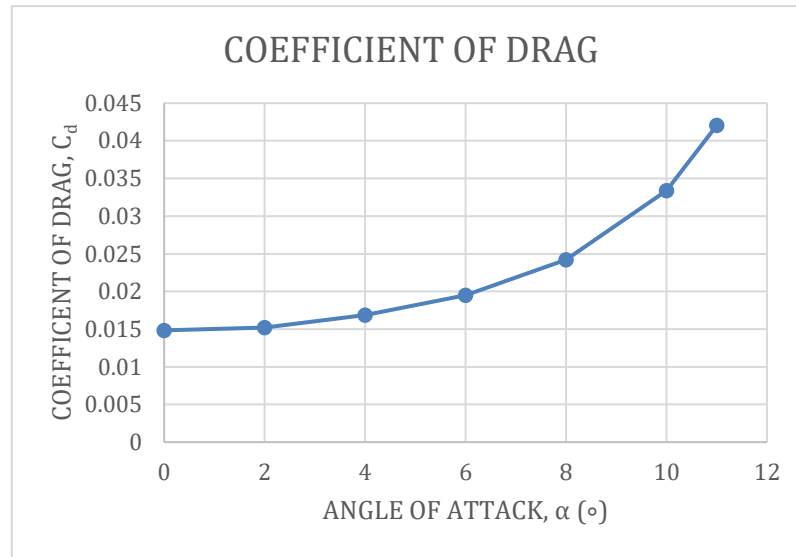


Fig.4 Variation of Coefficient of Drag vs Angle of Attack at $Re = 200,000$

From Fig.4, at 0° to about 6° , drag rises gradually from approximately $C_D = 0.014$ at 0° to approximately $C_D = 0.019$ at 6° . This is your predominantly attached-flow region as drag is associated with viscous effects and the pressure distribution around the airfoil. From 6° the drag rises noticeably and at 8° is $C_D = 0.024$. From 8° to 11° , the increase becomes steeper in nature. The highest drag value was obtained at 11° , where the $C_D = 0.042$. The sharp rise indicates that the adverse pressure gradient over the upper surface was becoming increasingly strong. This has likely caused boundary-layer thickening and an increasing tendency for upper surface separation, resulting in a larger wake and increased pressure drag. This result is consistent with the lift curve observed in Fig.1 where the rate of lift is reduced between 10° and 11° .

Although the lift coefficient did not show a clear post-stall reduction by 11° , the rapid drag increases indicating that the airfoil was approaching stall conditions. The pressure coefficient, skin friction coefficient, velocity contours, and pressure contours results should therefore be used to investigate whether the separation has begun near the trailing edge or progressed upstream.

3.1.3 DRAG POLAR AND AERODYNAMIC EFFICIENCY.

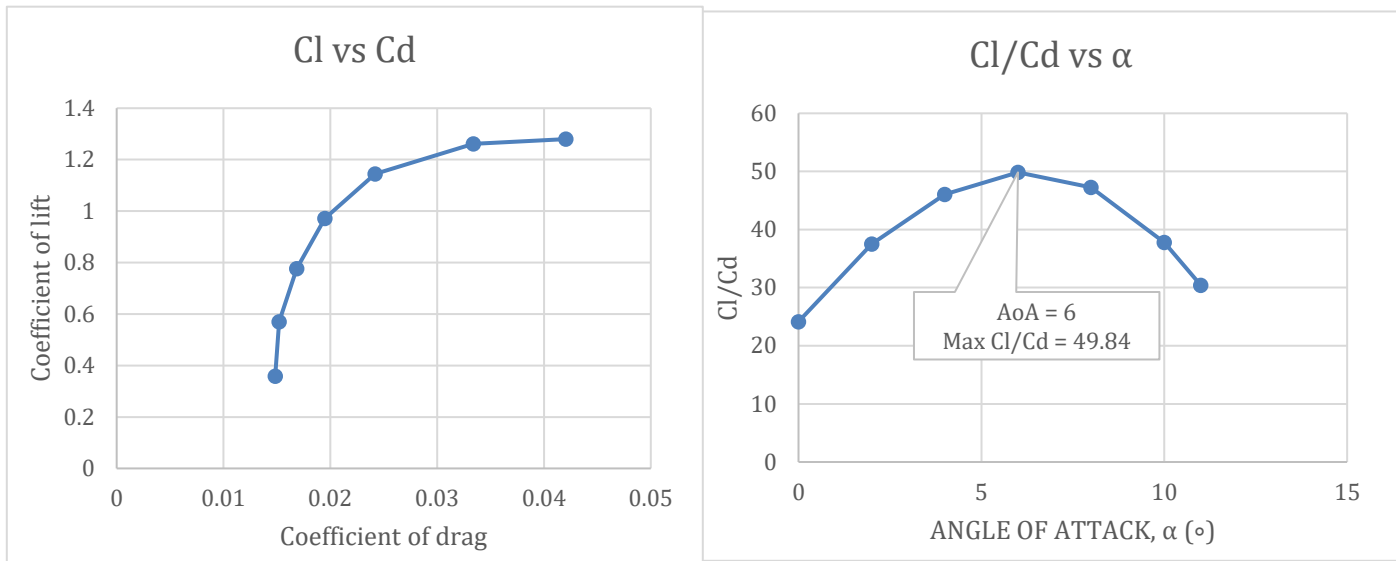


Fig. 5 Cl vs Cd and Cl/Cd vs Angle of Attack at $Re = 200,000$

From Fig.5 The drag polar (Cl vs Cd) and aerodynamic efficiency (Cl/Cd vs α) is plotted. The drag polar shows that the E387 airfoil produced lift efficiently at low and moderate angle of attacks. The lift to drag ratio rises strongly from 0° to 6° however, at higher angles of attacks between 8° and 11° , drag increases disproportionately relative to lift, indicating a reduction in aerodynamic efficiency and which may have caused onset of strong adverse pressure gradient and separation effects. The drag polar therefore showed that the E387 airfoil generated lift efficiently over the lower and moderate angle of attack. The 11° represents the highest stable lift condition obtained in the steady SST analysis, however, its reduced lift-to-drag ratio shows that it is no longer in operating conditions.

In the Cl/Cd vs α , 6° is considered the most aerodynamically efficient angle of attack for the $k-\omega$ SST model at $Re = 200,000$. Even though lift continues increasing at 8° and 10° , the drag rises too quickly, so the airfoil becomes less efficient.

Therefore, for the following analysis key angles that will be studied for the following understanding are as follows:

1. 0° : baseline attached flow
2. 6° : maximum C_L/C_D , best aerodynamic efficiency
3. 10° : high-lift condition before severe instability
4. 11° : last stable case
5. 12° or 14° : oscillatory case

3.1.4 VELOCITY AND PRESSURE CONTOURS K- Ω SST $Re = 200,000$

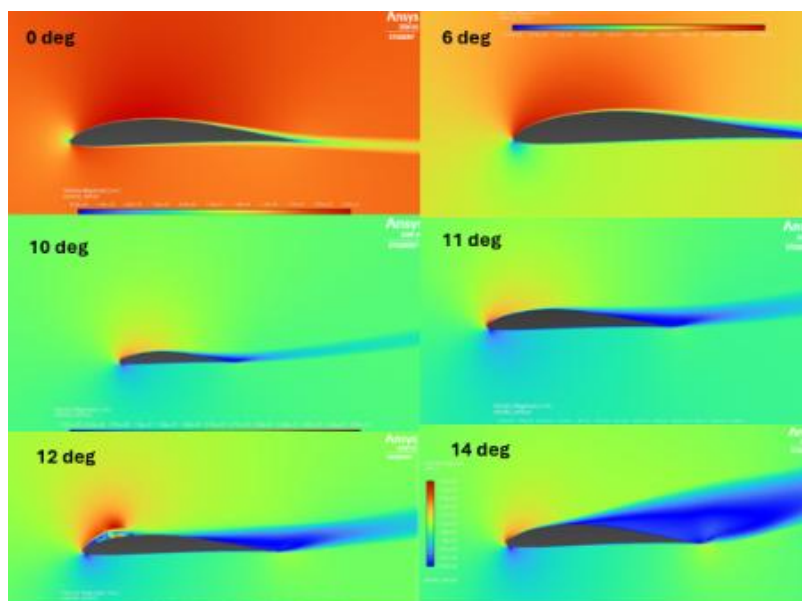


Fig. 6 Velocity Contours for $Re = 200,000$ at 0, 6, 10, 11, 12, 14 degrees.

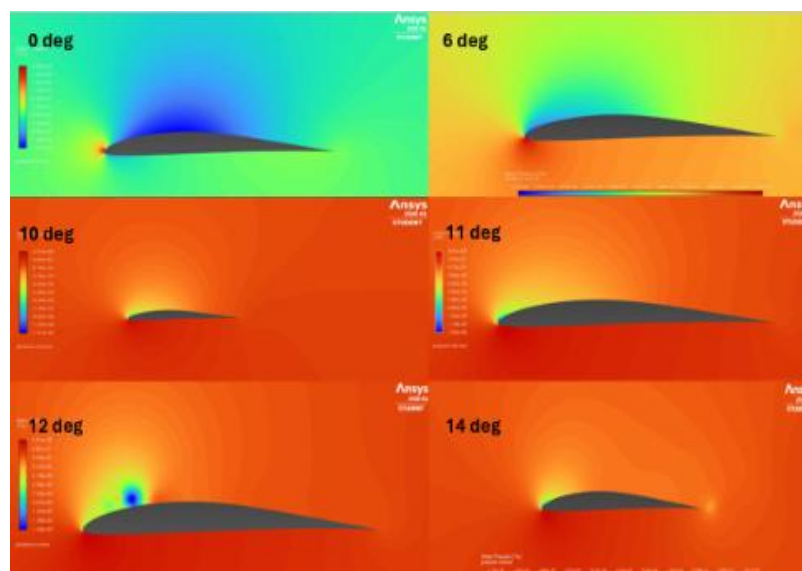


Fig. 7 Pressure Contours for $Re = 200,000$ at 0, 6, 10, 11, 12, 14 degrees.

The pressure and velocity contours at $Re = 200,000$ show a clear transition from attached flow at low angles of attack to increasingly separated flow at higher angles of attack. At 0° and 6° , the pressure distribution and narrow wake indicate largely attached flow with efficient lift generation. As the angle of attack increases to 10° and 11° , the upper surface adverse pressure gradient strengthened and the downstream wake became thicker, indicating increased boundary layer growth and drag. At 12° , a local upper surface flow disturbance and a substantially thicker wake suggested the onset of separation related behaviour. At 14° , the large low velocity wake and altered pressure recovery indicated significant flow separation and near stall or post stall aerodynamic behaviour. These results show that the airfoil experiences a progressive reduction in aerodynamic efficiency as the angle of attack increases beyond the moderate lift operating range.

3.1.5 COEFFICIENT OF PRESSURE K- Ω SST OVER THE AIRFOIL ANALYSIS AT $Re = 200,000$ AT 6° .

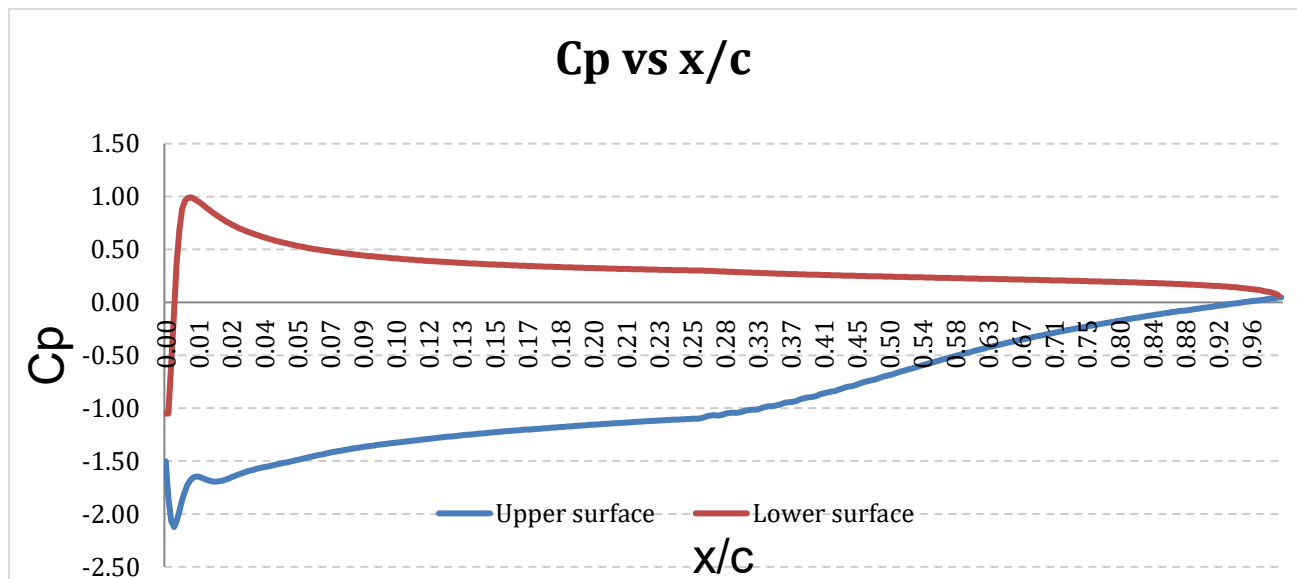


Fig. 8. C_p vs x/c for pressure coefficient distribution at $\alpha = 6^\circ$ and $Re = 200,000$

The above Coefficient of Pressure vs x/c chart at $\alpha = 6^\circ$. The trend shows a clear pressure difference between the two airfoil surfaces confirming effective lift generation in the pre-stall operating region. The upper surface exhibits a strong negative pressure coefficient near the leading edge, reaching approximately $C_p = -2$, which indicates the suction peak produced by accelerated flow over the suction side of the airfoil. Downstream of the leading edge, the upper surface pressure gradually recovers towards the trailing edge. The difference between the lower and upper surface pressure distribution produces a net upward aerodynamic force, explaining the positive lift coefficient obtained at this angle of attack.

The lower surface maintains a comparatively higher-pressure coefficient over most of the chords, with a positive pressure peak close to the leading edge followed by a gradual reduction towards the trailing edge. The pressure recovery over the upper surface is smooth, although a strong recovery is visible over the downstream portion of the chord, approximately $x/c = 0.45$. This indicates an adverse pressure gradient region. Therefore, the airfoil at $\alpha = 6^\circ$ is operating at predominantly in an attached flow lift condition.

3.1.6 SKIN FRICTION COEFFICIENT K- Ω SST OVER THE AIRFOIL ANALYSIS AT $Re = 200,000$ AT 6° .

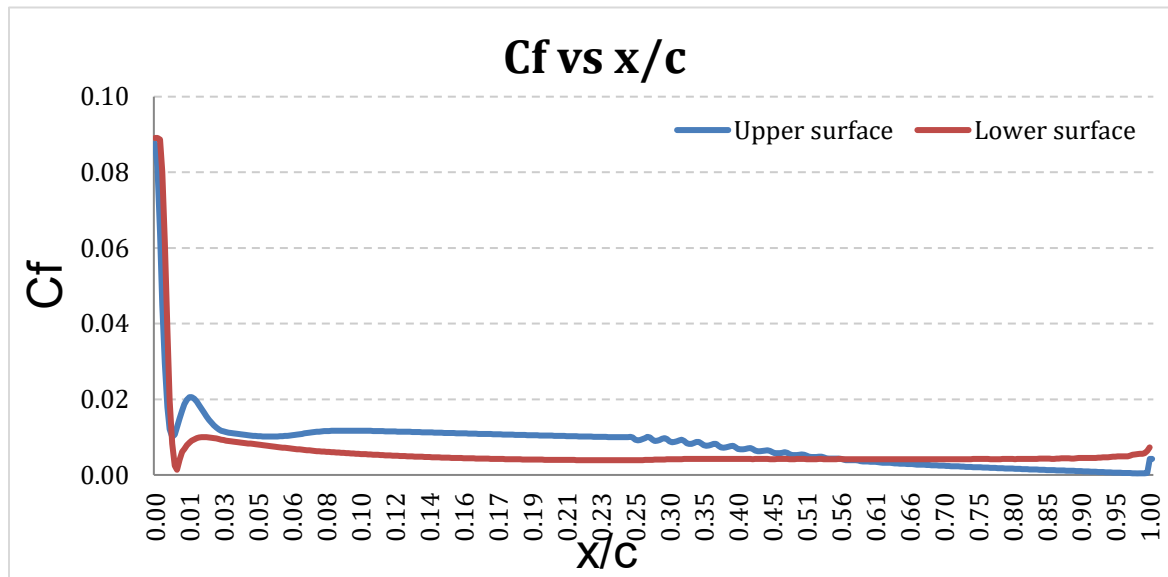


Fig. 9. C_f vs x/c for pressure coefficient distribution at $\alpha = 6^\circ$ and $Re = 200,000$

The skin friction coefficient distribution at $\alpha = 6^\circ$ and $Re = 200,000$, indicates that the flow remains predominantly attached over both the upper and lower surfaces of the airfoil. On both surfaces, the skin friction coefficient is higher near the leading edge, where the boundary layer is very thin and the wall shear stress is the greatest. This is followed by a rapid reduction in C_f over the forward portion of the chord as the boundary layer is developed downstream.

The upper surface shows a higher C_f level than the lower surface over much of the chord, which is consistent with strong suction effects acting on the upper side of the airfoil. Over the mid-chord region, both curves continue to decrease gradually, indicating boundary layer growth and a progressive reduction in wall shear stress. Since the C_f value does not cross below zero, there is no straightforward evidence of flow reversal or a confirmed laminar separation bubble from this plot alone. Instead, the distribution suggests attached flow with some weakening of wall shear stress towards the aft portion of the airfoil, particularly at the trailing edge.

3.1.7 $Re = 300,000$ K- Ω SST

Table 2. k- ω SST Coefficient of lift, drag, lift to drag ratio and Angle of attack.

ANGLE OF ATTACK	COEFFICIENT OF LIFT, C_l	COEFFICIENT OF DRAG, C_d	C_l/C_d
0	0.3632	0.01314	27.64
2	0.577	0.01418	40.691
4	0.78547	0.0156	50.350
6	0.98427	0.0180	54.682
8	1.162	0.02237	51.944
10	1.2932	0.03025	42.750
11	1.3314	0.03656	36.417
12	1.336	0.04624	28.892

At $Re = 300,000$ The velocity is of the flow field based on the chord length of the E387 airfoil is taken as follows.

$$Re = \frac{\rho \cdot V \cdot c}{\mu} \text{ where } \rho = 1.225 \frac{kg}{m^3}, C = 0.15 \text{ m}, \mu = 1.7894 \times 10^{-5} \frac{kg}{m.s}$$

$$V = \frac{Re * \mu}{\rho * c}$$

Therefore $V = 29.2147 \text{ m/s}$ at $Re = 300,000$

3.1.8 LIFT AND DRAG COEFFICIENT VARIATION WITH ANGLE OF ATTACK

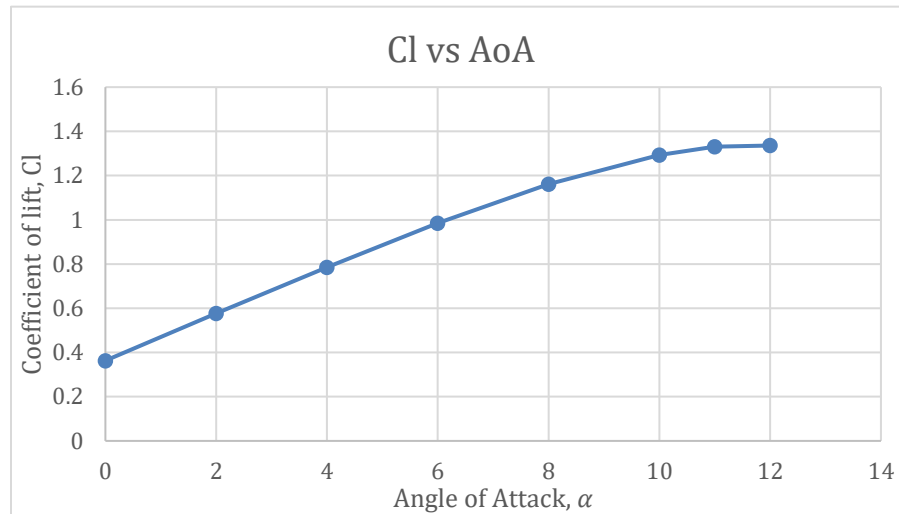


Fig.10 Variation of Coefficient of Lift vs Angle of Attack at $Re = 300,000$

Fig.10 presents the variation of lift coefficient with angle of attacks for the E387 Airfoil at Reynolds Number 300,000. The positive lift coefficient at 0° is consistent with cambered geometry of the E387 airfoil, which produces lift even when the geometric angle of attack is zero. The lift coefficient increased approximately at 0° to $C_L = 1.162$ at 8° . This near linear increase indicates that the flow remained predominantly attached over the Airfoil surface through the low to moderate angle of attack. Between 10° and 12° , the lift coefficient continued to increase, reaching approximately at $C_L = 1.3314$ at 11° . However, the lift coefficient increased only slightly to approximately $C_L = 1.336$ at 12° . This flattening of the lift curve suggests that the Airfoil was approaching a near-stall condition, where the increasing angle of attack produced a smaller increase in lift due to the onset of adverse pressure gradient effects and possible upper surface separation. Based on Fig 11. The lift coefficient plot versus iteration stabilized after approximately 450 iterations. The small remaining fluctuations were limited, indicating that a reliable steady state aerodynamic solution was obtained. Therefore, the corresponding C_L values were used in quantitative aerodynamic analysis.

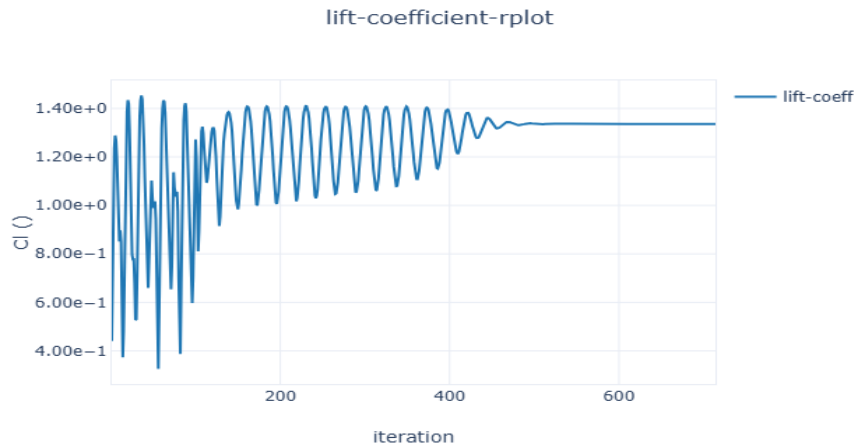


Fig 11. coefficient convergence history for the $k-\omega$ SST model at $Re = 300,000$ and $\alpha = 12^\circ$

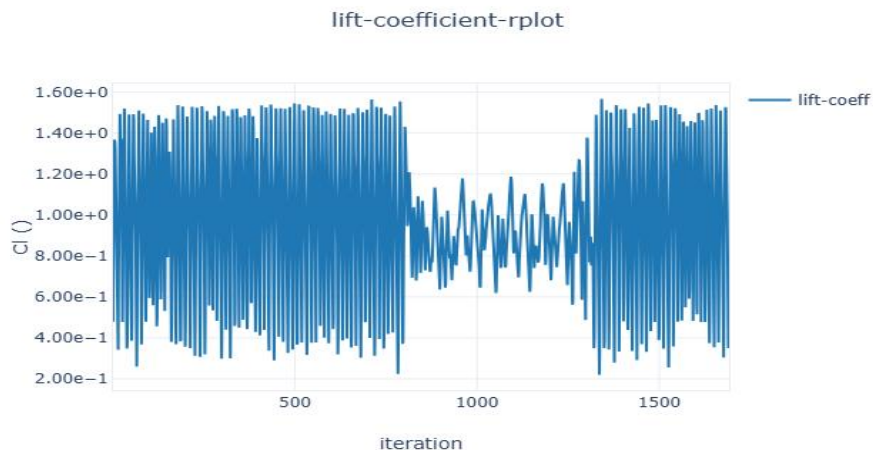


Fig.12 Lift-Coefficient monitor for $k-\omega$ SST model at $Re = 300,000$ for $\alpha = 14^\circ$

Based on Fig 12. simulations conducted at 14° did not produce stable lift coefficient trend. Severe oscillations were observed in the monitor, and therefore no dependable steady-state lift coefficient was reported for this case. This behavior indicated that the flow has become strongly separation dominated. At these higher angles of attacks, the upper surface boundary layer is likely to separate, producing enlarged wake and time dependent aerodynamic loading. Since the present simulations were performed using a steady RANS equation, a single reliable steady state C_L values could not be extracted.

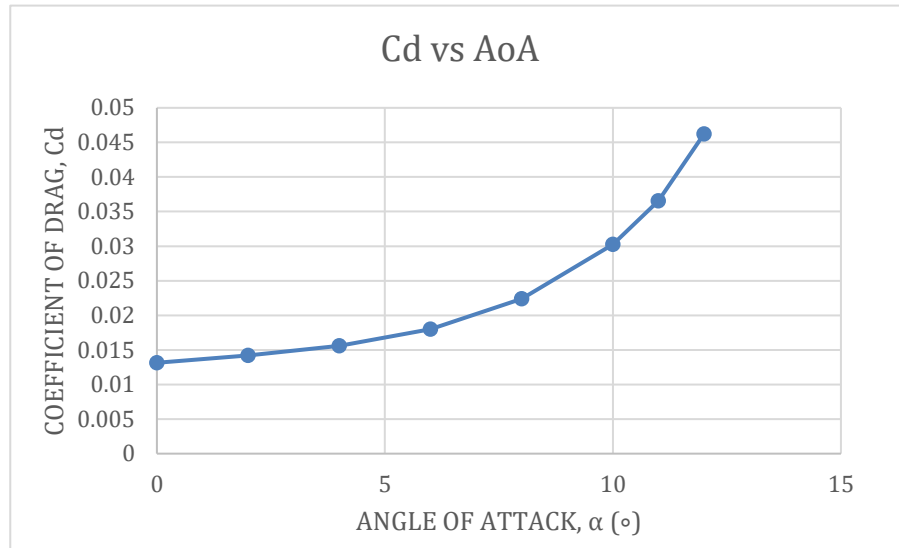


Fig.13 Variation of Coefficient of Drag vs Angle of Attack at $Re = 300,000$

From Fig.13, at 0° to about 6° , drag rises gradually from approximately $C_D = 0.01314$ at 0° to approximately $C_D = 0.018$ at 6° . This is your predominantly attached-flow region as drag is mainly associated with viscous effects and the pressure distribution around the airfoil. From 6° the drag rises noticeably and at 8° is $C_D = 0.02237$. From 8° to 12° , the increase becomes steeper in nature. The highest drag value was obtained at 12° , where the $C_D = 0.04624$. The sharp rise indicates that the adverse pressure gradient over the upper surface was becoming increasingly strong. This has likely caused boundary-layer thickening and an increasing tendency for upper surface separation, resulting in a larger wake and increased pressure drag. This result is consistent with the lift curve observed in Fig.8 where the rate of lift is reduced between 10° and 12° .

Although the lift coefficient did not show a clear post-stall reduction by 11° , the rapid drag increases indicating that the airfoil was approaching near stall conditions. The pressure coefficient, skin friction coefficient, velocity contours, and pressure contours results should therefore be used to investigate whether the separation has begun near the trailing edge or progressed upstream.

3.1.9 DRAG POLAR AND AERODYNAMIC EFFICIENCY

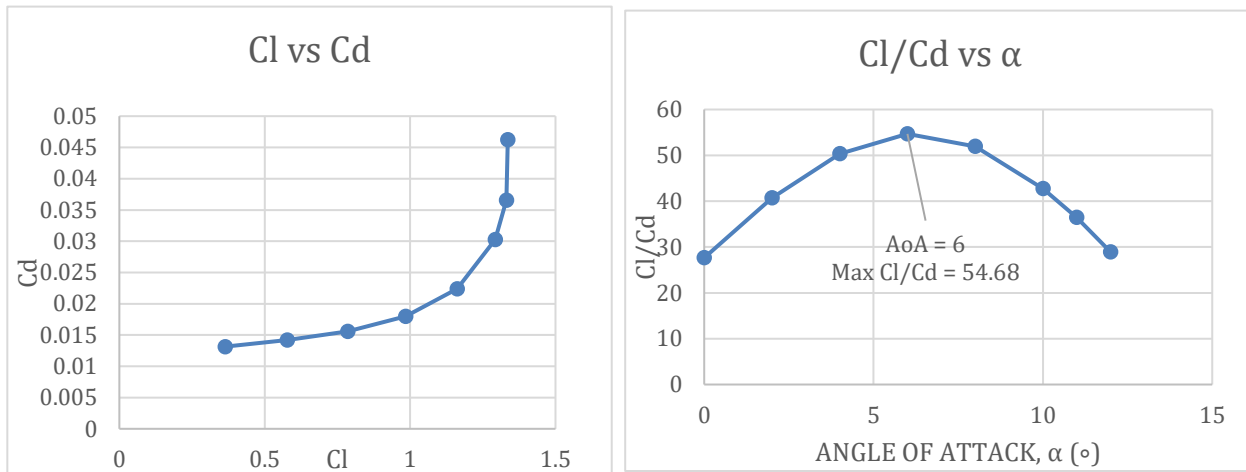


Fig. 14 C_L vs C_D and C_L/C_D vs Angle of Attack at $Re = 300,000$

From Fig.14 The drag polar (C_L vs C_D) and aerodynamic efficiency (C_L/C_D vs α) is plotted.

The drag polar shows that the E387 airfoil produced lift efficiently at low and moderate angle of attacks. The lift to drag ratio rises strongly from 0° to 6° however, at higher angles of attacks between 8° and 12° , drag increases disproportionately relative to lift, indicating a reduction in aerodynamic efficiency and which may have caused onset of strong adverse pressure gradient and separation effects. The drag polar therefore showed that the E387 airfoil generated lift efficiently over the lower and moderate angle of attack. The 12° represents the highest stable lift condition obtained in the steady SST analysis, however, its reduced lift-to-drag ratio shows that it is no longer in operating conditions.

In the C_L/C_D vs α , 6° is considered the most aerodynamically efficient angle of attack for the $k-\omega$ SST model at $Re = 300,000$. Even though lift continues increasing at 8° and 12° , the drag rises too quickly, so the airfoil becomes less efficient.

Therefore, for further analysis key angles that will be studied for the following understanding are as follows:

1. **0°** : baseline attached flow
2. **6°** : maximum C_L/C_D , best aerodynamic efficiency
3. **10°** : high-lift condition before severe instability
4. **11° and 12°** : last stable case
5. **14°** : oscillatory case

3.1.10 VELOCITY AND PRESSURE CONTOURS K- Ω SST $Re = 300,000$

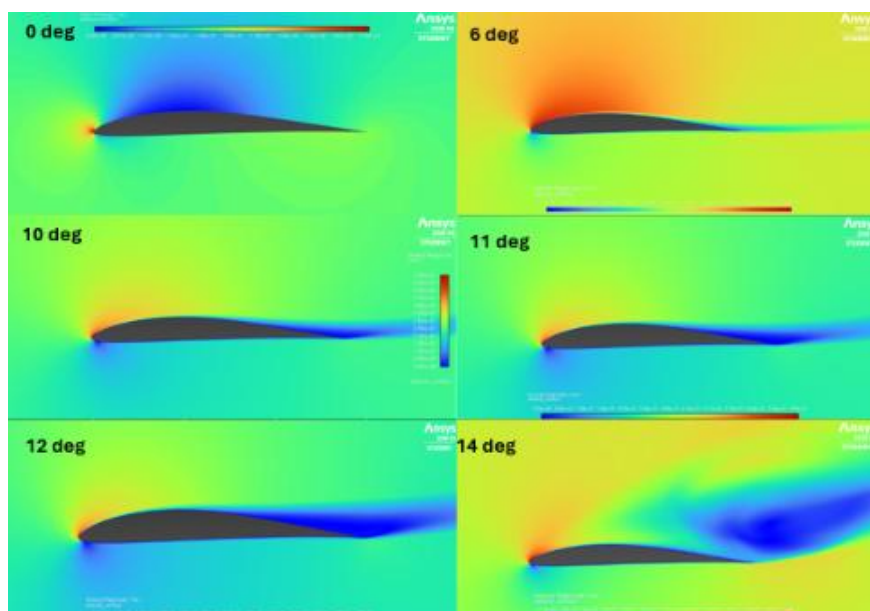


Fig. 15 Velocity Contours for $Re = 300,000$ at 0, 6, 10, 11, 12, 14 degrees.

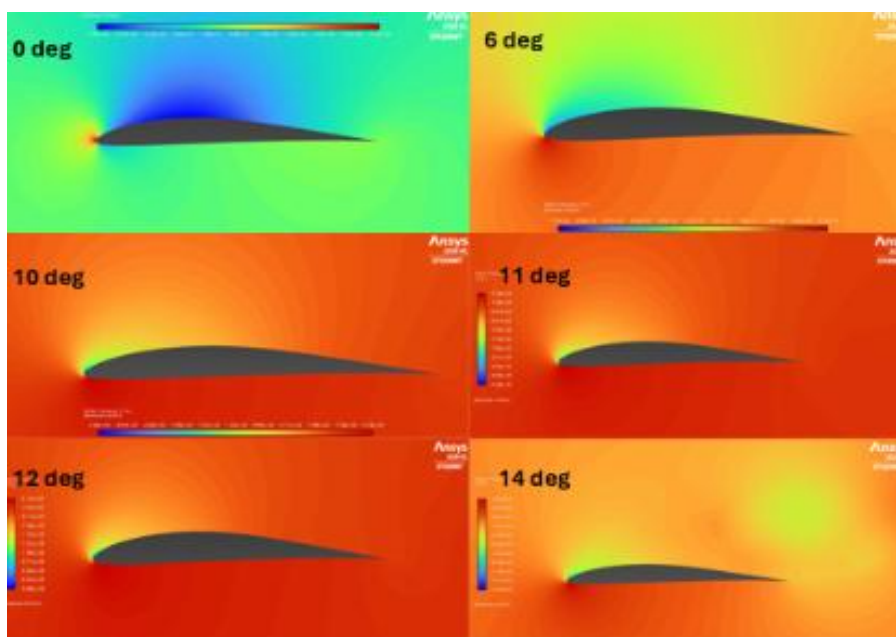


Fig. 16. Pressure Contours for $Re = 300,000$ at 0, 6, 10, 11, 12, 14 degrees.

The pressure and velocity contours at $Re = 300,000$ show the same overall aerodynamic progression observed at $Re = 200,000$. A clear transition from attached flow at low angles of attack to increasingly separated flow at higher angles of attack. At 0° and 6° , the pressure distribution and narrow wake indicate largely attached flow with efficient lift generation. As the angle of attack increases to 10° and 11° , the upper surface adverse pressure gradient strengthened and the downstream wake became thicker, indicating increased boundary layer growth and drag. At 12° , a local upper surface flow disturbance and a substantially thicker wake suggested the onset of separation related behaviour. At 14° , the large low velocity wake and altered pressure recovery indicated significant flow separation and near stall or post stall aerodynamic behaviour. These results show that the airfoil experiences a progressive reduction in aerodynamic efficiency as the angle of attack increases beyond the moderate lift operating range.

3.1.11 COEFFICIENT OF PRESSURE K- Ω SST OVER THE AIRFOIL ANALYSIS AT $Re = 300,000$ AT 6°

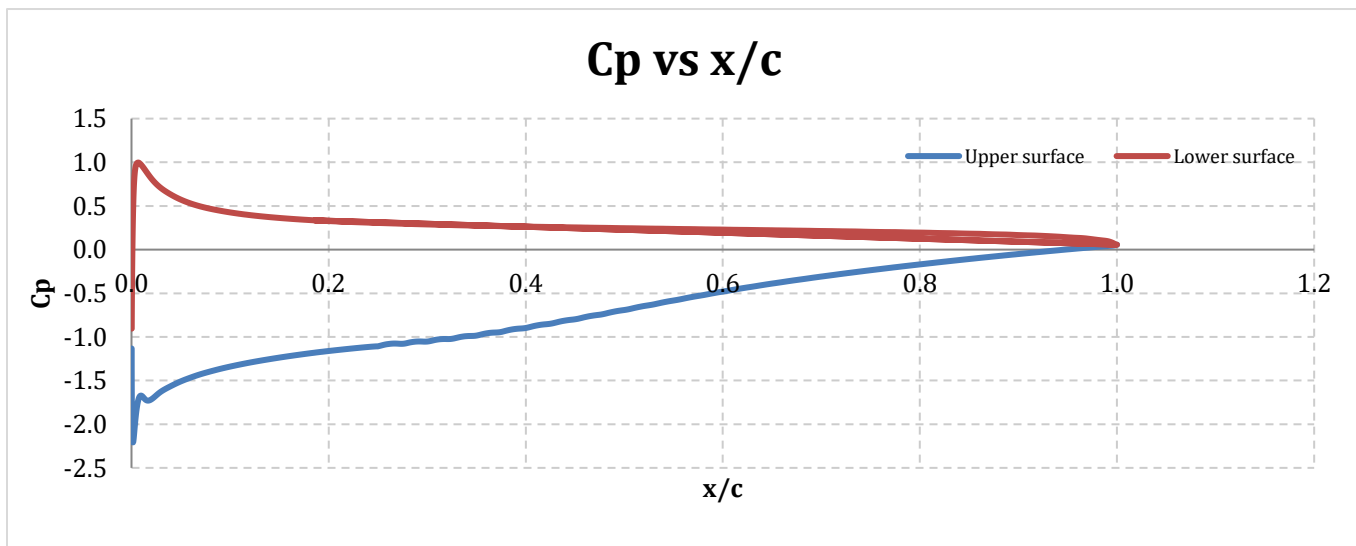


Fig. 17. C_p vs x/c for pressure coefficient distribution at $\alpha = 6^\circ$ and $Re = 300,000$

The pressure coefficient distribution around the airfoil at and $Re = 300,000$, $\alpha = 6^\circ$ shows a clear pressure difference between the upper and lower surfaces of the E387 airfoil. The upper surface exhibits a strong negative pressure coefficient close to the leading edge, indicating accelerated flow and the formation of a suction peak. This low-pressure region is responsible of the lift generated by the airfoil. Downstream the leading edge, the upper surface pressure recovers progressively towards the trailing edge. The recovery is smooth,

although the presence of an adverse pressure gradient. The lower surface maintains a comparatively higher pressure coefficient over most of the chord.

3.1.12 SKIN FRICTION COEFFICIENT K- Ω SST OVER THE AIRFOIL ANALYSIS AT $Re = 300,000$ AT 6°

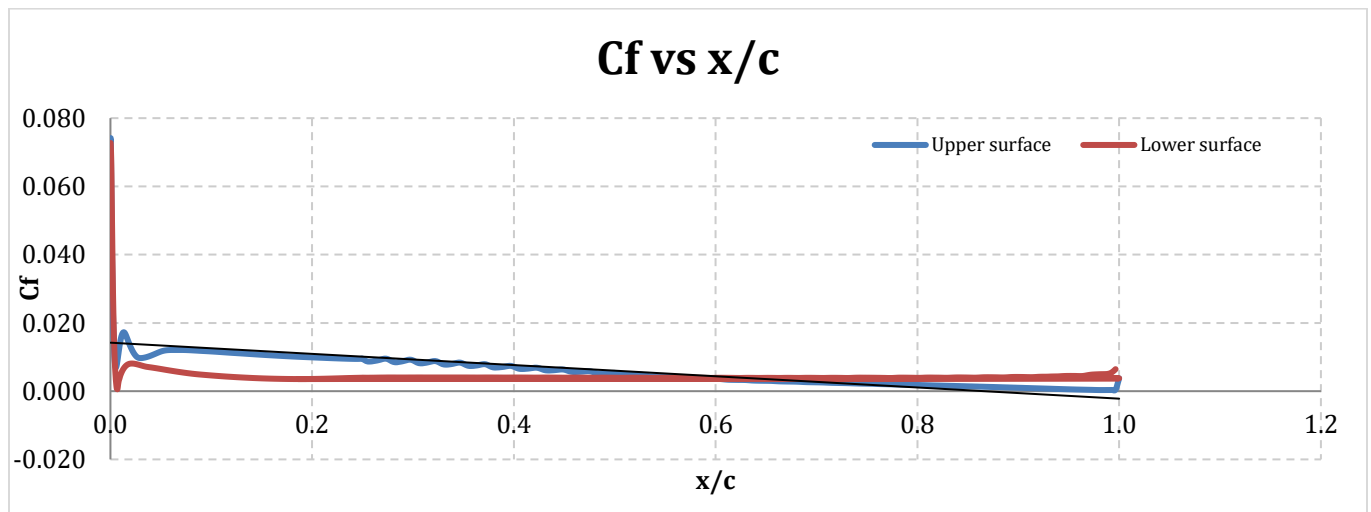


Fig. 18. C_f vs x/c for pressure coefficient distribution at $\alpha = 6^\circ$ and $Re = 300,000$

The skin friction coefficient distribution at and $Re = 300,000$, $\alpha = 6^\circ$ shows that the wall shear stress is greater near the leading edge and decreases along the airfoil chord as the boundary layer is developing. After the leading-edge region, both the upper and lower surfaces values gradually decreases towards the trailing edge indicating thickening of the boundary layer. The upper surface maintains a higher skin friction coefficient due to strong flow acceleration and suction effects observed in the pressure coefficient distribution. Since no negative C_f is observed, the result does not indicate a confirmed reversed flow region or a fully developed laminar separation bubble at this condition.

3.2 TRANSITION SST RESULTS AND ANALYSIS

This section presents the aerodynamic results obtained using the Transition SST turbulence model. The E387 airfoil was analyzed at Reynolds number of 200,000 and 300,000 over an angle of attack at a range from 0° to 14°. The results were evaluated using lift coefficient, drag coefficient, aerodynamic efficiency, pressure coefficient, skin friction coefficient, and flow field contours. Only the results from 0° to 10° are included in the quantitative Transition SST analysis because these cases reached stable aerodynamic coefficient values. The 12° and 14° cases exhibited oscillations persistently, indicating unsteady or possible flow separation related behavior. Hence 12° and 14° data points were excluded from the final coefficient points.

3.2.1 RE = 200,000 TRANSITION SST

Table 3. Transition SST Coefficient of lift, drag, lift to drag ratio and Angle of attack.

ANGLE OF ATTACK	COEFFICIENT OF LIFT, C_l	COEFFICIENT OF DRAG, C_d	C_l/C_d
0	0.366	0.01026	35.7065
2	0.581	0.0118	49.1688
4	0.793	0.0138	57.338
6	0.9998	0.0164	60.780
8	1.165	0.0233	49.974
10	1.240	0.03809	32.568

3.2.2 LIFT AND DRAG COEFFICIENT VARIATION WITH ANGLE OF ATTACK

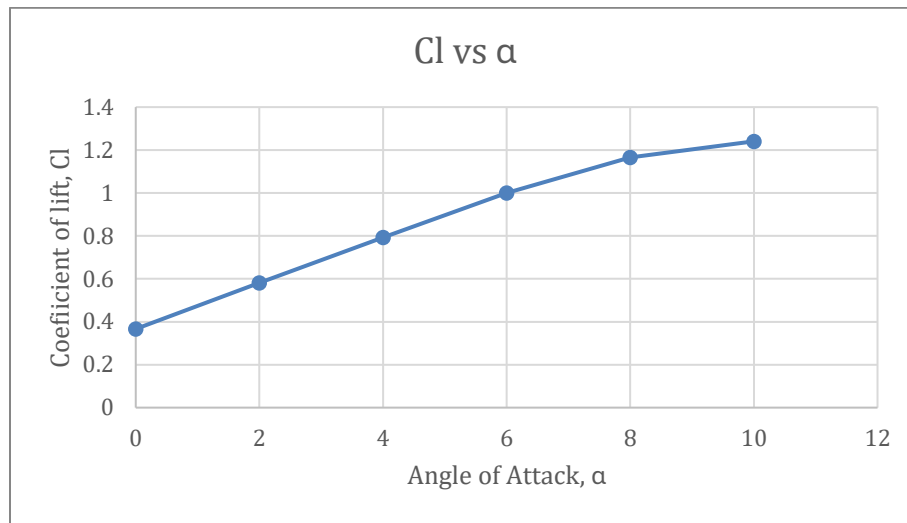


Fig.19 Variation of Coefficient of Lift vs Angle of Attack at $Re = 200,000$

As seen in Fig.19, from the 0° to 6° , the lift coefficient increases almost linearly from 0.366 to 0.998. lift coefficient increases steadily as the angle of attack rises to 10° . The lift curve is linear upto 6° , indicating an attached flow over the airfoil. The linear increase in C_l over the lower and moderate angle range indicates that the airfoil is generating lift while the flow remains largely attached. The transition SST model predicts a maximum lift coefficient of $C_{l} = 1.241$ at 10° . From 8° to 10° the the coefficient of lidt only increases from 1.165 to 1.240. However, since the 10° case remained stable, it can be concluded that the airfoil is still generating lift at that angle.

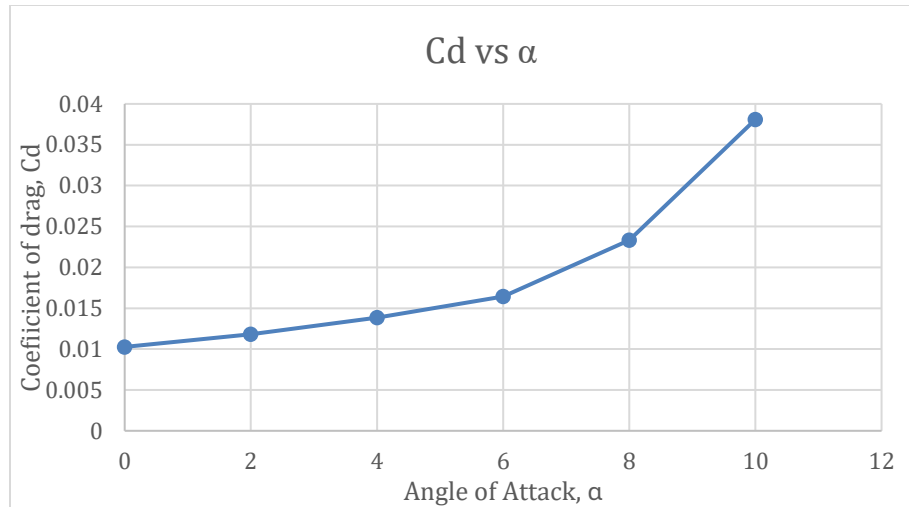


Fig.20 Variation of Coefficient of Drag vs Angle of Attack at $Re = 200,000$

The drag coefficient increased gradually at low and moderate angles of attack rising from $C_d = 0.0103$ at 0° to $C_d = 0.0164$ at 6° . This gradual increase is expected because pressure drag and viscous drag normally increase as lift and angle of attack increases. Beyond 6° , the drag increase becomes more pronounced. The drag coefficient rose to 0.0233 at 8° , 0.0381 at 10° . This rather growth in drag rise suggests substantial flow separation and a thicker wake, both of which increase pressure drag. Although the solution remained stable up to 10° , the drag trend shows that aerodynamic losses became more significant at higher angles of attack.

3.2.3 DRAG POLAR AND AERODYNAMIC EFFICIENCY

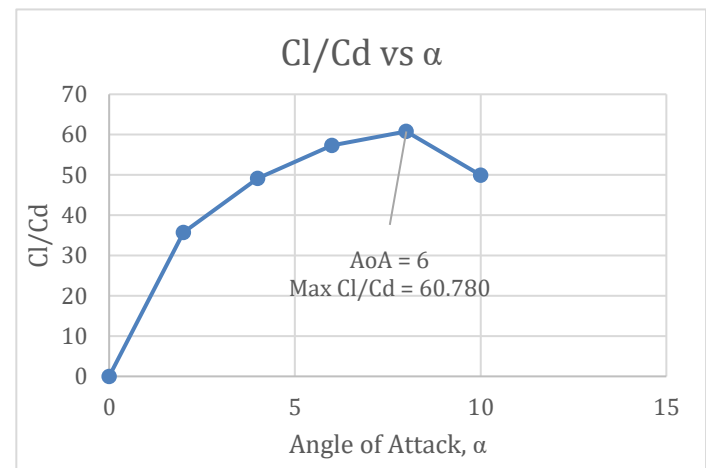
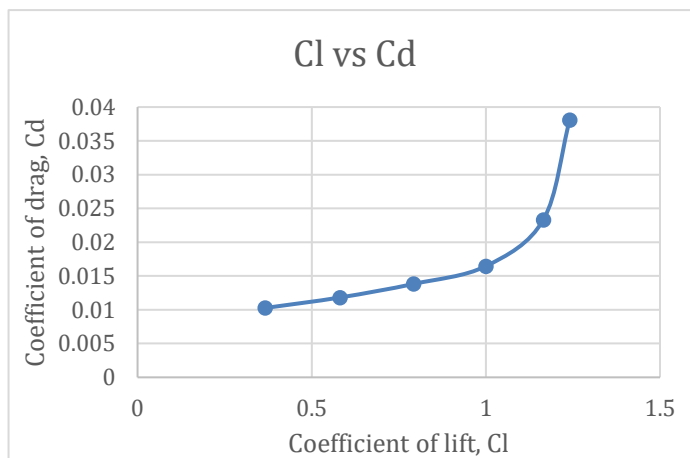


Fig. 21 C_l vs C_d and C_l/C_d vs Angle of Attack at $Re = 200,000$

The C_l vs C_d plot, commonly known as drag polar, shows how the lift produced by the airfoil changes relative to the aerodynamic drag. At lower drag values, the lift coefficient increased efficiently from approximately from 0.37 to 0.998 as C_d increased from 0.0103 to 0.0164. This corresponds to the efficient attached flow operating between the ranges of 0° and 6° . Beyond this region, further increases in lift with increasing angle of attack, the coefficient of drag increases disproportionately largely. As the lift increases from 0.998 at 6° to 1.24 at 10° , the drag coefficient rises sharply to 0.038. This indicates that the airfoil has exceeded its optimum operating range and is experiencing significant separation related losses in performance. Therefore, the drag polar confirms that the most efficient lift producing region occurs around 4° – 6° .

The C_l/C_d vs α commonly known as aerodynamic efficiency, the lift to drag ratio initially increases with angle of attack from 35.71 at 0° to a maximum value of 60.78 at 6° . This shows that the airfoil becomes more aerodynamically efficient as angle of attack increases within the lower and moderate operating ranges. The highest efficiency at 6° indicates that this was the optimum operating condition among the tested angles, as it produced the highest lift and relatively lowest drag. Beyond 6° , C_l/C_d declines rapidly. It reduces from 49.97 at 8° and 32.57 at 10° . This sharp reduction occurred because drag increased much faster than lift at higher angles of attack. Therefore, although lifts continued to increase up to 10° , the airfoil becomes progressively less efficient.

In the C_l/C_d vs α , 6° is considered the most aerodynamically efficient angle of attack for the Transition SST model at $Re = 200,000$. Even though lift continues increasing at 8° and 10° , the drag rises quickly, so the airfoil becomes less efficient.

3.2.4 VELOCITY AND PRESSURE CONTOURS FOR TRANSITION SST $Re = 200,000$ AT 6°

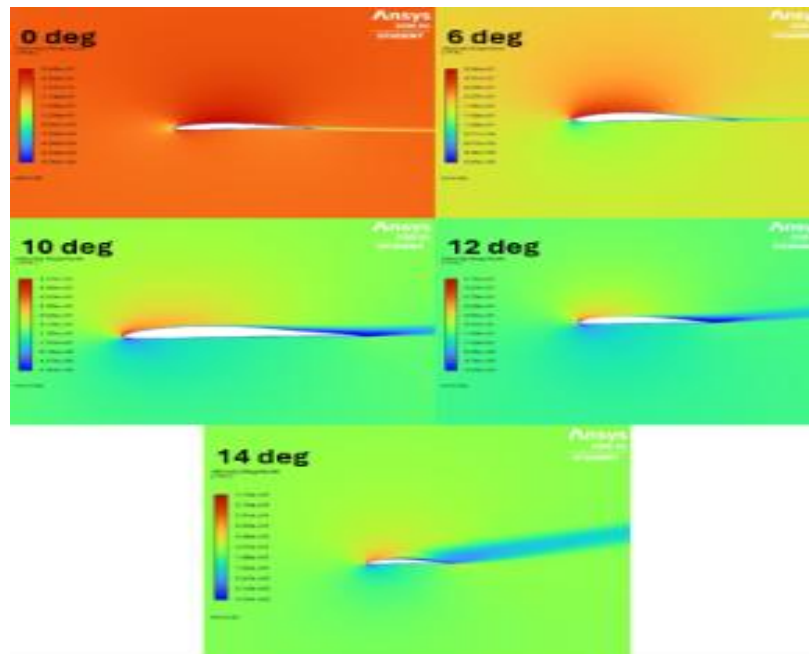


Fig. 21 Velocity Contours for $Re = 200,000$ at 0, 6, 10, 11, 12, 14 degrees.

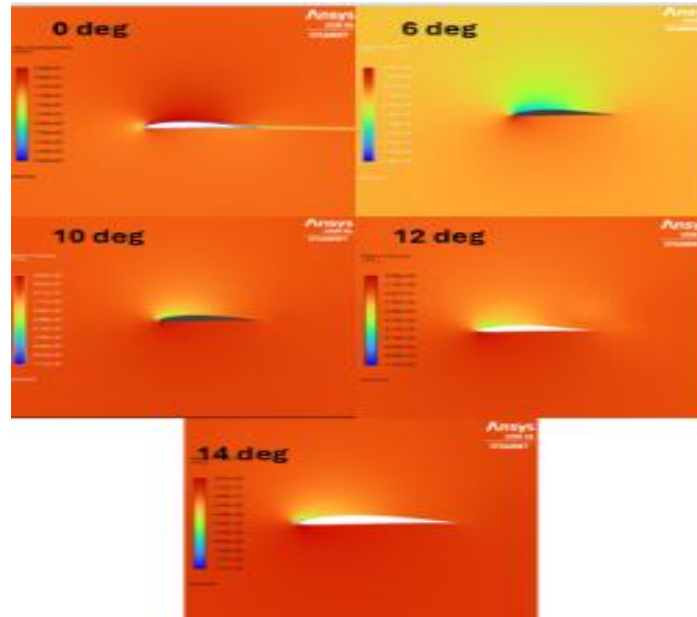


Fig. 22 Pressure Contours for $Re = 200,000$ at 0, 6, 10, 11, 12, 14 degrees.

3.2.5 COEFFICIENT OF PRESSURE TRANSITION SST OVER THE AIRFOIL ANALYSIS AT $Re = 200,000$ AT 6°

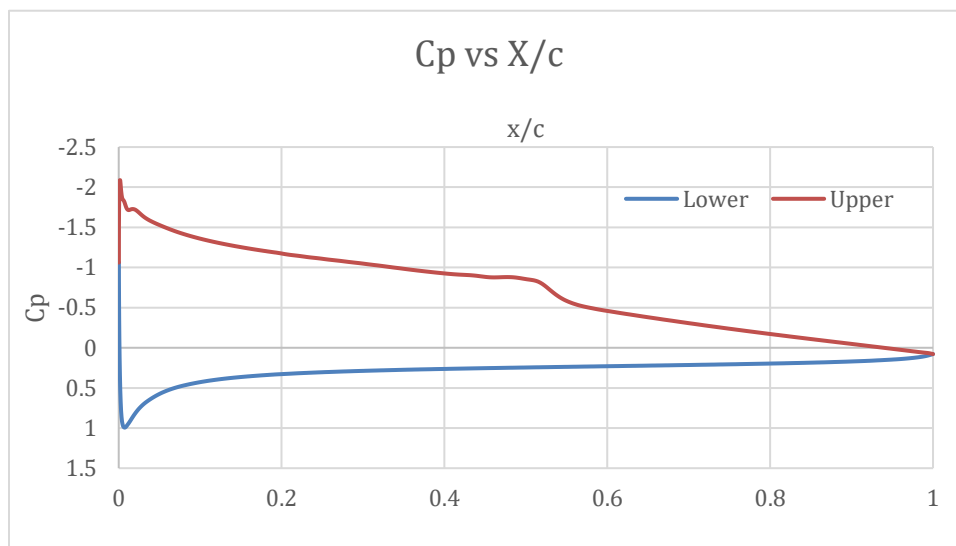


Fig. 23 C_p vs x/c for pressure coefficient distribution at $\alpha = 6^\circ$ and $Re = 200,000$

At $\alpha = 6^\circ$ and $Re = 200,000$, the pressure coefficient distribution shows a clear difference between the upper and lower surface, confirming effective lift generation. The upper surface has a strong negative C_p value near the leading edge, indicating a suction peak

caused by accelerated flow over the upper surface. The pressure then recovers gradually along the chord towards the trailing edge. The lower surface maintains a comparatively higher-pressure coefficient over most of the chord. This pressure difference between the lower and upper surface produces the upwards force responsible for lift. A noticeable pressure recovery region is visible around the $x/c = 0.5$ to 0.6 on the upper surface indicating an increase in adverse pressure gradient in the mid to aft chord region. The pressure distribution remains generally smooth, and both curves converge near the trailing edge.

3.2.6 SKIN FRICTION COEFFICIENT TRANSITION SST OVER THE AIRFOIL ANALYSIS AT $Re = 200,000$ AT 6°

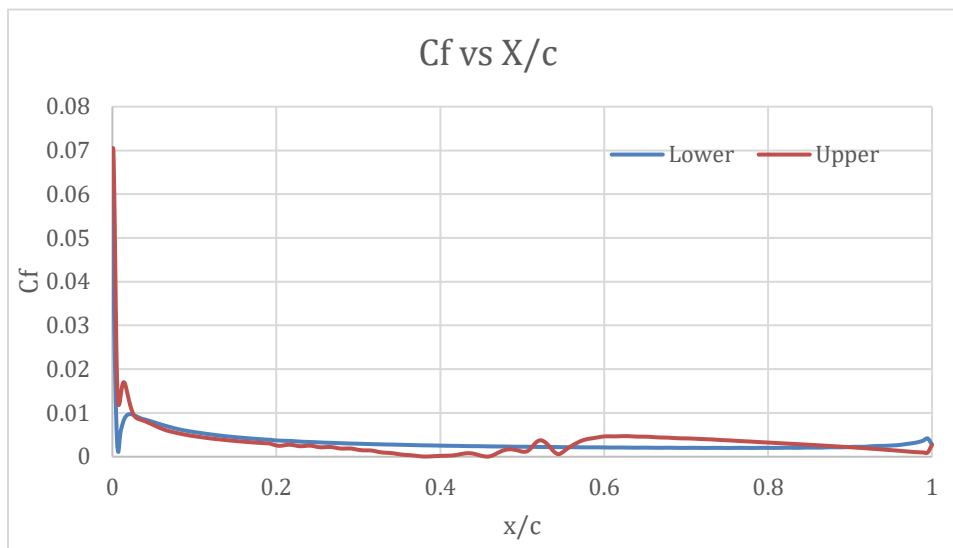


Fig 24. C_f vs x/c for pressure coefficient distribution at $\alpha = 6^\circ$ and $Re = 200,000$

The skin friction coefficient distribution at $\alpha = 6^\circ$ and $Re = 200,000$ shows that wall shear stress is highest close to the leading edge and decreases downstream along both the surfaces. This behavior is expected because the boundary layer is initially thin near the leading edge and grows progressively along the airfoil chord. The upper surface shows a large initial C_f value than the lower surface which is consistent with the strong flow acceleration and suction effect observed in the pressure coefficient plot.

3.2.7 $Re = 300,000$ TRANSITION SST

Table 4. Transition SST Coefficient of lift, drag, lift to drag ratio and Angle of attack.

ANGLE OF ATTACK	COEFFICIENT OF LIFT, C_l	COEFFICIENT OF DRAG, C_d	C_l/C_d
0	0.3701	0.00862	42.8975
2	0.587	0.00999	58.837
4	0.802	0.0118	67.886
6	1.012	0.0143	70.4821
8	1.166	0.0225	51.727
10	1.282	0.0335	38.159

3.2.7 LIFT AND DRAG COEFFICIENT VARIATION WITH ANGLE OF ATTACK

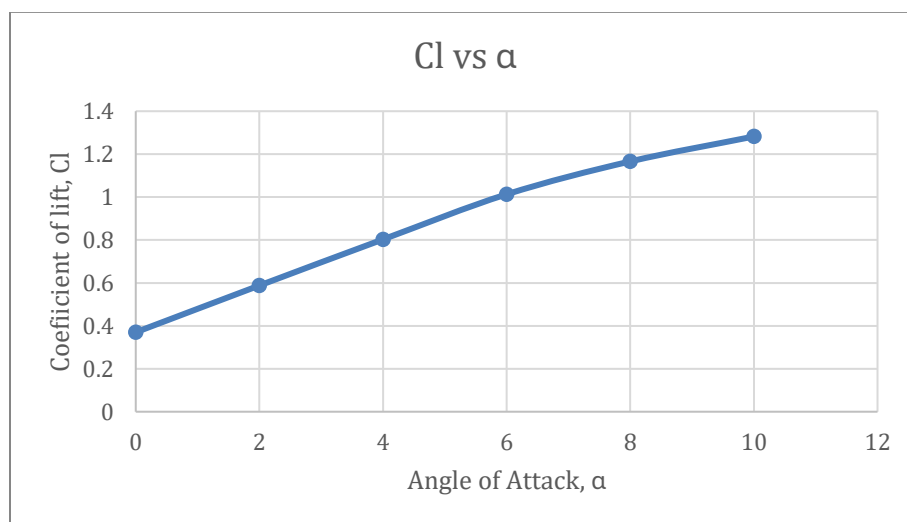


Fig.25 Variation of Coefficient of Lift vs Angle of Attack at $Re = 200,000$

As seen in Fig.25, from the 0° to 6° , the lift coefficient increases almost linearly from 0.37 to 1.012. lift coefficient increases steadily as the angle of attack rises to 10° . The lift curve is linear up to 6° , indicating an attached flow over the airfoil. The linear increase in C_l over the lower and moderate angle range indicates that the airfoil is generating lift while the flow remains largely attached. The transition SST model predicts a maximum lift coefficient of $C_{l} = 1.282$ at 10° . From 8° to 10° the the coefficient of lift only increases from 1.166 to 1.282. However, since the 10° case remained stable, it can be concluded that the airfoil is still generating lift at that angle.

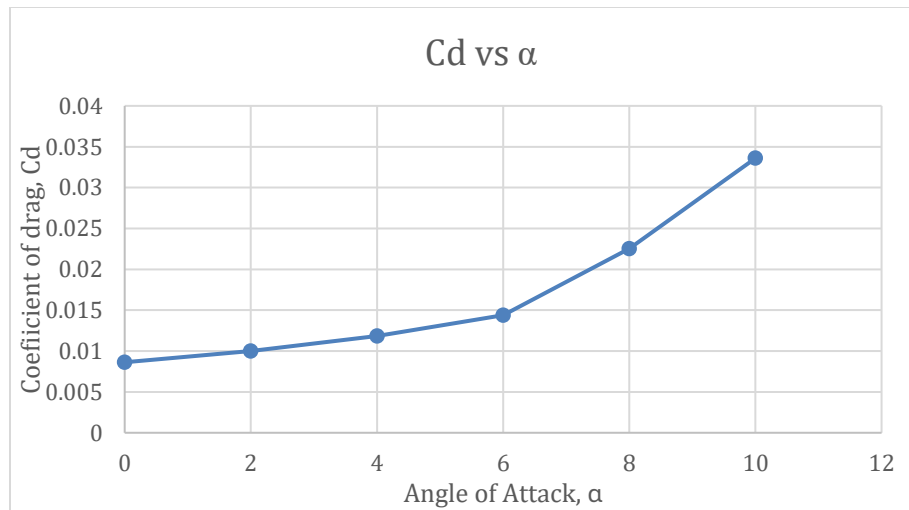


Fig.26 Variation of Coefficient of Drag vs Angle of Attack at $Re = 200,000$

The drag coefficient increased gradually at low and moderate angles of attack rising from $C_d = 0.0086$ at 0° to $C_d = 0.0143$ at 6° . This gradual increase is expected because pressure drag and viscous drag normally increase as lift and angle of attack increases. Beyond 6° , the drag increase becomes more pronounced. The drag coefficient rose to 0.0225 at 8° , 0.0335 at 10° . This rather growth in drag rise suggests substantial flow separation and a thicker wake, both of which increase pressure drag. Although the solution remained stable up to 10° , the drag trend shows that aerodynamic losses became more significant at higher angles of attack.

3.2.8 DRAG POLAR AND AERODYNAMIC EFFICIENCY

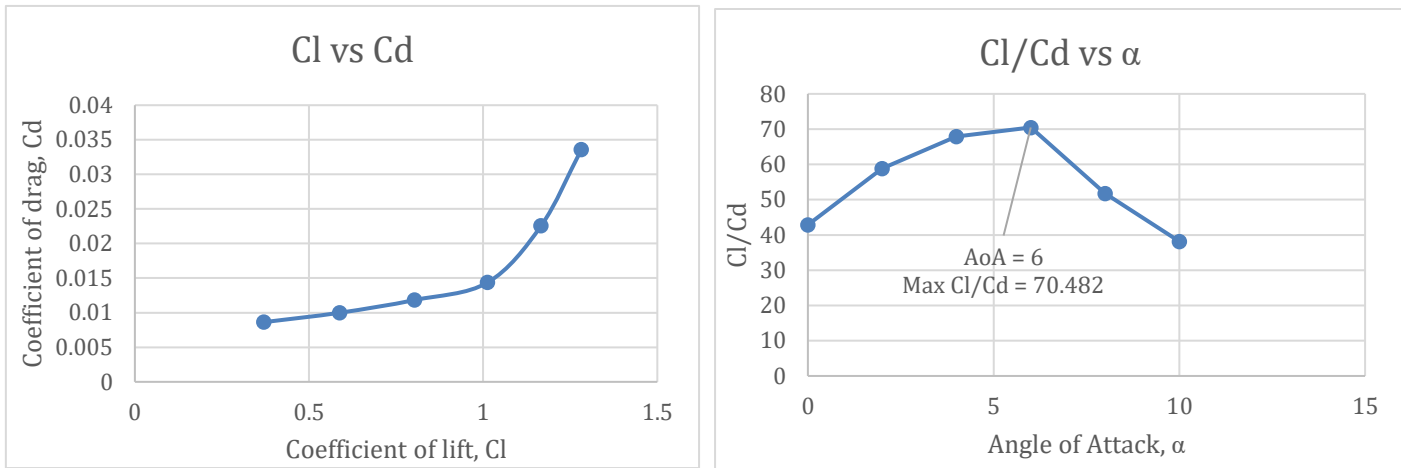


Fig. 27 C_l vs C_d and C_l/C_d vs Angle of Attack at $Re = 300,000$

The C_l vs C_d plot, commonly known as drag polar, shows how the lift produced by the airfoil changes relative to the aerodynamic drag. At lower drag values, the lift coefficient increased efficiently from approximately from 0.3701 to 1.012 as C_d increased from 0.0086 to 0.014. This corresponds to the efficient attached flow operating between the ranges of 0° and 6° . Beyond this region, further increases in lift with increasing angle of attack, the coefficient of drag increases disproportionately largely. As the lift increases from 1.012 at 6° to 1.282 at 10° , the drag coefficient rises sharply to 0.0335. This indicates that the airfoil has exceeded its optimum operating range and is experiencing significant separation related losses in performance. Therefore, the drag polar confirms that the most efficient lift producing region occurs around 4° – 6° .

The C_l/C_d vs α commonly known as aerodynamic efficiency, the lift to drag ratio initially increases with angle of attack from 42.8975 at 0° to a maximum value of 70.482 at 6° . This shows that the airfoil becomes more aerodynamically efficient as angle of attack increases within the lower and moderate operating ranges. The highest efficiency at 6° indicates that this was the optimum operating condition among the tested angles, as it produced the highest lift and relatively lowest drag. Beyond 6° , C_l/C_d declines rapidly. It reduces from 51.72 at 8° and 38.159 at 10° . This sharp reduction occurred because drag increased much faster than lift at higher angles of attack. Therefore, although lifts continued to increase up to 10° , the airfoil becomes progressively less efficient.

In the C_l/C_d vs α , 6° is considered the most aerodynamically efficient angle of attack for the Transition SST model at $Re = 300,000$. Even though lift continues increasing at 8° and 10° , the drag rises quickly, so the airfoil becomes less efficient.

3.2.9 VELOCITY AND PRESSURE CONTOURS

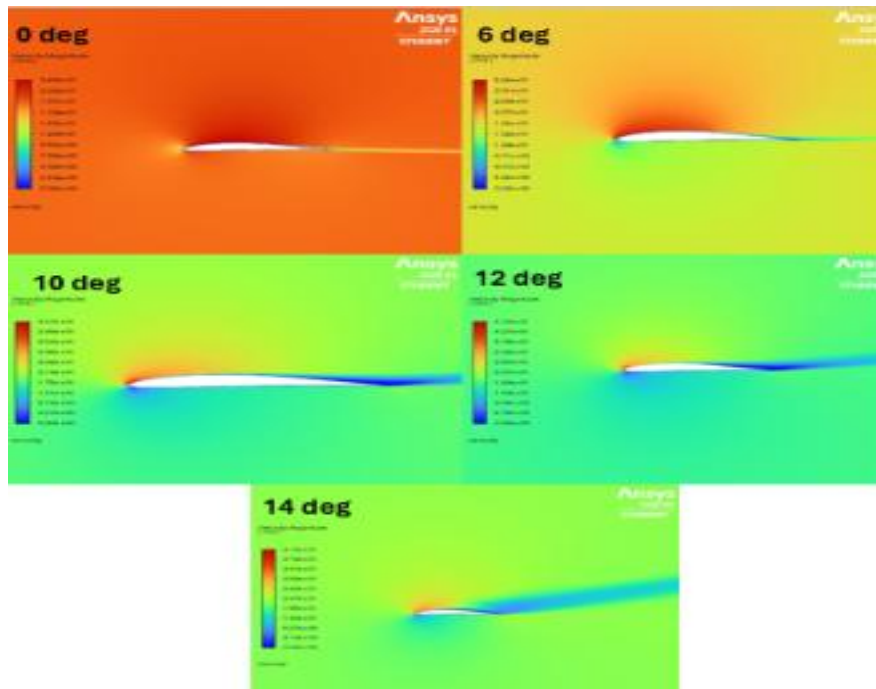


Fig. 28 Velocity Contours for $Re = 200,000$ at 0, 6, 10, 11, 12, 14 degrees.

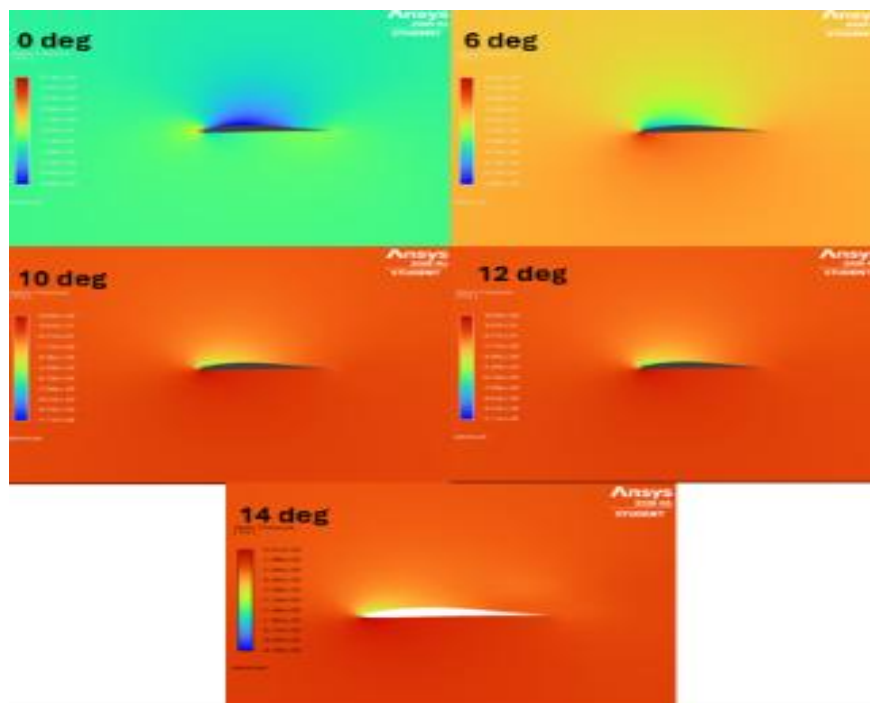


Fig. 29 Pressure Contours for $Re = 300,000$ at 0, 6, 10, 11, 12, 14 degrees.

3.2.10 COEFFICIENT OF PRESSURE TRANSITION SST OVER THE AIRFOIL ANALYSIS AT $Re = 300,000$ AT 6°

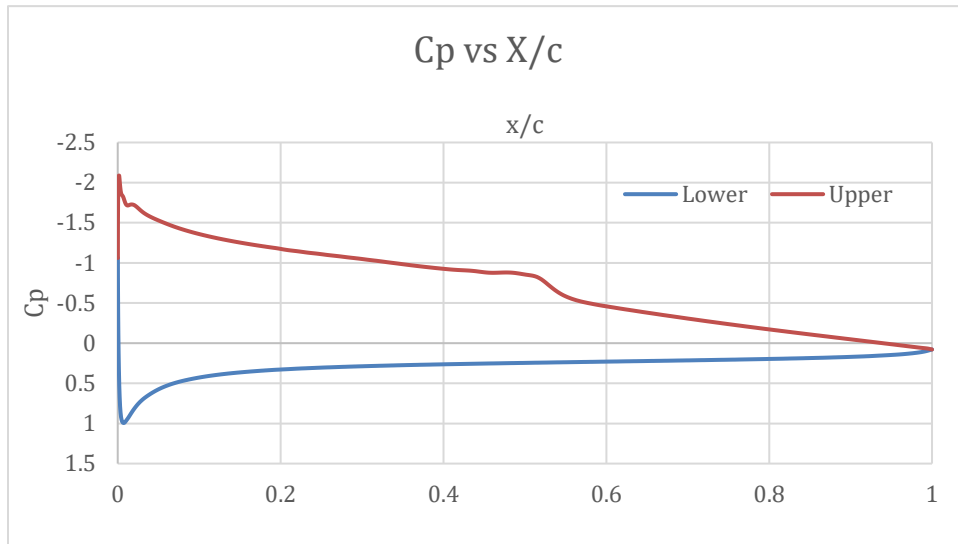


Fig. 30 C_p vs x/c for pressure coefficient distribution at $\alpha = 6^\circ$ and $Re = 300,000$

At $\alpha = 6^\circ$ and $Re = 300,000$, the pressure coefficient distribution shows a clear difference between the upper and lower surface, confirming effective lift generation. The upper surface has a strong negative C_p value near the leading edge, indicating a suction peak caused by accelerated flow over the upper surface. The pressure then recovers gradually along the chord towards the trailing edge. The lower surface maintains a comparatively higher-pressure coefficient over most of the chord. This pressure difference between the lower and upper surface produces the upwards force responsible for lift. A noticeable pressure recovery region is visible around the $x/c = 0.5$ to 0.6 on the upper surface indicating an increase in adverse pressure gradient in the mid to aft chord region. The pressure distribution remains generally smooth, and both curves converge near the trailing edge.

3.2.11 SKIN FRICTION COEFFICIENT TRANSITION SST OVER THE AIRFOIL ANALYSIS AT $Re = 300,000$ AT 6°

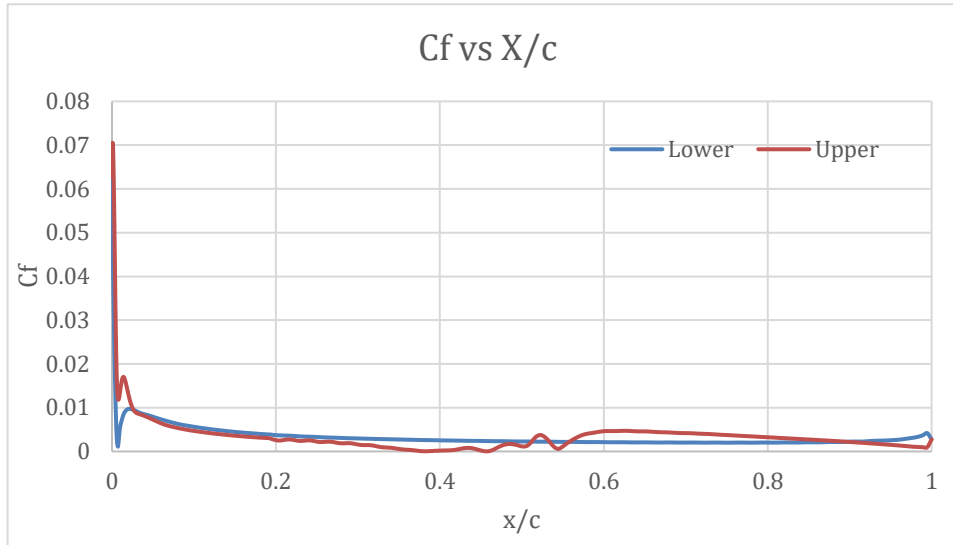


Fig. 31 Cf vs x/c for pressure coefficient distribution at $\alpha = 6^\circ$ and $Re = 300,000$

The skin friction coefficient distribution at $\alpha = 6^\circ$ and $Re = 300,000$ shows that wall shear stress is highest close to the leading edge and decreases downstream along both the surfaces. This behavior is expected because the boundary layer is initially thin near the leading edge and grows progressively along the airfoil chord. The upper surface shows a large initial Cf value than the lower surface which is consistent with the strong flow acceleration and suction effect observed in the pressure coefficient plot.

3.3 UIUC DATA COMPARISON VS 1 K- Ω SST AND TRANSITION SST

The CFD results for the E387 airfoil was validated against clear airfoil wind tunnel data from the UIUC low speed airfoil test programme. The UIUC data sets contain experimental lift and drag data at average Reynolds numbers of $Re = 199,879$ and $Re = 300,289$, which were selected as the closest available conditions to present CFD cases of $Re = 200,000$ and $Re = 300,000$. (University of Illinois Urbana-Champaign Applied Aerodynamics Group, 2003)

The higher angle transition SST cases were excluded from steady validations because their coefficient values remained oscillatory. This behavior is consistent with increasingly separated and unsteady flow near stall, for which a single steady-state aerodynamic coefficient may not be represented. Similar limitations of stall airfoil measurements and comparisons were noted in the E387 win tunnel study (Selig and McGranahan, 2004)

The experimental angles of attack do not exactly match every CFD angle, so the nearest available UIUC points were used in each comparison.

The percentage error was calculated separately for C_l and C_d :

$$\% \text{ Error} = \left| \frac{\text{CFD value} - \text{UIUC value}}{\text{UIUC value}} \right| \times 100$$

For example, at $Re = 200,000$, $\alpha = 6^\circ$, for the Transition SST lift prediction:

$$\% \text{ Error}_{C_l} = \left| \frac{0.9998 - 1.017}{1.017} \right| \times 100$$

$$\% \text{ Error}_{C_l} = 1.69\%$$

For the corresponding Transition SST drag prediction:

$$\% \text{ Error}_{C_d} = \left| \frac{0.01645 - 0.0140}{0.0140} \right| \times 100$$

$$\% \text{ Error}_{C_d} = 17.50$$

3.3.1 RE = 200,000 COMPARISON

Table 5: Validation comparison at $Re \approx 200,000$

AoA	Cl (k-omega)	Cd (k-omega)	Cl (Transition SST)	Cd (Transition SST)	AoA (UIUC)	Cl (UIUC)	Cd (UIUC)	Percentage Error Cl (k-omega)	Percentage Error Cd (k-omega)	% error Cl (transition SST)	% error Cd (Transition SST)
0	0.35842	0.01484	0.366	0.01026	-0.05	0.381	0.0104	5.9%	42.7%	3.8%	1.3%
2	0.5705	0.0152	0.581	0.0118	2.01	0.593	0.0118	3.8%	28.8%	2.0%	0.1%
4	0.776	0.01685	0.793	0.0138	4.09	0.806	0.0134	3.7%	25.7%	1.6%	3.2%
6	0.9718	0.0195	0.9998	0.0164	6.09	1.017	0.0140	4.4%	39.3%	1.7%	17.5%
8	1.1442	0.0242	1.165	0.0233	8.15	1.191	0.0181	3.9%	33.7%	2.2%	28.8%
10	1.2613	0.0334	1.240	0.03809	10.12	1.234	0.0358	2.2%	6.7%	0.5%	6.4%
11	1.2798	0.0421	-	-	11.18	1.227	0.0550	4.3%	23.5%	-	-

At $Re = 200,000$, both CFD models reproduce the overall lift trend well. The k- ω SST model produced lift errors between approximately 2.2% and 5.9% while the transition SST model produced lower lift errors over its stable range, between approximately 0.5% and 3.8%. The average lift coefficient error was approximately 4.0% for the k- ω SST and 2.0% for transition SST.

The main difference between the models appears in drag prediction. The fully turbulent $k-\omega$ SST model consistently overpredicted C_d in the lower and moderate angles of attack range, with drag errors ranging from 6.7% to 42.7%. This is expected because the model assumes fully turbulent boundary layer increasing predicted skin-friction drag compared with a lower Reynolds number airfoil that retains laminar flow over parts of its surface. The transition SST model produced lower drag errors in general. And at 6° and 8° , it is still overpredicting drag, which may be associated with sensitivity to predict transition position, local recovery, and wall resolution.

Overall, the transition SST model provided the closest coefficient of lift and drag values compared with the UIUC data within the stable operating range.

3.3.2 $Re = 300,000$ COMPARISON

Table 6: Validation comparison at $Re \approx 300,000$

AoA	Cl (k- ω)	Cd (k- ω)	Cl (Transition SST)	Cd (Transition SST)	AoA (UIUC)	Cl (UIUC)	Cd (UIUC)	% Error Cl (k- ω)	%Error Cd (k- ω)	% error Cl (transition SST)	% error Cd (Transition SST)
0	0.332	0.01314	0.3701	0.00862	-0.04	0.381	0.0104	7.6%	58.3%	5.8%	3.9%
2	0.5770	0.01418	0.5870	0.00998	1.96	0.593	0.0118	5.4%	54.1%	3.8%	8.6%
4	0.7846	0.01560	0.8020	0.01180	4.00	0.806	0.0134	4.0%	48.36%	1.8%	12.4%
6	0.9843	0.01807	1.0120	0.01430	6.11	1.017	0.0140	5.4%	57.1%	2.8%	24.3%
8	1.1620	0.02237	1.1660	0.02250	8.14	1.191	0.0181	3.2%	27.1%	2.9%	27.8%
10	1.2932	0.03025	1.2820	0.03350	10.12	1.234	0.0358	3.5%	11.8%	2.6%	2.3%
11	1.3314	0.03656	-	-	11.16	1.227	0.0550	6.9%	31.9%	-	-
12	1.3360	0.04624	-	-	12.17	-	-	8.1%	40.3%	-	-

At $Re = 300,000$, both models again captured the increase in lift with angle of attack. The $k-\omega$ SST model produced lift errors between 3.2% and 8.1%, while Transition SST produced lower lift errors over the stable range, from approximately 1.8% and 5.8%. The average stable range error was approximately 5.5% for $k-\omega$ SST and 3.3% for transition SST.

The drag comparison shows a clearer distinction between the models. The $k-\omega$ SST model overpredicted drag strongly at lower and moderate angles of attack, with errors ranging from 27.1% to 58.3% between 0° and 8° . This is consistent with assumption of a fully turbulent boundary layer, which tends to increase skin friction drag.

Transition SST significantly improved the drag prediction, 2° , 4° , and particularly 10° , where the drag error was only approximately 2.4% on average. However, the model still overpredicts drag at 6° and 8° . This indicates that, although the transition model better represents the physics of laminar to turbulent boundary layer development, drag remains sensitive to transition, turbulence intensity, and mesh resolution near the wall.

4.CONCLUSION

This project investigated the aerodynamic performance of the Eppler 387 airfoil at Reynolds numbers of 200,000 and 300,000 using the $k-\omega$ SST and Transition SST turbulence models in ANSYS fluent. The study compared lift, drag, aerodynamic efficiency, pressure distribution, skin-friction behavior, and separation trends against available experimental reference data.

At Reynolds number 200,000 the average lift coefficient error was approximately 4% for the $k-\omega$ SST model and approximately 2 % for the transition SST model. At Reynolds number 300,000, the average stable lift error was approximately 5.5% for $k-\omega$ SST model and 3.3% for transition SST. The transition model also produced generally lower drag errors, particularly at low and moderate angles of attack, where the fully turbulent $k-\omega$ SST model tends to overpredict drag.

The difference is important because the E387 airfoil operates in a low Reynolds number regime where the boundary layer may remain laminar over part of the chord before transitioning to turbulence. The $k-\omega$ SST model assumes a fully turbulent boundary later from the leading edge, which can increase predicted skin-friction drag and reduce its ability to represent transitional flow behavior accurately. In comparison, the transition SST model accounts for laminar to turbulent transition and therefore produced lift and drag values that were generally closer to the experimental referenced data.

For the $k-\omega$ SST model at Reynolds number of 200,000, the maximum aerodynamic efficiency occurred at 6° , where the lift to drag ratio reached approximately 49.84. This condition provided the best balance between lift generation and drag. Although lift continued to increase at higher angles of attack, drag increased more rapidly. At 11° , the highest stable lift coefficient was obtained, but the lift to drag ratio has reduced to approximately 30.44, showing that the airfoil was operating in a high drag less efficient condition.

Overall, the study concludes that the $k-\omega$ SST model is useful as a simpler and more easily implemented baseline model. However, for detailed low Reynolds number UAV airfoil analysis, where the lift and drag prediction is required, the transition SST model should be preferred. Its additional setup sensitivity and computational efforts are justified by its improved experimental aerodynamic data as it better represents the transition sensitive flow behavior.

5. FUTURE WORK AND RECOMMENDATIONS

1. Future work should apply unsteady RANS simulations to the 2° – 14° cases and beyond, where severe coefficient oscillations were observed. This would allow time dependent separation to develop and fluctuate aerodynamic forces to be studied more realistically.
2. A more detailed mesh independence study particularly near the leading edge, trailing edge, boundary layer, and wake region. These areas are important because they strongly influence the predicted pressure gradients, skin friction values, transition location, lift and drag coefficients as well as separation points. A finer mesh near the airfoil wall would improve the velocity gradient and wall shear stress, especially for the sensitivity of the transition SST model.
3. The present study investigated Reynolds' numbers of 200,000 and 300,000. These values are relevant to low-speed UAV flights, however future work should investigate a wider range of Reynolds' numbers. Additional cases would include Reynolds numbers between 100000, 150000, 350000, and 400000. Lower Reynolds number would represent slower flight, but higher Reynolds numbers would represent faster cruise conditions or large wing chords.
4. Future work should include direct wind tunnel testing of the E387 airfoil, where possible. Although published UIUC and NASA data provided useful validation, direct experimental testing would allow the exact CFD geometry, Reynolds number, and angle of attack range to be matched more closely. Flow visualization methods such as smoke or oil flow visualization could also be used to identify separation, reattachment, and transition behavior. These techniques would provide stronger evidence for validating the flow features predicted by the transition SST model.
5. The present study only used a 2-D airfoil model. This is appropriate for analyzing profile drag and boundary layer development. However, an actual UAV wing is three-dimensional and experiences additional aerodynamic effects. This would allow us to investigate induced drag, wing tip vortices, spanwise flow, lift distribution and three-dimensional separation behavior.
6. The transition SST model should be investigated further by varying inlet turbulence intensity, turbulence viscosity ratio, and transition related boundary conditions. At low Reynolds numbers, these parameters may significantly affect the predicted transition location and the drag/separation behavior.
7. Since the original project identifies leading edge surface irregularities as an important practical condition to experiment. Due to time constraints, surface irregularity effects could not be studied. Hence, A dedicated study would therefore

provide useful recommendations on how surface finish may also diminish UAV wings performance.

6. CHALLENGES AND DECISION MAKING

Several challenges were encountered during the CFD analysis of the E387 airfoil, particularly when investigating high angles of attack conditions and comparing the two turbulence modelling approaches. The challenges were important because low Reynolds airfoil flows are extremely sensitive to transition and boundary layer development.

1. One of the main challenges that was encountered during the CFD study was achieving stable steady-state convergence at higher angles of attack, particularly between 12° and 14° . At these conditions, the lift and drag coefficient showed large and persistent oscillations. Since at 12° and 14° were unstable, a single final C_l and C_d values could not be considered dependable. These cases were not included in quantitative lift, drag, drag polar and aerodynamic efficiency plots but only velocity and pressure contours, skin friction and coefficient of pressure were analyzed for the flow.
2. Convergence was not assessed using residual values alone. Residual reduction is important, but it does not always guarantee that the aerodynamic coefficients have reached stable values. Therefore, the convergence study included both residual, lift and drag coefficient.
3. The selection of turbulence models was another important engineering decision in this study. The $k-\omega$ SST model was selected as a fully turbulent reference model because it is commonly used for aerodynamic flows involving adverse pressure gradients, near wall effects and flow separation. It provides a practical baseline for assessing how a conventional fully turbulent RANS model predicts low Reynolds number airfoil behavior.

The E387 airfoil operates in Reynolds number range where the boundary layer may not be fully turbulent across the entire surface. At low Reynolds numbers, the flow may remain laminar over part of the airfoil before transitioning to turbulence. The location of this transition can influence drag, pressure recovery, separation, and stall behavior. For this reason, the transition SST model was selected.

4. Although the transition SST model was expected to provide a more physically representative prediction of low Reynolds number airflow, it included additional numerical sensitivity. The model is more dependent on mesh quality, inlet turbulence conditions, near wall resolution and boundary layer treatment. Slight changes in turbulence intensity, turbulence viscosity ratio, mesh density or first layer height may influence the predicted transition location. This means that Transition SST results must be interpreted carefully. A difference between the $k-\omega$ SST and Transition SST models may reflect genuine transition flow physics, but it may also be influenced by sensitivity of the transition to the numerical model.

5. A further challenge in this project was the learning curve associated with using ANSYS fluent for low aerodynamic number aerodynamic simulations. Although the basic CFD workflow appears straightforward, obtaining reliable airfoil results require careful decision at each stage, including geometry preparation, mesh generation, turbulence model selection, and solver settings. At the beginning of the study, considerable amount of time was required to understand how changes in the numerical setup may affect the predicted lift and drag values. Mesh generation was also an important learning challenge. The mesh needed to be sufficiently refined around the airfoil surface, leading edge, trailing edge, and boundary layer region while remaining computationally manageable. Another challenge was interpreting fluent results correctly. Initially it was tempting to judge convergence only from the residual plots. However, it became clear that stable aerodynamic coefficients were equally important. Overall learning ANSYS was a significant challenge, but it became one of the main outcomes of the project. The process developed practical understanding of CFD setup, turbulence modelling, and aerodynamic post processing.

7. AKNOWLEDGMENTS

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including average Reynolds numbers of 199,879 and 300,289
https://m-selig.ae.illinois.edu/pd/pub/lsat/vol4/e387_c_drg.txt

9. APPENDICES

RESEARCH ETHICS APPROVAL FORM

SECTION 1 : GENERAL PROJECT INFORMATION

Research by students and staff at the University must receive ethical approval before any data collection commences. Applications should be made via this approval form for all individual and group projects completed on the EAU engineering degree programmes.

Applicants must complete this Research Ethics Approval Form, which include the Risk Checklist.

For student project classified as Risk Category 1 (low risk, eg literature review), these can be approved on this Research Ethics Approval Form by the research supervisor and filed with the programme coordinator.

Applicants whose research studies are classified as Risk Category 2 (medium risk, eg laboratory based) must also complete section 2 of this form and can be approved by the research supervisor and filed with the programme coordinator.

Applicants whose research studies are classified as Risk Category 3 (high risk) must talk to their supervisor to agree specific ethical mitigation, which will require significant changes to their project scope.

APPLICANT DETAIL	
Your Name (if a group project, include all names)	Sagar PrakashKumar Tailor, Hasan Mohammed Al-Ezzi
Student ID	10153641, 10178015
Course Title	Aerospace Applications
Student email	
STATUS	
• Undergraduate Student	☒ Y
• Taught Postgraduate Student	☐
• Research Postgraduate Student	☐
• Other (please specify)	☐
PROJECT DETAILS	
Research Supervisor's Name	Dr. Walid Abou Hweij
Project Title	CFD analysis and model comparison for E387 airfoil
Project Start Date	8 th June 2026

Expected Completion Date of Project	5 th July 2026
Project Summary – please give a brief summary of your research project (maximum 100 words)	
This project investigated the aerodynamic performance of the E387 low-Reynolds-number airfoil using ANSYS Fluent CFD at Reynolds numbers of 200,000 and 300,000. The study compared the fully turbulent $k-\omega$ SST model with the Transition SST $\gamma - Re_\theta$ model across multiple angles of attack. Lift coefficient, drag coefficient, lift-to-drag ratio, pressure distribution, skin-friction behaviour, and flow contours were analysed. CFD predictions were validated against UIUC wind-tunnel data. Results showed that the Transition SST model generally provided more accurate lift and drag predictions in the stable pre-stall range, while $k-\omega$ SST tended to overpredict drag. The project recommends Transition SST for low-Reynolds-number UAV airfoil analysis.	
CONFIRMATION STATEMENTS	
Please tick this box to confirm that you have read the Research Ethics Policy/Procedures and that you will adhere to these in the conduct of this project	☐ Y

RISK CHECKLIST – Please answer ALL questions in each of the sections below by ticking YES or NO.		YES	NO
Will your research project			
1	Involve direct and/or indirect contact with human participants?	☐	☒ N
2	Involve analysis of pre-existing data which contains personal or sensitive information not in the public domain?	☐	☒ N
3	Collect / contain sensitive personal data?	☐	☒ N
4	Involve respondents where participants may be identified?	☐	☒ N
5	Use any information OTHER than that which is freely available in the public domain?	☐	☒ N
6	Include a financial incentive to participate in the research?	☐	☒ N
7	Involve participants who are particularly vulnerable or at risk?	☐	☒ N
8	Involve any participants who are unable to give informed consent?	☐	☒ N
9	Involve any deliberate deception or covert data collection?	☐	☒ N
10	Involve the use of an external research instrument (eg questionnaire, interview)	☐	☒ N
11	Involve laboratory, workshop or off-campus construction/testing (eg machining, construction, testing)	☐	☒ N
12	Involve a clinical trial?	☐	☒ N
13	Involve the possibility of incidental findings related to health status?	☐	☒ N
14	Is there any significant risk that the project may lead to physical harm to the participants?	☐	☒ N
15	Fit into any of the following security-sensitive categories: concerns terrorist or extreme groups; commissioned by the military; involve the acquisition of security clearances? If yes, seek guidance from your supervisor	☐	☒ N

RISK CHECKLIST – Please answer ALL questions in each of the sections below by ticking YES or NO. Will your research project	Tick the box which applies to your project
If you answered NO to all the above questions, your project is provisionally classified as Risk Category 1 (low risk – literature reviews will be Risk Category 1)	Y
If you answered YES to question 11 and NO to all other questions, your project is provisionally classified as Risk Category 2a (medium risk – engineering laboratory based will be Risk Category 2). Please complete the Section 2a of this form to demonstrate mitigation of risk)	☐
If you answered YES to any question from 1-10 and NO to all questions 11-15, your project is provisionally classified as Risk Category 2 (medium risk – project utilising questionnaires will be Risk Category 2). Please complete the Section 2b of this form.	☐
If you answered YES to question 11 AND YES to any question from 1-10 and NO to all questions 12-15, your project is provisionally classified as Risk Category 2 (medium risk – engineering laboratory and project utilising questionnaires will be Risk Category 2). Please complete the Section 2a and 2b of this form.	☐
If you answered YES to any question from 12-15, your project is provisionally classified as Risk Category 3 (high risk –talk to your supervisor, at this stage your project can not be approved)	☐

10. PROJECT PROPOSAL

AEROSPACE APPLICATIONS – EG118 / 302SE

RESEARCH PROPOSAL FORM

Please fill in your details as provided on this form, however, please answer the questions in Part B on separate sheets that are clearly typed and attached to this form.

PART A – DETAILS

Student Name & No. (Group leader): Sagar PrakashKumar Tailor

Student ID: 10153641

Course: Aerospace Applications

Email: 10153641@eau.ac.ae

Supervisor: **Dr. Walid Abou Hweij**

PART B – PROJECT DETAILS

1. Project Title

NUMERICAL INVESTIGATION OF TURBULENCE MODEL ACCURACY AND SURFACE ROUGHNESS EFFECTS ON LOW-REYNOLDS-NUMBER E387 AIRFOIL AERODYNAMICS

2. Abstract/ Summary

The rapid growth in small unmanned aerial vehicles (UAVs) across various sectors has created a growing demand for accurate aerodynamic performance prediction tool capable of operating reliably in low Reynold number flow regimes. Unlike conventional manned aircraft like commercial, cargo or military, small, fixed wing UAVs typically operate at chord Reynolds number between 100000 and 500000, a transitional regime where boundary layers behavior is neither fully laminar nor fully turbulent, making it one of the most complex and poorly predicted aerodynamic flight conditions in modern aerospace engineering.

At Reynolds numbers between 150000 and 300000, the boundary layer over an airfoil tends to begin as laminar near the leading edge and then undergoes transition or separation depending on the angle of attack, surface conditions often create a laminar separation bubble that may significantly alter lift, drag and other stall characteristics. During the manufacturing process of UAVs, some modelling irregularities within the leading edge may be seen, other instances include flies, dust, tape that may also create some aerodynamic inefficiencies within the airfoil itself. Some of the conventional models like RANS (Reynolds Averaged Navier Stokes) models such as K- ω SST model assumes fully turbulent flow, have been widely shown to misinterpret the transitional behavior, yet they are used often due to its computational efficiency and familiarity. The k- ω SST model solves 2 transport equations - One for turbulent kinetic energy (k) and one for specified dissipation rate ω . While this model performs well in fully turbulent attached flows, it assumes turbulence production begins from the leading edge and contains no mechanism to represent a laminar region making it inappropriate for transitional Reynolds numbers that small UAVs operate within. The transition SST model, also known as the $\gamma - Re_{\theta}$ model, was developed by Menter and Langtry specifically to resolve this limitation by augmenting the k- ω SST model with 2 additional transport equations - one for intermittency γ , ranging from 0 in fully laminar flow to 1 fully turbulent flow and directly scales turbulence depending on local flow conditions, and one for transition momentum thickness Reynolds number $Re_{\theta t}$ predicting where the transition will be triggered based on pressure gradients and freestream turbulence intensity. Together these 4 equations allow the $\gamma - Re_{\theta}$ model to onset the progression of laminar to turbulent transitions, formation of laminar separation bubbles and stall behavior capabilities.

This project is a study using ANSYS Fluent, with 2-D steady-state RANS simulations performed on E387 airfoil across a range of angles of attack from 0 deg to beyond stall, at Reynolds numbers of 150000 and 300000 with both smooth and irregular leading-edge surfaces.

3. *List overall objectives of the project and the activities, approach and mechanism to achieve each objective. Both Objects and activities should be measurable, realistic, and will be used in part assessment in determining the level of achievement of the project.*

OBJECTIVE 1: ESTABLISH THEORETICAL FOUNDATION THROUGH LITERATURE REVIEW

The first objective includes conducting a comprehensive review of low Reynolds numbers aerodynamics, transition flow behavior, turbulence modelling approaches, and existing CFD validation studies for UAV airfoils.

ACTIVITIES:

- Review literature on laminar separation bubble formation, transitional flow regimes, and their significance between 150000 and 300000.
- Study and document the theoretical knowledge based on $k-\omega$ SST and Transition SST ($\gamma-Re\theta$) turbulence models, including its governing equations, assumptions and known limitations at low Reynolds numbers.
- Review UIUC datasets for the E387 airfoil to understand and validate against the ANSYS CFD results and produced a structured literature review report.

OBJECTIVE 2: DEVELOP AND VALIDATE COMPUTATIONAL MODELS

Construct verified and validated 2D CFD model of the Eppler E387 airfoil in ANSYS Fluent to accurately replicate the experimental data and observe aerodynamic behavior before any further study is conducted.

ACTIVITIES:

- Obtain E387 airfoil coordinates and import geometry to ANSYS.
- Construct a C-type computational domain with appropriate boundaries.
- Design inflation layer mesh with wall $y^+ \leq 1$ for better viscous sublayer resolution.
- Conduct a mesh independence study confirming that C_l and C_d values vary by less than 1% between refinements.

- Validate all three modeling approaches including laminar, k- ω SST, and Transition SST independently against published UIUC E387 wind tunnel data between AoA 0° to stall.

OBJECTIVE 3: COMPARE TURBULENCE MODEL PERFORMANCE ACROSS REYNOLDS NUMBERS

Evaluate and compare the predictive accuracy of laminar, $\gamma - R_{e\theta}$ and k- ω SST transition modelling for E387 airfoil.

ACTIVITIES:

- Run steady-state RANS simulations using all three modelling approaches at Reynolds numbers of 150000 and 300000 across AoA from 0° to stall.
- Extract tabulated Cl, Cd, Cl/Cd ratio, drag polar, and Cl/Cd vs AoA for all three models.
- Quantify each model relative to UIUC experimental data and identify where and why each model fails in the transition regime.
- Comparative results report with lift curve, drag polar, Cl/Cd plots for all three models against UIUC benchmark data with percentage error data.

OBJECTIVE 4: ASSESS THE EFFECT OF LEADING-EDGE SURFACE IRREGULARITIES

Investigate how leading-edge surface irregularities as a representation of manufacturing process defects and how it may affect predicted aerodynamic performance at low Reynolds numbers.

ACTIVITIES:

- Develop a set of leading-edge irregularities of varying distribution to represent surface roughness.
- Apply geometric modifications and re-mesh for each case.
- Run transition SST simulation for one case at Reynolds number of 150000 across the full range of AoA from 0° to stall.
- Compare Cl, Cd, Cl/Cd and stall angle against the smooth baseline geometry to quantify performance degradation.

OBJECTIVE 5: SYNTHESIS FINDINGS AND PRODUCE ENGINEERING RECOMMENDATIONS

Analyse all simulation results and produce evidence-based conclusions and recommendations applicable to UAV aerodynamic design.

ACTIVITIES:

- Compare turbulence models across all tested conditions and determining which model provides the most reliable predictions for low-Reynolds number UAV airfoil analysis.
- Assess practical implications of leading-edge roughness findings in UAV component fabrication.
- Document all methodology, results, and analysis in structured final project report

4. Project Outcomes

The outcomes of the project based on the objectives are as follows:

- A structured comprehensive literature review documenting the knowledge on low Reynolds-number aerodynamics, laminar separation bubble behavior, transitional flow theory and known limitations of $k-\omega$ SST and Transition SST turbulence models for UAV airfoil applications.
- A fully verified and validated 2D ANSYS fluent computational model of the Eppler E387 airfoil, with mesh independence study, validation plots comparing all three-modelling approaches against datasets between Re of 150000 and 300000.
- A detailed comparative analysis report presenting lift coefficient, drag coefficient, Cl/Dd ratio, drag polar and stall angle predictions for laminar, $k-\omega$ SST and Transition SST turbulence models. The report will include plots and errors table compared to UIUC benchmark datasheet. Also include a critical discussion of where and why each model succeeds or fails the transitional flow regime.
- A study report of the sensitivity of the E387 aerodynamic performance to leading edge surface irregularity and varying lengths. This report will also measure how

surface irregularities may affect lift, drag, and stall behavior and compare with the other models.

- A proper engineering recommendations document providing evidence-based guidance on which turbulence model should be used as the minimum standard for low-Reynolds number UAV airfoil CFD analysis without significant aerodynamic performance degradation.
- A complete summarized presentation of the entire project covering objectives, methodology, key findings, and recommendations while demonstrating the ability to communicate CFD and analysis findings both clearly to people with technical and non-technical backgrounds.

5. How can this project be beneficial in the real world?

Currently there is a large adoption of small UAV over many sectors such as agriculture, surveillance, and defense applications. As mentioned earlier, the small UAV operates at a chord Reynolds number between 100,000 and 500,000, this causes conventional fully turbulent CFD models to give many inaccurate predictions. Even though the predictions given are inaccurate UAV developers tend to rely on the standard $k-\omega$ SST model because of familiarity with the model and because of its low computational cost, these reasons lead university groups and small companies to use it overlooking or not understand the inaccuracies it introduces.

This project is beneficial in the real world because it aims to quantify the accuracy that is lost when using the $k-\omega$ SST model and ignoring the transitional physics, with validation of results against UIUC wind tunnel data, and investigating whether additional computational cost is justified by using the $\gamma - Re_\theta$ model. Through quantifying the effects of realistic leading-edge surface imperfections due to many factors such as tape, molding defects, and 3D printing layer lines, which leads to a degradation in drag, lift, and stall behavior, the $\gamma - Re_\theta$ model informs manufacturing tolerance and surface finish requirements.

Target Audience	Benefit
UAV manufacturers	Providing guidance supported by evidence on the minimum acceptable turbulence model for low Reynold airfoil analysis, this leads to reduction of the usage of wind tunnel testing reducing costs.
Production and Quality Engineers	Providing Quantified tolerance data on how leading-edge surface irregularities degrade performance, informing maintenance and fabrication standards.
University Students	Through the completed report they will gain access to a documented and repeatable ANSYS Fluent methodology for low Reynolds airfoil simulation
Researchers and Academics	Access to a validated E387 benchmark case comparing $k-\omega$ SST and $\gamma - R_{e\theta}$ models.

6. What are the key questions the project attempts to answer?

These should be testable in some way.

1- How much does the Transition SST $\gamma - R_{e\theta}$ model compare to the $k-\omega$ SST model in terms of the improvement on prediction accuracy of lift coefficient, drag coefficient, and stall angle of the E387 airfoil, when the results are validated against the UIUC wind tunnel data at Reynolds numbers 150,000 and 300,000? (Objectives 2 and 3)

2- How does the formation and behavior of the laminar separation bubble on the E387 airfoil change with Reynolds number and angle of attack, and which modelling approach correctly captures its location, length, and effect on the pressure distribution? (Objectives 1 and 3)

3- Relative to the smooth baseline geometry, what is the quantitative effect of the leading-edge surface irregularities on the lift, drag, lift to drag ratio and stall angle of the E387 airfoil at $Re = 150,000$ and $300,000$. (Objective 4)

4- What turbulence modelling standard is recommended for low Reynolds number UAV airfoil CFD analysis, based on the measured prediction error through the experiment and the computational cost of each model? (Objective 5)

7. What Research Methods as well as primary and/or secondary data sources do you intend to use?

Stage 1 – Secondary research (understanding existing work):

A structured literature review of low-Reynolds-number aerodynamics, laminar separation bubble physics, and transition modelling theory using peer-reviewed journal papers, conference papers, and textbooks (e.g., Menter & Langtry's original Transition SST publications, Selig et al.'s low-speed airfoil studies).

Secondary data sources:

UIUC Low-Speed Airfoil Tests (LSATs) database: experimental C_l , C_d , and drag polar data for the E387 at $Re = 150,000$ – $300,000$, used as the validation benchmark.

UIUC Airfoil Coordinates Database: For geometry coordinates of E387 to use in model construction.

NASA Langley E387 experimental datasets for cross-checking benchmark quality.

Stage 2 – Model development and verification:

Construction of the 2D E387 Geometry in ANSYS and conducting a mesh independence study.

Stage 3 – Validation:

Run laminar, $k-\omega$ SST, and Transition SST $\gamma - R_{e\theta}$ models independently and validate predicted C_l , C_d , and drag polars against the UIUC experimental data from 0° AoA to stall. Quantify agreement using percentage error and error tables/plots.

Stage 4 – Primary data generation:

Primary data source: CFD results generated by the author steady state 2D RANS simulations across AoA 0° to beyond stall at $Re = 150,000$ and $300,000$, for all three modelling approaches, on both smooth and modified leading-edge geometries. Also,

extraction of C_l , C_d , C_l/C_d , drag polars, pressure coefficient distributions, skin friction distribution, and flow field contours.

Stage 5 – Analysis, evaluation, and iteration:

Comparative quantitative analysis of model error relative to experiment; sensitivity analysis of surface irregularity cases against the smooth baseline. Where discrepancies are found, refine mesh, boundary conditions, or solver settings and repeat continuing the measure analyze evaluate loop until no further meaningful improvement is achievable within the project constraints.

8. Please provide draft chapter heading for your report

Chapter 1 – Introduction

Chapter 2 – Literature Review

Chapter 3 – Computational Methodology

Chapter 4 – Model Validation Against UIUC Experimental Data

Chapter 5 – Turbulence Model Comparison Across Reynolds Numbers

Chapter 6 – Effect of Leading-Edge Surface Irregularities

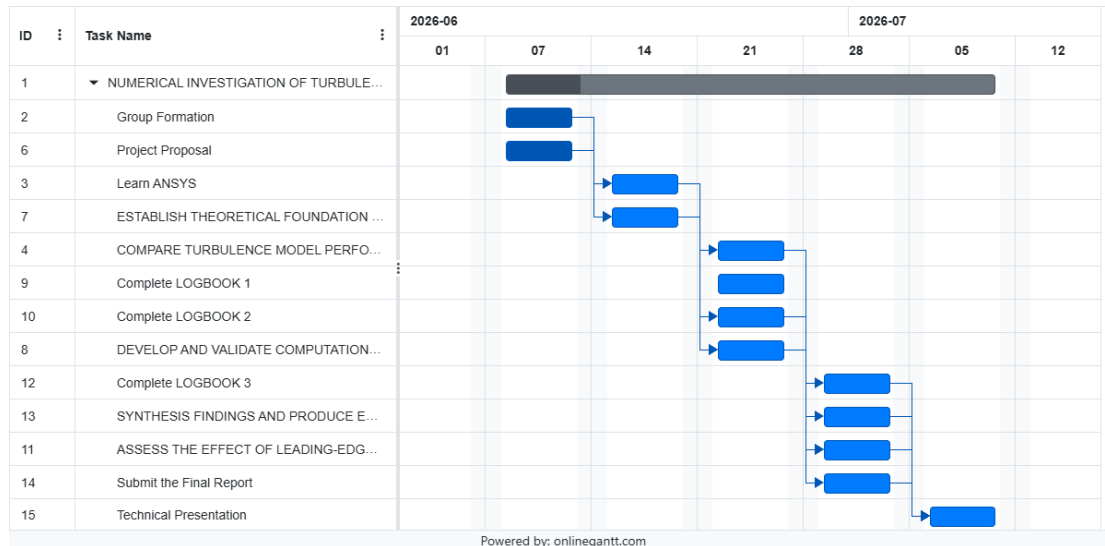
Chapter 7 – Synthesis, Discussion, and Engineering Recommendations

Chapter 8 – Conclusions and Future Work

References

Appendices

PART C – GANTT CHART



PART D – ACKNOWLEDGEMENT

I hereby acknowledge that I have reviewed and accepted the proposal the group has provided for their project specification.

Supervisor

Name: Dr. Walid Abou Hweij

Signature: _____

Date: _____

Student

Name: Sagar PrakashKumar Tailor

ID: 10153641

Date: 12/06/202

(Group leader)

Signature: Sagar.T

6

LECTURER USE ONLY

Date received:

Supervisor Comments:

10. AI USAGE INTERACTION LOG

The following examples show how AI was used during the learning stage of the experiment to support understanding of ANSYS. A user interaction log is attached, including selected conversations and advice obtained from ChatGPT.

User Interaction Log

Date: 2026-07-05

Time (approx.): 7:00 PM (UTC+4)

Platform: ChatGPT for Web (desktop)

Tool Used: GPT-5.5 Thinking (vision-enabled)

Uploaded File

- File Name: Screenshot(s) of ANSYS Fluent lift-force monitor, scaled residuals, and velocity-magnitude contour results for the E387 airfoil.
- Content: Transition SST simulation results at $Re = 200,000$ and 0° angle of attack, including convergence behaviour, wall y^+ checks, and flow-visualisation output.

User Actions

1. Updated the CFD study condition from $Re = 150,000$ to $Re = 200,000$ and requested a practical sequence for obtaining stable Transition SST convergence.
2. Asked where to locate and adjust pressure, momentum, turbulence, intermittency, and transition Reynolds-number under-relaxation factors in ANSYS Fluent.
3. Requested a step-by-step Hybrid Initialization and post-100-iteration procedure, including when to move individual equations from first-order to second-order discretisation.
4. Shared lift-force and scaled-residual plots for a 0° E387 Transition SST case and asked whether the solution had converged.
5. Asked how to show flow separation visually on the E387, then shared a velocity-magnitude contour and asked how to stop a pathline calculation that was not displaying.

AI Responses

- Recommended a staged solution approach: begin with lower under-relaxation factors and first-order schemes, then change momentum, k , and ω to second order only after the force monitors stabilise; move intermittency and $Re\theta$ to second order last.
- Specified Hybrid Initialization as the preferred starting method and advised against reinitialising when only solution methods or relaxation factors are changed.
- Interpreted the lift-force history as iteration-stable after approximately 120-150 iterations, while noting that drag-coefficient stability and residual behaviour must also be checked before declaring full convergence.
- Assessed the residual plot as stable overall: continuity remained around 10^{-4} - 10^{-3} while the momentum, turbulence, intermittency, and transition residuals remained low and bounded; small isolated spikes were judged acceptable because they decayed.
- Explained that the 0° velocity-magnitude contour showed largely attached flow with a narrow trailing-edge wake, rather than clear separation; recommended x -velocity contours, streamlines/pathlines, and signed wall-shear or skin-friction data to identify reverse flow and separation.
- Provided safe interruption methods for an unresponsive pathline display: press Esc, try Ctrl+C in the Fluent console, and use Task Manager only as a final option; advised using fewer inlet seeds and simple line pathlines for the next attempt.

User Response / Applications

- Will retain the converged 0° Transition SST solution as the attached-flow baseline at $Re = 200,000$, subject to confirming that CL and CD remain flat over the final iteration range.
- Will apply the staged first-order-to-second-order procedure consistently for subsequent angles of attack rather than changing all discretisation methods at once.
- Will use velocity-magnitude contours for overall flow-field presentation, but will support any separation claim using x -velocity, streamline/pathline, and wall-shear or C_f evidence.
- Will treat high-angle cases with persistent force oscillations separately as non-converged steady-state or separation-sensitive cases, rather than presenting a single unstable iteration as the final aerodynamic result.

User Interaction Log

Date: 2026-07-05

Time (approx.): 7:00 PM (UTC+4)

Platform: ChatGPT for Web (Desktop)

Tool Used: GPT-5.5 Thinking (vision-enabled)

Uploaded File

- Filename: IMG_655D5E61-376B-47DA-BA87-B79C32E7C4C4.jpeg
- Content: Screenshot from ANSYS Fluent showing the Solution Controls settings for an E387 airfoil CFD simulation. The panel included the flow Courant number, explicit relaxation factors, and turbulence-variable under-relaxation factors.

User Actions

1. Uploaded a screenshot of the ANSYS Fluent Solution Controls panel during the CFD convergence review.
2. Requested guidance on whether the numerical settings were suitable for E387 airfoil simulations, particularly where lift and drag coefficients oscillated at higher angles of attack.
3. Asked whether the settings could explain an unusually high lift coefficient at 0° angle of attack.

AI Responses

- Reviewed the displayed settings: Flow Courant Number = 200; Momentum = 0.5; Pressure = 0.5; Turbulent Kinetic Energy = 0.8; Specific Dissipation Rate = 0.8.
- Explained that a Courant number of 200 may be too aggressive for near-stall or separated-flow cases, where lift and drag monitors can oscillate.
- Recommended reducing the Courant number to approximately 30-50 for difficult high-angle-of-attack cases.
- Suggested reducing the turbulence-variable relaxation factors to around 0.5-0.6 if instability persisted.
- Clarified that solution controls alone would not explain a very high lift coefficient at 0°.
- Recommended verifying inlet velocity components, airfoil orientation, reference area, lift and drag directions, and the airfoil-wall selection in the report definitions.

User Response / Application

- Used the guidance to distinguish potential numerical instability from physically unsteady separated-flow behaviour near stall.
- Retained the original settings for stable low-angle-of-attack simulations.
- Considered lower Courant-number and turbulence-relaxation settings for simulations above approximately 10°-12°, where lift and drag plots showed sustained oscillations.
- Treated the high C_l result at 0° as a setup and force-reporting verification issue rather than attempting to correct it only by changing numerical relaxation settings.

User Interaction Log

Date: 2026-07-05

Time (approx.): 7:00 PM (UTC+4)

Platform: ChatOPT for Web (desktop)

Tool Used: GPT-3.5 Thinking (vision-enabled)

Uploaded File

- File Name: Screenshot(s) of ANSYS Fluent lift-coefficient monitors and velocity-contour results for the E387 airfoil.
- Content: CFD results for k- ω SST and Transition SST at $Re = 200,000$, including convergence behaviour at 4° , 11.5° , and 12° angle of attack.

User Actions

1. Asked whether the solution should be initialized when moving between angles of attack and whether previous contours would be overwritten.
2. Shared lift-coefficient histories showing stable convergence at lower angle of attack and persistent oscillations after changing to 11.5° and 12° .
3. Asked how to interpret velocity contours, particularly in relation to the UIUC E387 reference data at $Re = 200,000$ and $N_{crit} = 5$.
4. Checked inlet velocity specification method, velocity-component sign convention, lift/drag report vectors, turbulence intensity, and turbulent-viscosity ratio.
5. Confirmed that both k- ω SST and Transition SST exhibited persistent oscillations from 12° upward.

AI Responses

- Explained that previously converged solutions can be used as initial fields for the next angle of attack, but that each angle must be saved/exported before results are overwritten in Fluent.
- Identified the high-amplitude, repeating lift-coefficient oscillations at 11.5° – 12° as non-converged steady-state behaviour; advised against using a single final iteration as a valid aerodynamic coefficient.
- Clarified that the UIUC reference at $Re = 200,000$ and $N_{crit} = 5$ represents transition-sensitive low-Reynolds-number flow, whereas standard k- ω SST is a fully turbulent model.
- Recommended reporting oscillating cases as non-converged steady solutions and using lift/drag monitor histories, skin-friction contours, and upper-surface C_f versus x/c plots for qualitative separation analysis.
- Advised that a negative or near-zero skin-friction coefficient can indicate reversed wall shear or separation, while cautioning that an oscillatory steady solution does not establish one fixed separation point or prove full stall.
- Recommended keeping turbulence boundary conditions consistent across the angle sweep: intensity 0.5% and turbulent viscosity ratio 10, rather than changing them solely to force convergence.

User Response / Applications

- Recognised that lower-angle cases, such as 4° , can produce stable coefficient monitors, while 12° and above require separate treatment because both turbulence-model solutions remain oscillatory.
- Will avoid labelling the 12° condition as confirmed stall solely from the monitor history; instead, it will be described as a separation-sensitive, non-converged steady-state condition.
- Will use converged angles normally in the CL -AoA and CD -AoA polars, while presenting high-angle oscillatory cases with explicit non-convergence labels or iteration-averaged values only where required.
- Will support the high-angle discussion with skin-friction results and C_f versus x/c plots to assess developing suction-side separation qualitatively.